

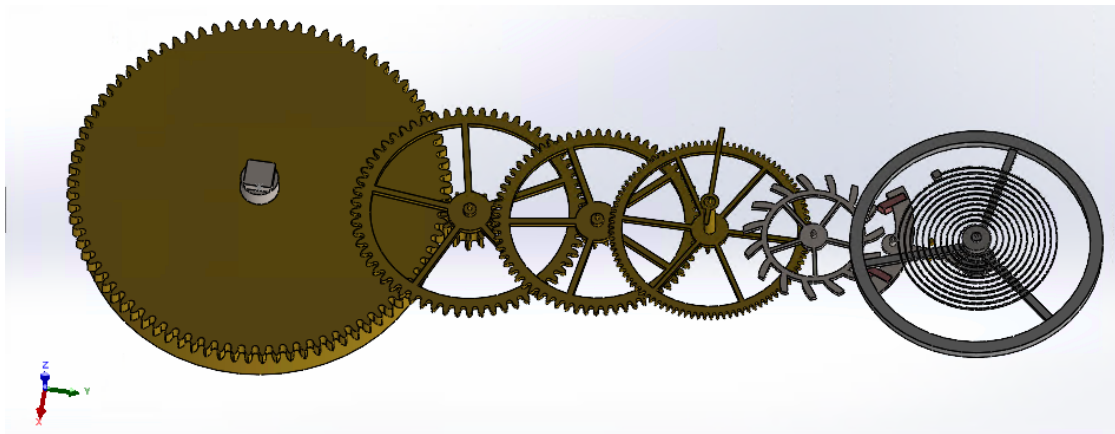
Design, Analysis and Simulation of a Mechanical Watch Going Train with a Swiss Lever Escapement

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The complete assembly, from the barrel (left) to the balance (right): thirteen unique parts on one line of centres.

Videos. Complete train running (Study C): https://www.youtube.com/watch?v=dhshj_wPEz8

Escapement close-up: https://youtu.be/M3oqJB_sfAk

Abstract

This report presents the design, analysis and simulation of the basic mechanism of a mechanical watch: a mainspring barrel, a three-wheel going train, a Swiss lever escapement and a balance with its hairspring. The mechanism comprises thirteen unique parts, modelled parametrically in SolidWorks from a single file of 393 linked equations. A train ratio of 6720 gives a rate of 28 800 beats per hour, with the second wheel turning once per hour and the fourth once per minute. All teeth follow the ogival NIHS 20-30 form; the eight profiles were generated with an external open-source tool and reconstructed as constrained sketches that reproduce the generated flanks to better than a nanometre and the roots to within $1.5\text{ }\mu\text{m}$. A contact analysis and a 1000-sample Monte Carlo study of each mesh fixed the tooth widths and the centre-distance tolerances. Checking the design against first principles led to several revisions. The original barrel held the balance above 150° for only about 19 hours, against a predicted 185° at 40 hours after redesign. A contact pressure of 913 MPa required the barrel tooth width to be doubled, a plate analysis required a thicker, clamped barrel lid, and a press-fit analysis showed that the wheels must be riveted. The mechanism was then run as a rigid-body contact simulation, first as an isolated escapement and then as a complete train driven from the barrel, in which the balance period was 0.2498 s against a design value of 0.2500 s and the train followed the escapement at its exact ratios. Every calculation is worked step by step in the appendices, and the requirements for manufacture are set out.

Keywords: mechanical watch, going train, Swiss lever escapement, ogival gearing, NIHS 20-30, mainspring, hairspring, multibody contact simulation, tolerance analysis.

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Nomenclature

Lengths are in millimetres, forces in newtons, torques in N mm and stresses in MPa throughout, which is the unit set used by the CAD model. Names in **typewriter** are variables of the equations file.

Gear train and tooth form (Sections 5 to 7)

Z, p	teeth of a wheel, leaves of a pinion
m	module, pitch diameter divided by tooth count
a	centre distance of a mesh, $a = m(Z + p)/2$
R_{b4}, R_{4e}	speed ratio barrel to fourth wheel, fourth wheel to escape wheel
r_p, r_a, r_f	pitch, tip and root radius
h_a, h_f, s^*	addendum, dedendum and pitch-circle thickness factors, in modules
R_o, R_t, ρ	ogive radius, pinion tip radius, root fillet radius
η_i, η_t	efficiency of mesh i , efficiency of the whole train
p_H	maximum Hertz contact pressure

Springs and oscillator (Sections 8 and 9)

R, r	barrel inner radius, arbor core radius
b, t, L	height, thickness and length of a spring strip
K, k	mainspring stiffness, hairspring stiffness (per radian)
$M(n), M_{\max}, M_0$	barrel torque after n turns, at full wind, when let down
σ_d	design bending stress of the mainspring at full wind
N	barrel turns between let-down and full wind
I, f	balance moment of inertia, balance frequency
θ	balance amplitude (half swing from rest)
Q, η_e	oscillator quality factor, escapement efficiency
T_E	torque at the escape wheel
φ	escape wheel rotation per balance period, $2\pi/Z_e$

Strength and machine elements (Section 10)

$R_{p0.2}, R_m$	0.2 % proof strength, tensile strength
S_e, K_f	endurance strength at the stated cycle count, fatigue notch factor
SF	safety factor
E^*, R'	contact modulus, relative radius of curvature
δ	diametral interference of a press fit
D	flexural rigidity of a plate, $Et^3/12(1 - \nu^2)$
n_g	shock level as a multiple of g

Motion simulation (Section 11)

$k_c, e, c_{\max}, d$	contact stiffness, force exponent, maximum damping, full-damping penetration
μ_k, v_k	kinetic friction coefficient, friction transition velocity
c	viscous damper on the balance standing for pivot and air losses
$h_{\max}$	maximum integrator step

1 Introduction

1.1 Project Overview

A mechanical watch keeps time without electricity. A coiled spring stores a fraction of a joule (about 0.15J in this design), a train of gears carries that energy to an oscillator, and a small mechanism called the escapement lets the gears advance by exactly one step for every swing of the oscillator. Everything else in a watch, from the hands to the calendar, is driven from this chain.

This project models that chain and nothing else: the mainspring barrel, the going train, a Swiss lever escapement and a balance with its hairspring. It comprises thirteen unique parts, built in SolidWorks and assembled on a single line of centres so that the whole energy path can be seen at once. The aim was not a wearable movement. It was to understand why each dimension of a watch is what it is, by deriving the dimensions rather than copying them, and then to check that the resulting mechanism actually runs.

The concept and the starting requirements come from Bartosz Ciechanowski's illustrated article on the mechanical watch [1], which explains the same chain of barrel, train, lever escapement and balance through interactive models. I used the article as reading and as design inspiration only. Its models cannot be downloaded and none of its geometry, models or code is reused here: every part in this report was dimensioned from the calculations that follow and modelled from an empty file. What the article supplied is the target that those calculations had to meet, a watch beating 28 800 times per hour that runs for about 40 hours and shows seconds, minutes and hours.

1.2 Scope

The following are modelled, analysed and simulated:

- the barrel drum with its lid, arbor and mainspring;
- the second, third and fourth wheels, each with its pinion and arbor, and a seconds hand on the fourth arbor;
- the escape wheel with its pinion, the pallet fork with its two ruby stones and guard dart, and the banking pins;
- the balance with its staff, impulse roller, roller jewel and safety roller, and the hairspring with its collet and stud.

The following are deliberately outside the model. They matter for a real watch and are discussed where they limit a conclusion:

- the main plate, bridges, jewels and end stones that carry the pivots. Every arbor in the model turns about an ideal axis;
- shock protection for the balance;
- the winding and setting mechanism, the click, and the motion works that drive the hour and minute hands. The tooth counts of the motion works are derived in Section 5, but the parts are not built;
- finite element analysis of the two springs. The spring behaviour used here comes from beam theory.

1.3 Methodology

Three decisions shaped the work.

One parametric source. Every dimension of every part is a named variable in one text file that all parts link to. Changing the module of one mesh, or the height of the mainspring, updates

the tooth form, the centre distances, the barrel size and the positions of every downstream part. This made it practical to revise the design when a calculation showed that it had to change, which happened several times.

Derive, then check. The tooth counts, modules, spring dimensions and escapement angles were first derived from the requirements. Each was then checked as a machine element: tooth bending, contact pressure, pivot stress under running load and under shock, joints, plates and wear. Section 4.3 lists the cases where the check failed and what was changed.

Simulate at the smallest useful scale. Three rigid-body motion studies were run in SolidWorks Motion: a kinematic check of the gear ratios (Study A), the escapement alone under a constant torque (Study B), and the whole train driven from the barrel (Study C). The escapement study was done in a separate four-part assembly because impacts between a tooth and a ruby stone last microseconds and force the integrator to very small steps.

1.4 Organisation of the Report

The body of the report is written to be read straight through. Section 2 explains the mechanism and its vocabulary for a reader who knows mechanical engineering but not watches. Sections 3 and 4 describe the model and the requirements. Sections 5 to 7 cover the gears, Sections 8 and 9 the springs, escapement and oscillator, Section 10 the strength checks, Section 11 the motion studies and Section 12 manufacture. Section 13 states the limitations and the future work, Section 14 the position with respect to standards, and Section 15 concludes. Each calculation in the body states the governing formula first, then the inputs, the result and what the result meant for the design.

The appendices repeat every calculation with the numbers substituted line by line, so that any result can be followed by hand. Appendix F groups the variables of the equations file by subsystem.

All numerical results were recomputed independently from the equations file while writing this report. Values taken from literature rather than calculated, such as material strengths and the efficiency of the escapement, are identified as such, together with the source they were checked against (Section 4.2). Results of the motion studies are those exported from Studies A and C; quantities that were not exported are identified in Section 11.5.

2 Principle of Operation

This section introduces the mechanism and the terms used in the rest of the report. Figure 1 shows the energy path and Figure 2 the escapement and balance; the full train is on the title page.

2.1 Energy Transmission Path

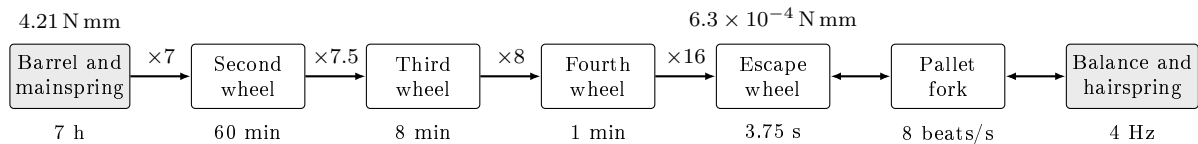


Figure 1: Energy path of the mechanism. The speed multiplication of each mesh is shown above the arrows and the period of rotation of each arbor below it. Torques are the full-wind values without losses. Energy flows from left to right through the train; between the escape wheel, the pallet fork and the balance the interaction is intermittent and in both directions, since the balance unlocks the train and the train then gives impulse to the balance.

Barrel and mainspring. The mainspring is a long, thin strip coiled inside a toothed drum, the *barrel*. Its inner end hooks onto the *barrel arbor*, which is held still while the watch runs, and its outer end bears on the drum wall. As the spring unwinds it turns the drum, and the teeth on the drum's rim drive the first pinion of the train. The barrel is slow and strong: in this design it turns once in seven hours and delivers about 4 Nmm when fully wound.

Going train. Each *wheel* (the large gear, in brass) drives a *pinion* (the small gear, in steel, whose teeth are called *leaves*) on the next arbor. Because the wheel always drives the pinion, speed is multiplied and torque divided at every stage. The second wheel turns once per hour and would carry the minute hand, the third wheel is an intermediate, and the fourth wheel turns once per minute and carries the seconds hand. By the time the motion reaches the escape wheel, the speed has been multiplied 6720 times and the torque has fallen to a few ten-thousandths of a newton millimetre.

Balance and hairspring. The *balance* is a small flywheel on a *staff*. A flat spiral spring, the *hairspring*, is fixed to the staff at its inner end (through a *collet*) and to the frame at its outer end (the *stud*). Together they form a torsional oscillator whose frequency depends only on the inertia of the balance and the stiffness of the spring. Here the frequency is 4 Hz. The *amplitude* is the angle the balance swings to either side of rest; a healthy watch runs between about 220° and 300°.

2.2 Swiss Lever Escapement

If the train were connected directly to nothing it would unwind in seconds. The escapement holds it and releases it one step at a time, and uses each release to give the balance a small push.

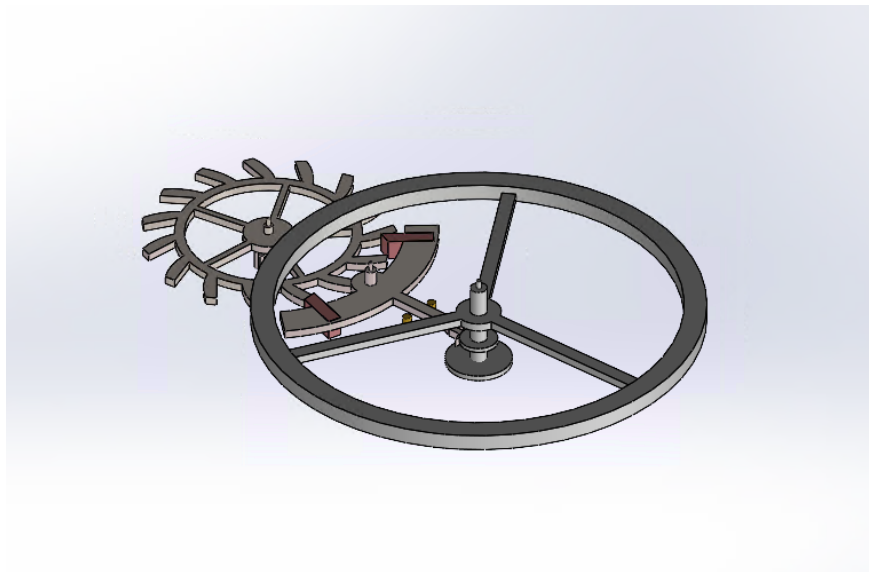


Figure 2: The escapement and balance as modelled. The escape wheel (left) has 15 club-shaped teeth. The pallet fork carries two ruby stones that alternately lock a tooth, and its forked end engages the roller jewel on the balance staff. The two banking pins limit the travel of the fork.

The *escape wheel* has 15 teeth of a characteristic club shape. The *pallet fork* (or lever) pivots between the escape wheel and the balance. It carries two ruby *pallet stones*, the entry stone and the exit stone, and ends in a fork. A ruby pin on the balance staff, the *roller jewel* (or impulse pin), enters the fork once per swing. One swing of the balance in one direction is a *beat*, and proceeds as follows.

1. *Locked.* A tooth of the escape wheel rests against the side face (the *locking face*) of one stone. The lever rests against a *banking pin*. The balance is free and swings on its own;

this is why the lever escapement is called a detached escapement.

2. *Unlocking*. As the balance returns through its rest position, the roller jewel enters the fork and pushes the lever. The stone slides along the tooth until the tooth's corner reaches the end of the locking face.
3. *Impulse*. The tooth now slides across the inclined end face of the stone (the *impulse face*), and then the stone slides across the inclined top of the tooth. The escape wheel pushes the lever, and the fork pushes the roller jewel, so the balance receives energy.
4. *Drop*. The tooth leaves the stone. The wheel is momentarily free and turns until another tooth lands on the locking face of the other stone.
5. *Run to banking*. The lever continues a little further until it rests on the other banking pin. The balance swings on, reverses, and the sequence repeats mirrored.

Two beats make one full oscillation, and in two beats the escape wheel advances by one tooth. A 4 Hz balance therefore makes 8 beats per second, or 28 800 beats per hour (bph).

A recording of the escapement running in the motion study described in Section 11 is available at https://youtu.be/M3oqJB_sfAk [2].

2.3 Terminology

The escapement is described by a small set of angles, which become design variables in Section 9.

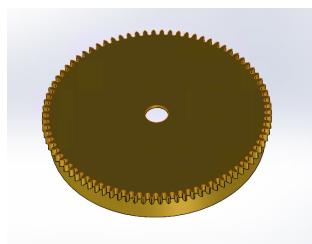
Lock	the angle the lever must turn before a locked tooth is released. It is the depth by which the tooth overlaps the stone.
Draw	the inclination of the locking face that makes the tooth pull the stone inwards, so that the lever is held against its banking pin and cannot drift.
Impulse	the lever angle during which the tooth drives the stone.
Drop	the free rotation of the escape wheel between leaving one stone and landing on the other. It is needed for clearance and it is lost energy.
Slide	the lever travel after the drop, until it reaches the banking pin; also called run to banking.
Lift angle	the angle of balance rotation during which the roller jewel is engaged in the fork.
Safety action	a guard pin (<i>dart</i>) on the lever and a notched disc (<i>safety roller</i>) on the balance staff, which together prevent the lever from crossing to the wrong side while the balance is away.

Two terms from gearing are specific to horology. An *ogival* tooth has flanks that are circular arcs meeting in a point, like a Gothic arch, instead of involutes. The *module* keeps its usual meaning, pitch diameter over tooth count, but watch modules are of the order of 0.1 mm. Finally, a *pivot* is the thin end of an arbor that runs in a bearing, in a real watch a pierced ruby *jewel*.

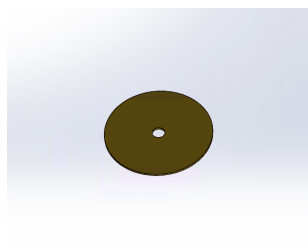
3 System Description and Parametric Model

3.1 Component Description

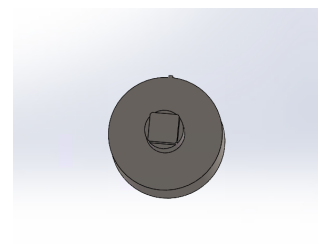
Figure 3 shows the thirteen unique parts and Table 1 lists them with their material and principal dimensions. Several parts are multi-body: the pallet fork is a steel lever with two ruby stones, and the balance is a copper-beryllium wheel with a ruby roller jewel. Each wheel is modelled in one piece with its arbor, pinion and pivots, which is how the equations file organises them, although in manufacture the brass wheel and the steel pinion are separate (Section 12).



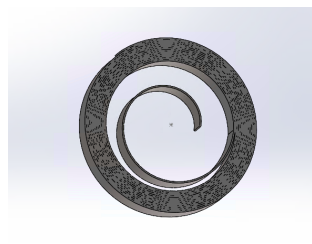
(a) Barrel drum



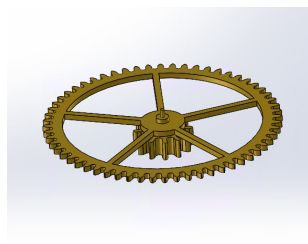
(b) Barrel lid



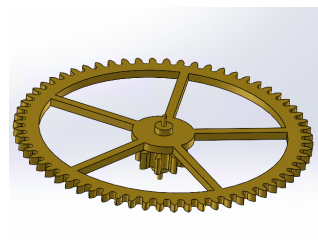
(c) Barrel arbor



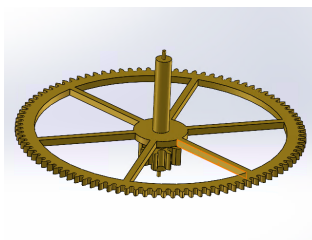
(d) Mainspring



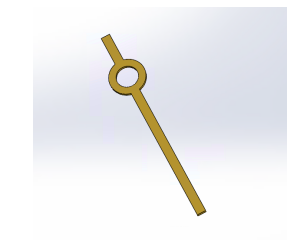
(e) Second wheel



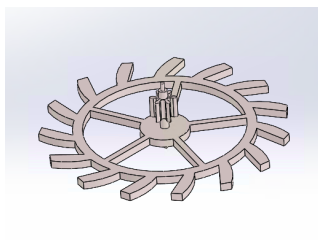
(f) Third wheel



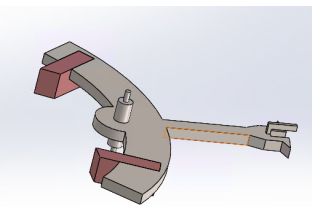
(g) Fourth wheel



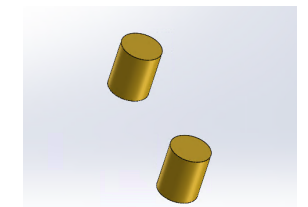
(h) Seconds hand



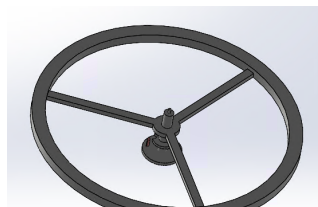
(i) Escape wheel



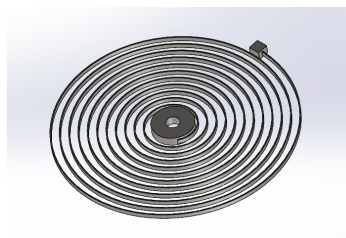
(j) Pallet fork



(k) Banking pins



(l) Balance



(m) Hairspring

Figure 3: The thirteen unique parts of the mechanism, rendered from the SolidWorks models. Parts are not shown to a common scale.

Table 1: Parts list. Dimensions in mm at scale factor $S = 1$.

Part (file)	Material	Principal dimensions
Barrel drum (<code>barrel</code>)	Brass	84 teeth, $m = 0.140$, tip $\varnothing 12.152$, height 1.88, teeth 0.50 wide
Barrel lid (<code>barrel_lid</code>)	Brass	thickness 0.20, snapped into a groove in the drum
Barrel arbor (<code>barrel_arbor</code>)	Steel	core radius 1.761, shaft $\varnothing 1.20$, winding square 0.90
Mainspring (<code>mainspring</code>)	Nivaflex 45/5	strip $0.095 \times 1.40 \times 410.1$
Second wheel (<code>wheel_2</code>)	Brass, steel	60 teeth, $m = 0.120$; pinion 12 leaves, $m = 0.140$; pivots $\varnothing 0.12$
Third wheel (<code>wheel_3</code>)	Brass, steel	64 teeth, $m = 0.100$; pinion 8 leaves, $m = 0.120$; pivots $\varnothing 0.10$
Fourth wheel (<code>wheel_4</code>)	Brass, steel	96 teeth, $m = 0.065$; pinion 8 leaves, $m = 0.100$; pivots $\varnothing 0.08$
Seconds hand (<code>seconds_hand</code>)	Steel	length 3.5, on the fourth arbor
Escape wheel (<code>escape_wheel</code>)	Steel	15 club teeth, radius 2.20; pinion 6 leaves, $m = 0.065$; pivots $\varnothing 0.07$
Pallet fork (<code>pallet_fork</code>)	Steel, ruby	centre distance to escape wheel 2.540, fork length 2.20, stones 0.25×0.80
Banking pins (<code>banking_pins</code>)	Steel	two pins $\varnothing 0.16$
Balance (<code>balance</code>)	Glucydur, ruby	rim $\varnothing 8.0$ outside, $\varnothing 7.2$ inside, three arms, inertia 3.60 mg cm^2
Hairspring (<code>hairspring</code>)	Nivarox	strip $0.025 \times 0.10 \times 108.8$, 11.35 turns

3.2 Assembly Layout

A real movement folds its train into a circle to save space. Here all seven axes lie on one straight line in plan, and each mesh sits on its own level, 0.5 mm above the previous one (Table 2). The arrangement wastes space but makes every mesh visible, keeps every centre distance equal to its theoretical value, and means that the position of each axis is simply the running sum of the centre distances before it. The total length from the barrel axis to the balance axis is 22.93 mm.

Table 2: Position of each axis along the line of centres and the level of the mesh it drives.

Axis	X (mm)	Level Z (mm)	What sits at that level
Barrel	0	0	barrel teeth and second pinion
Second wheel	6.720	0.5	second wheel and third pinion
Third wheel	10.800	1.0	third wheel and fourth pinion
Fourth wheel	14.400	1.5	fourth wheel and escape pinion
Escape wheel	17.715	2.0	escape wheel teeth
Pallet fork	20.255	2.19	pallet plate, just above the escape wheel
Balance	22.926	2.19	fork, roller jewel and safety roller

3.3 Parametric Equation File

The model is driven by one text file, `watch_global.txt.txt`, which every part and assembly links to. It holds 393 named variables. Of these, 192 are independent inputs (tooth counts, modules, spring cross-sections, escapement angles, thicknesses); the remaining 201 are derived from them by formulas that SolidWorks evaluates. No dimension in any sketch is typed as a number: each is linked to a variable or to an expression of variables. Three consequences were important in practice.

- *Changes propagate.* When the barrel module was raised from 0.120 to 0.140 mm (Section 4.3), the barrel diameter, the inner radius available to the spring, the spring length, the first centre distance and the plan position of every other part followed without any sketch being edited.
- *Calculations live with the geometry.* Quantities that are not dimensions, such as the main-spring torque at 24 hours or the predicted balance amplitude, are also variables in the file. They update with the geometry and serve as check values.
- *The design scales.* Every length is multiplied by a scale factor S , set to 1. Tooth counts, angles and ratios are not scaled.

Built-in checks guard against inconsistent edits. The variable `bph_check`, defined as $60 R_{4e} Z_e \cdot 2$, must read 28 800, and `T_2_min` must read 60, the period of the second wheel in minutes.

3.4 Material Selection

Table 3 lists the materials. SolidWorks has brass and alloy steel in its library; the four watch alloys were entered as custom materials. Three sets of values have to be kept apart: what the CAD library holds (used only for mass and inertia), what the real alloy offers, and what the strength calculations in this report assume.

Table 3: Materials. Density ρ in kg/m³, moduli in GPa, strengths in MPa.

Material and use	In the CAD model	Real alloy	Used in calculations
Brass: wheels, barrel, lid	ρ 8500, E 100	CuZn39Pb2 (CW612N), hard rolled. Temper R490 of EN 1652, as tabulated for the closely related CuZn39Pb3, gives $R_{p0.2} \geq 420$, R_m 490 to 570 [3]	$R_{p0.2}$ 400, R_m 550, S_e 150 at 10^8 cycles
Steel: pinions, arbors, pivots, escape wheel, lever, pins	ρ 7700, E 210	hardened and tempered carbon steel, about 700 HV	$R_{p0.2}$ 1600, R_m 2000, S_e 800 (burnished pivots)
Glucydur: balance	ρ 8300, E 130	age-hardened copper-beryllium, non-magnetic, $\alpha \approx 17 \times 10^{-6} \text{ K}^{-1}$	$R_{p0.2}$ 1000
Ruby: stones, roller jewel	ρ 3980, E 345	synthetic corundum, E 345 to 430 by orientation	E 400 in contact calculations
Nivarox: hairspring	ρ 8150, E 190	Fe-Ni-Cr-Ti-Be alloy with a low thermoelastic coefficient	E 190, $R_{p0.2}$ 1300, S_e 450
Nivaflex 45/5: mainspring	ρ 8500, E 220, ν 0.34	Co-Ni-Cr alloy, cold worked and aged; E 220, ρ 8500, strength up to about 3000 [4, 5]	E 220, $R_{p0.2}$ 2475, design stress 2000

Only one result in the whole model depends on a library density: the inertia of the balance, which fixes the hairspring stiffness and so its length. The stiffness of both springs comes from variables in the equations file, not from the material library.

The brass values require qualification, because the barrel teeth turn out to be limited by contact pressure (Section 7). A proof strength of 400 MPa is realistic for hard-rolled strip, from which wheels are blanked, and is consistent with temper R490 (the table located is for CuZn39Pb3, a closely related leaded brass). It is not realistic for brass bar in a softer temper, where published values for the same alloy range from 140 to 320 MPa. The barrel drum is turned from bar, so its material has to be ordered in a hard temper for the margins in this report to hold.

4 Design Requirements, Assumptions and Principal Findings

4.1 Design Requirements

Table 4 lists the requirements. The rate, the power reserve and the time display were given at the outset, and are those of the reference watch described by Ciechanowski [1]. The strength and life targets were set during the analysis, so that every check had a stated pass criterion.

Table 4: Design requirements.

Requirement	Value
Rate	28 800 bph, a 4 Hz balance
Power reserve	at least 40 hours, with an amplitude of at least 150° at 40 hours (taken as the practical running limit) and a target of 180°
Amplitude	at most 315° at full wind, to keep a margin against rebanking; at least 200° after 24 hours
Time display	fourth wheel once per minute (seconds hand); second wheel once per hour (minute hand)
Tooth form	NIHS 20-30 ogival profiles on every mesh of the train
Strength	the safety factors of Table 19: 2.0 on yield for running loads, 1.5 to 2.0 on yield under a $5000g$ shock, 1.5 in fatigue at the ten-year cycle count, and 1.2 on first yield in contact
Service life	ten years between services, which is 1.26×10^9 oscillations of the balance

The amplitude limits need a word of explanation. Below about 150° a watch still ticks but its rate becomes poor and it is close to stopping. Above about 315° , with a lift angle of 52° , the roller jewel swings so far round that it strikes the outside of the fork, which is called rebanking or knocking. The design therefore has to deliver enough torque to stay above the lower limit when the spring is nearly run down, without exceeding the upper limit when it is fully wound.

4.2 Design Assumptions and Their Verification

Several inputs cannot be derived from geometry. Table 5 lists them with the value used and the outcome of checking each against a published source. Where the check could only narrow a value to a range, the consequences of the range are carried through in the sensitivity study of Section 9.2.

Two classes of input could not be verified from public sources and are stated as such where they are used: endurance strengths for the watch alloys at 10^8 to 10^9 cycles, for which I found no published S-N data, and the contents of the NIHS standards, which are sold by the Swiss watch industry federation and were not obtained.

4.3 Design Revisions Resulting from the Analysis

The first complete version of the model was geometrically consistent and would have assembled, but it would not have functioned as a timekeeper. Table 6 summarises the defects that the calculations exposed, in the order in which they were found, and the change that resolved each. They are the main result of the project, and each is derived in the section indicated.

Table 5: Assumptions, and the result of checking each one.

Assumption	Value used	Check
Quality factor of the balance, Q	250	A Swatch Group patent gives “200 for a usual watch” [6]. The nominal value is therefore somewhat optimistic; the design still meets the reserve requirement at $Q = 200$ (Table 18).
Efficiency of the Swiss lever escapement, η_e	0.30	Confirmed: patent literature from ETA states “around 30 %” and attributes the loss to drops and sliding on inclined planes [7].
Efficiency of the train, η_t	0.70	Partly confirmed. The four meshes together give 0.862 in the contact simulation (Section 7). The remaining factor of 0.81 is attributed to pivot friction and is an estimate.
Mesh friction coefficient	0.20	Default of the simulator, on the high side for oiled brass on polished steel. It makes the efficiency and contact results conservative.
Mainspring alloy	$E = 220$ GPa, $\rho = 8500$, strength up to ≈ 3000 MPa	Confirmed against the manufacturer’s data for Nivaflex 45/5 [4, 5].
Brass strength	$R_{p0.2} =$ 400 MPa	Consistent with hard-rolled strip: temper R490 gives $R_{p0.2} \geq 420$ in the table found, which is for the closely related CuZn39Pb3 [3]; the value for CuZn39Pb2 itself was not located. Not valid for soft bar (Section 3.4).
Yield in line contact	$p_H =$ $1.67 R_{p0.2}$	Confirmed. In Hertz line contact the maximum shear stress is $0.30 p_H$, and the Tresca criterion $\tau_{\max} = R_{p0.2}/2$ gives $p_H = R_{p0.2}/0.60$ [8].
Shock level	5000 g	Consistent with the figure commonly quoted for the ISO 1413 pendulum test, a 3 kg hammer at 4.43 m/s simulating a 1 m fall [9]. The standard itself was not purchased, so the value remains an engineering assumption.
Mainspring torque law	linear in barrel turns, capped by bending stress	Follows classical strip-spring theory [10]. It has not been checked by finite elements or by test (Section 8.1).

Table 6: Defects found by analysis and the design changes that resolved them.

Subject	Finding	Change	Section
Power source	With the original barrel the predicted amplitude was 196° at full wind and 135° after 24 hours; it fell below 150° after about 19 hours, not 40.	Barrel module $0.120 \rightarrow 0.140$; spring height $1.00 \rightarrow 1.40$. Prediction 252° , 214° , 185° at 0, 24 and 40 hours.	8.1, 9.2
Mainspring model	The torque model set the full-wind torque to $2\pi KN$, which implies zero torque when let down, used $E = 200$ GPa, and did not check the coils wrapped on the arbor.	Stress-capped linear model with $E = 220$ GPa; a required preset radius for the inner coils.	8.1
Time display	With 54 teeth on the second wheel it turned once in 54 minutes, so no arbor could carry a minute hand.	$Z_2 = 54 \rightarrow 60$. The second wheel turns once per hour and the barrel once in 7 hours.	5
Tooth form	One estimated set of tooth proportions was shared by all meshes, with dedendums deeper than the standard.	One NIHS 20-30 set per mesh, taken from generated profiles.	6
Barrel teeth	The contact simulation gave 913 MPa at full wind on 0.25 mm wide brass teeth, above first yield at about 670 MPa.	Tooth width $0.25 \rightarrow 0.50$ and second pinion length $0.40 \rightarrow 0.60$, giving about 646 MPa.	7
Second pivot	The stronger spring raised the running load on the second arbor by 59 %.	Pivot $\varnothing 0.10 \rightarrow 0.12$.	10.3
Barrel lid and base	Under a 5000 <i>g</i> shock the weight of the spring would bend a 0.10 mm lid resting on its step to 1745 MPa, a safety factor of 0.23.	Lid 0.20 and snapped into a groove (2.53); base $0.15 \rightarrow 0.18$ (2.05).	10.8
Wheel joints	A press fit of the second wheel on its arbor needs between 1.32 and $1.68 \mu\text{m}$ of interference, which cannot be machined.	Train wheels to be riveted on a pinion shoulder.	10.6
Balance pivots	At 5000 <i>g</i> the balance pivots reach 829 MPa, a safety factor of 1.93 with rigid bearings.	Not resolved in the model: shock-absorbing jewel settings are required.	10.3

5 Gear Train

5.1 Transmission Ratio Requirement

The train has to connect three fixed speeds. The balance beats at 28 800 bph, the fourth wheel must turn once per minute to carry the seconds hand, and the second wheel must turn once per hour to carry the minute hand. The barrel should turn slowly enough that the spring lasts 40 hours within its available turns.

Each tooth of the escape wheel accounts for two beats, so the speed of the escape wheel follows from the beat rate and its tooth count Z_e :

$$n_e = \frac{\text{bph}}{2 Z_e} \text{ rev/h}, \quad R_{4e} = \frac{n_e}{n_4} = \frac{\text{bph}}{2 Z_e \cdot 60}. \quad (1)$$

With $Z_e = 15$, the classical tooth count for a Swiss lever, the escape wheel turns at 960 rev/h or 16 rev/min, so $R_{4e} = 16$. Between the second and the fourth wheel the ratio must be exactly 60. Between the barrel and the second wheel the ratio is free; it sets the hours per barrel turn.

5.2 Selection of Tooth Numbers

For a train in which wheels drive pinions, the angular speed and the ideal torque change at each mesh as

$$\omega_{i+1} = \omega_i \frac{Z_i}{p_{i+1}}, \quad T_{i+1} = T_i \frac{p_{i+1}}{Z_i} \eta_i, \quad (2)$$

where Z_i is the tooth count of the driving wheel, p_{i+1} the leaf count of the driven pinion and η_i the efficiency of the stage. The chosen counts are given in Table 7.

Table 7: Tooth counts, modules and centre distances of the four meshes.

Mesh	Z	p	Ratio	m (mm)	Pitch $\varnothing$, wheel / pinion (mm)	a (mm)
1. Barrel / second pinion	84	12	7	0.140	11.760 / 1.680	6.720
2. Second wheel / third pinion	60	8	7.5	0.120	7.200 / 0.960	4.080
3. Third wheel / fourth pinion	64	8	8	0.100	6.400 / 0.800	3.600
4. Fourth wheel / escape pinion	96	6	16	0.065	6.240 / 0.390	3.315

The overall ratios are

$$R_{b4} = \frac{84}{12} \cdot \frac{60}{8} \cdot \frac{64}{8} = 7 \times 7.5 \times 8 = 420, \quad R_{4e} = \frac{96}{6} = 16, \quad R_{b4} R_{4e} = 6720, \quad (3)$$

and the rate check closes exactly:

$$\text{bph} = 60 R_{4e} Z_e \cdot 2 = 60 \times 16 \times 15 \times 2 = 28\,800. \quad (4)$$

The reasoning behind the individual choices is as follows.

Meshes 2 and 3 must multiply to 60. The pair 7.5×8 uses two 8-leaf pinions, which keeps the pinions of the two middle stages identical in form (they share one set of NIHS proportions, Section 6) and avoids ratios above 8, which would need either a very large wheel or a pinion of 6 leaves.

The second wheel originally had 54 teeth. The first version of the train used $54/8 \times 64/8 = 54$, which gave the right rate at the fourth wheel but a second wheel that turned once in 54 minutes. No arbor then turned once per hour. Changing Z_2 to 60 corrected this at the cost of a larger second wheel (pitch diameter $6.48 \rightarrow 7.20$ mm) and a slower barrel.

Mesh 1 sets the barrel period. With a ratio of 7 the barrel turns once in $R_{b4}/60 = 7$ hours. A reserve of 40 hours then needs $40/7 = 5.71$ turns of the barrel, comfortably within the 8.75 turns

that the spring provides (Section 8.1). The 12-leaf pinion was chosen for the first mesh because it carries the highest tooth load in the train, and a pinion with more leaves has a stronger core and smoother action.

Mesh 4 uses a 6-leaf pinion. A ratio of 16 in one stage is high. With 96 teeth on the fourth wheel a 6-leaf escape pinion keeps the wheel at about the same diameter as the third wheel. A 6-leaf pinion engages largely before the line of centres, where friction is highest, and this mesh does show the lowest efficiency and the largest torque ripple in Section 7. The alternatives 112/7 and 128/8 give the same ratio with better action and are discussed in Section 12.

5.3 Modules and Centre Distances

The centre distance of an ogival mesh without profile shift is

$$a = \frac{m(Z + p)}{2}. \quad (5)$$

The module falls from 0.140 mm at the barrel to 0.065 mm at the escape pinion. The torque falls at every stage, by a factor of 7 to 8 at the first three meshes and of 16 at the last, so the teeth can become finer in step, and finer teeth allow the high tooth counts needed for the ratios within wheels of 6 to 7 mm diameter. The barrel module is the one module that was set by something other than tooth strength: it fixes the diameter of the barrel and therefore the volume available to the mainspring. It was raised from 0.120 to 0.140 mm to make room for a spring that could deliver 40 hours (Section 8.1).

5.4 Angular Speed, Torque and Power

Table 8 gives the speed of each arbor, the torque it carries when the spring is fully wound and no losses are assumed (the load case for strength), and the torque and power at half reserve with losses (the load case for the escapement). The half-reserve figures apply to each stage its simulated mesh efficiency from Section 7 and an equal share of the pivot loss, so that the product over the four stages is the assumed $\eta_t = 0.70$. The working is in Appendix A.

Table 8: Speed, torque and power at each arbor.

Arbor	Period	ω (rad/s)	Torque (N mm)		Power (μ W)
			full wind, no loss	half reserve, with loss	half reserve
Barrel	7.00 h	2.493×10^{-4}	4.212	3.248	0.81
Second wheel	60 min	1.745×10^{-3}	0.6017	0.430	0.75
Third wheel	8 min	1.309×10^{-2}	0.0802	0.0523	0.68
Fourth wheel	1 min	0.1047	0.01003	0.00598	0.63
Escape wheel	3.75 s	1.676	6.27×10^{-4}	3.38×10^{-4}	0.57

The whole mechanism runs on about a microwatt: 0.81μ W at half reserve and 1.05μ W at full wind. The torque at the escape wheel, $T_E = 3.38 \times 10^{-4}$ N mm at half reserve, is the quantity that links the two halves of the design: the springs and gears exist to deliver it, and the amplitude of the balance depends on it (Section 9.2).

5.5 Hour and Minute Display Train

The hour hand needs a further reduction of 12 from the second arbor. The two-stage gear train that provides it is known in horology as the motion works. Its parts are not modelled, but the tooth counts are fixed by a geometric constraint that is recorded here. The hour wheel turns freely on the cannon pinion that carries the minute hand, so the two stages of the reduction

share the same pair of axes and must have equal centre distances. With one module for both stages this requires equal tooth sums:

$$R_{mw} = \frac{Z_{mw}}{z_{cp}} \cdot \frac{Z_{hw}}{z_{mp}} = 12, \quad z_{cp} + Z_{mw} = z_{mp} + Z_{hw}. \quad (6)$$

The solution adopted is a cannon pinion of 10 leaves driving a minute wheel of 30 teeth, whose pinion of 8 leaves drives an hour wheel of 32 teeth: $3 \times 4 = 12$ and $10 + 30 = 8 + 32 = 40$. With $m = 0.12$ mm the centre distance is 2.400 mm. In these two meshes the pinion drives the wheel, the opposite of the going train, so NIHS 20-30 profiles for them would need to be generated and checked in that drive direction before the parts are built.

6 Tooth Profile: NIHS 20-30 Standard and Its Implementation

6.1 Rationale for the Ogival Profile

Power transmission gears use the involute form, and the standard method for rating them, ISO 6336 [11], assumes it. Watch trains do not. Their pinions have 6 to 12 leaves, far below the count at which an involute pinion can be cut without undercut, and they multiply speed instead of reducing it, so the wheel drives the pinion. In that direction the friction during the approach to the line of centres is what matters most, and the ogival form with its radial pinion flanks is designed to keep that approach short. ISO 6336 is restricted to involute teeth with a contact ratio of 1 to 2.5 and does not apply here. The tooth form in this project follows the Swiss watch industry standard for ogival gearing, NIHS 20-30 [12], and strength is checked directly on the resulting profile (Section 10.2).

6.2 Profile Generation

NIHS standards are sold by the Fédération de l'industrie horlogère suisse and I did not have access to the text of NIHS 20-30. The profiles were instead generated with the NIHS Generator of the Watchmaking Gear Simulator, an open-source package by hdrpano [13]. This is third-party software that I used and did not write. For a given tooth count and module it writes the tooth as a list of points and circular arcs. I generated the eight profiles of the four meshes in the “pratique” version of the standard, and they are archived with the project under `docs/references/NIHS`.

This has one consequence that should be stated plainly. The geometry in this report is verified against the generator's output, and that agreement is exact to within rounding. Whether the generator implements the standard faithfully is taken on trust.

6.3 Profile Definition

Figure 4 shows the generated profiles of the first mesh.

Wheel tooth. Each flank is a single circular arc, the ogive, which runs from the pointed tip on the tip circle through the pitch point and blends into a root fillet. The profile is fixed by five numbers, all expressed in modules: the addendum h_a , the dedendum h_f , the tooth thickness on the pitch circle s^* , the ogive radius R_o and the fillet radius.

Pinion leaf. Each flank is a straight radial line. It is closed at the top by a tip arc of radius R_t that is tangent to the flank and meets its mirror image in a point on the tip circle, and at the bottom by a root fillet tangent to the flank. It is fixed by the addendum, the dedendum, a flank factor p_l that sets the half angle of the leaf, and R_t .

Table 9 gives the factors read from the eight generated files. Meshes 2 and 3 share one set because both pinions have 8 leaves.

The trends are systematic. As the pinion gets smaller the wheel addendum falls and its dedendum rises (1.40/1.20 with 12 leaves, 1.30/1.30 with 8, 1.20/1.40 with 6), and the pinion tip

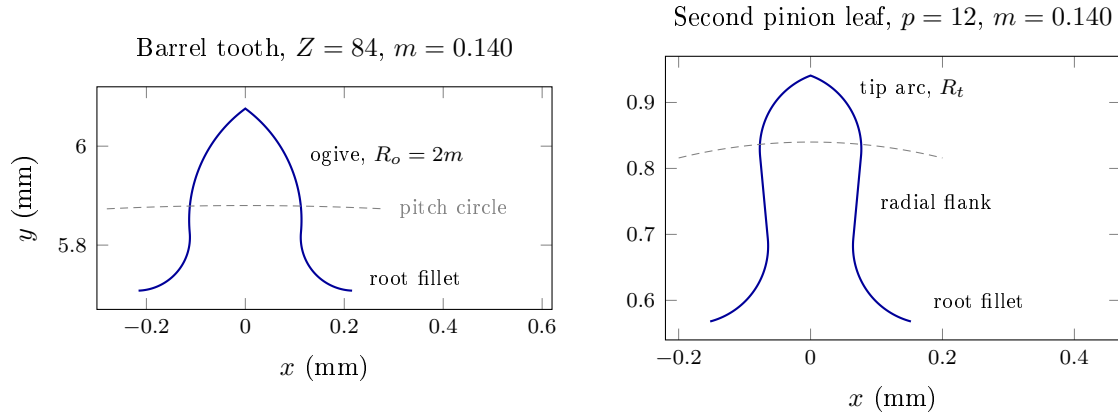


Figure 4: NIHS 20-30 profiles of the first mesh, plotted from the generated arc data. The dashed line is the pitch circle. The wheel tooth is a pointed arch of two circular arcs; the pinion leaf has straight radial flanks closed by two tip arcs.

Table 9: NIHS 20-30 factors per mesh, in multiples of the module unless marked. They are the variables $n1_*$ to $n4_*$ and k_og of the equations file.

Mesh (Z/p)	Wheel				Pinion				
	h_a	h_f	s^*	R_o	h_a	h_f	p_t	flank angle	R_t
1 (84/12)	1.40	1.20	1.6026	2.00	0.72	1.80	1.11823	5.339°	0.85
2 (60/8)	1.30	1.30	1.5801	2.00	0.82	1.70	1.12323	8.045°	0.95
3 (64/8)	1.30	1.30	1.5794	2.00	0.82	1.70	1.12323	8.045°	0.95
4 (96/6)	1.20	1.40	1.5405	2.00	0.82	1.60	1.16605	11.135°	1.05

radius grows. The ogive radius of $2.00\,m$ and the leaf thickness on the pitch circle, about $1.10\,m$, stay constant. The first version of the model had used a single estimated set for all meshes, with a wheel dedendum of $1.75\,m$ and pinion dedendums of 1.9 to $2.1\,m$, deeper than any value in the table. Replacing it thickened every pinion core and, because the barrel root circle moved outwards, also enlarged the space for the mainspring.

6.4 Parametric Reconstruction in CAD

A profile imported as a list of arcs would be dead geometry: it would not follow a change of module or tooth count. The profiles were therefore rebuilt in SolidWorks as sketches whose only inputs are the factors of Table 9, so that the tooth regenerates from the equations file. The constructions are given here and proved numerically in Appendix B.

Wheel. The sketch is drawn about the centreline of a tooth gap. With the tooth count Z ,

$$r_p = \frac{mZ}{2}, \quad r_a = r_p + h_a m, \quad r_f = r_p - h_f m, \quad \delta = \frac{\pi - s^*}{Z}, \quad (7)$$

where δ is the angle from the gap centreline to the pitch point, since the gap occupies the part of the angular pitch $2\pi/Z$ that the tooth thickness s^*m leaves free. Two points define the flank: the pitch point P at radius r_p and angle δ , and the tip T at radius r_a on the centreline of the tooth, at angle π/Z . The ogive is the arc of radius $R_o = 2m$ through both. Its centre C lies on the perpendicular bisector of PT , at a distance

$$\sqrt{R_o^2 - \frac{1}{4}|PT|^2} \quad (8)$$

from the midpoint, on the side away from the gap. The root fillet has its centre on the gap centreline at height y_c , touches the root circle there, and is tangent to the ogive. Tangency of

two circles on the same side requires $|C - (0, y_c)| = R_o + \rho$ with $\rho = y_c - r_f$; squaring and cancelling y_c^2 gives the closed form

$$y_c = \frac{|C|^2 - (R_o - r_f)^2}{2(C_y + R_o - r_f)}, \quad \rho = y_c - r_f, \quad (9)$$

with C_y the component of C along the gap centreline. In the sketch only δ and R_o are entered as dimensions. Everything else follows from coincidence and tangency relations, which is exactly what the algebra above expresses. The half gap is mirrored, cut through the blank and patterned Z times.

Pinion. The sketch is drawn about the centreline of a leaf. With n leaves, the flank lies on a radial line at the angle

$$\alpha = \frac{pl}{n} \text{ (radians)} \quad (10)$$

from the leaf centreline. The tip arc of radius R_t is tangent to this line and passes through the tip $T = (0, r_a)$. If t is the distance from the pinion centre to the point of tangency, the condition that T lies on the arc reduces, after the R_t^2 terms cancel, to a quadratic in t whose smaller root is

$$t = r_a \cos \alpha - \sqrt{2r_a R_t \sin \alpha - r_a^2 \sin^2 \alpha}. \quad (11)$$

The root fillet is centred on the centreline of the gap between leaves, at angle π/n , and is tangent to the flank and to the root circle:

$$y_c = \frac{r_f}{1 - \sin(\pi/n - \alpha)}, \quad \rho = y_c - r_f. \quad (12)$$

6.5 Reconstruction Accuracy

I compared the rebuilt geometry with the generated arc files for all eight profiles. The results are in Table 10.

Table 10: Difference between the sketch constructions and the generated NIHS 20-30 profiles.

Mesh	Ogive centre (wheel)	Flank end (pinion)	Pinion fillet radius	Wheel fillet, built / NIHS
1	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$0.755 m / 0.771 m$
2	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$0.744 m / 0.771 m$
3	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$0.745 m / 0.771 m$
4	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$< 0.001 \mu\text{m}$	$0.751 m / 0.771 m$

The tips, the flanks of wheels and pinions and the pinion roots coincide with the generated profiles to within the five decimal places of the stored factors. The one deliberate difference is the wheel root. The generated profile uses a fillet of $0.771 m$ whose two arcs, from the two sides of a gap, overlap very slightly past the gap centreline. A sketch cannot be closed that way, so the construction above makes the fillet tangent to the root circle exactly on the centreline, which gives a radius of 0.744 to $0.755 m$. The root outline then differs from the generated one by at most $1.5 \mu\text{m}$, in a region that no mating tooth ever reaches: the smallest clearance between a tip and the opposing root is $26 \mu\text{m}$ (Table 11).

6.6 Clearances and Backlash

With centre distances at their nominal value $m(Z + p)/2$, the radial clearance between each tip and the opposing root and the circumferential backlash on the pitch circle follow directly from the factors:

$$c_w = a - r_{a,w} - r_{f,p}, \quad c_p = a - r_{a,p} - r_{f,w}, \quad j = \pi m - s_w^* m - s_p^* m. \quad (13)$$

Table 11: Radial clearances and backlash at nominal centre distance (mm).

Mesh	Wheel tip to pinion root	Pinion tip to wheel root	Backlash
1. Barrel / second pinion	0.056	0.067	0.062
2. Second / third pinion	0.048	0.058	0.055
3. Third / fourth pinion	0.040	0.048	0.046
4. Fourth / escape pinion	0.026	0.038	0.032

The clearances scale with the module, as they should. The smallest, 0.026 mm at the fourth mesh, is the reason that mesh receives the tightest centre-distance tolerance in Section 7. A further rule had to hold on every arbor: in the CAD model the arbor passes through the pinion, so it must be thinner than the pinion root circle. The shallower NIHS dedendums helped. The escape pinion root is $\varnothing 0.182$ mm and its arbor $\varnothing 0.16$ mm.

7 Contact Analysis and Tolerance Study

7.1 Methodology

The geometry of Section 6 says that the teeth fit. It does not say how well they transmit torque, how hard they press on each other, or how much error in manufacture they tolerate. These questions were answered with the contact analysis of the Watchmaking Gear Simulator [13], again an external tool. For a pair of profiles it rolls the wheel through slightly more than one pitch in 301 positions, constructs the contact point geometrically at each position, and from the contact normal, a friction coefficient and the input torque it evaluates the instantaneous efficiency, the torque delivered to the pinion and the Hertz pressure. Its Monte Carlo mode repeats the analysis with random errors in the centre distance and in the two pitch diameters. Figure 5 shows the output for the barrel mesh.

Table 12: Settings of the contact analysis and the Monte Carlo study.

Setting	Value	Reason
Drive mode	wheel drives pinion	the going train multiplies speed
Positions per pitch	301 over 1.2 pitches	resolves the change of contact from one tooth pair to the next
Friction coefficient	0.20	simulator default; high for oiled brass on steel, so conservative
Input torque	4.2117, 0.6017, 0.0802, 0.0100 N mm	full-wind torque at each wheel with no losses, the worst case for pressure
Face width	0.25, 0.18, 0.18, 0.15 mm	the design at the time of the study
Monte Carlo samples	1000 per mesh	all 4000 samples solved; none failed
Centre distance error	± 0.015 mm, uniform	wider than the tolerance finally specified, to find the limit
Wheel / pinion diameter error	± 0.010 / ± 0.005 mm, uniform	typical of cut brass wheels and turned steel pinions

7.2 Nominal Results

For a mesh with input torque T_{in} on a wheel of Z teeth and output torque T_{out} on a pinion of p leaves, the instantaneous efficiency is

$$\eta = \frac{T_{out} Z}{T_{in} p}, \quad (14)$$

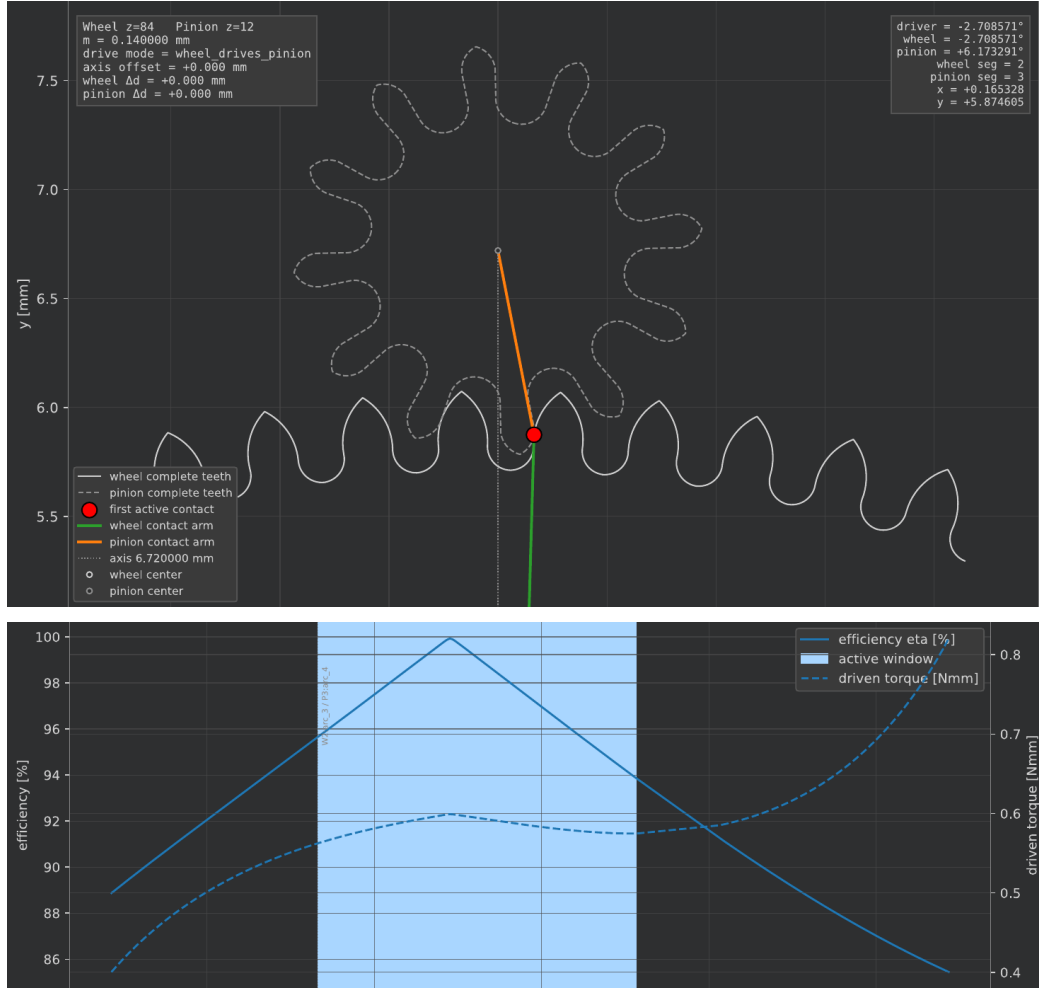


Figure 5: Contact analysis of the barrel mesh, as output by the simulator [13]. Top: the barrel teeth (solid) and the second pinion (dashed) at the first position of active contact, with the contact point marked and the centre distance of 6.720 mm shown dotted. Bottom: instantaneous efficiency (solid) and torque delivered to the pinion (dashed) as the wheel turns; the shaded band is one pitch of the wheel, 4.286° , over which this tooth pair carries the load.

and the *torque variation* is the peak-to-peak variation of T_{out} over one pitch relative to its mean. Table 13 gives the results, with the efficiency and ripple averaged over the Monte Carlo samples and the worst values found among them.

Table 13: Results of the contact analysis: Monte Carlo mean and worst case over 1000 samples per mesh, recomputed from the archived data.

Mesh	Efficiency (%)		Torque variation (%)		Hertz pressure (MPa)	
	mean	worst instant	mean	worst sample	nominal	worst sample
1 (84/12)	97.5	92.3	6.2	10.6	913	917
2 (60/8)	96.2	90.2	13.9	22.0	542	545
3 (64/8)	96.3	90.0	13.5	23.4	230	232
4 (96/6)	95.4	85.5	15.4	32.2	108	110

Three observations follow.

Efficiency. Within one pitch the efficiency is not constant. It reaches 100 % where the contact point crosses the line of centres, since there is no sliding there, and falls to about 94 % at either end of the engagement (Figure 5); the delivered torque varies accordingly. The product of the

four mean efficiencies is

$$\eta_{\text{gears}} = 0.975 \times 0.962 \times 0.963 \times 0.954 = 0.862. \quad (15)$$

The train efficiency assumed elsewhere in this report is 0.70, which leaves $0.70/0.862 = 0.81$ for the friction of the eight pivots. That split is plausible, but only the gear part is supported by a calculation.

Leaf count. Efficiency falls and ripple rises as the pinion loses leaves. The 12-leaf pinion delivers its torque within $\pm 3\%$ at nominal geometry, the 6-leaf escape pinion within about $\pm 7\%$. Since the fourth mesh feeds the escapement directly, its ripple appears as a variation of impulse from beat to beat.

Pressure. Hertz pressure falls quickly down the train because the load falls by a factor of seven or more per stage while the teeth shrink only slowly. The barrel mesh is the one that matters, and it failed.

7.3 Face Width of the Barrel Teeth

The contact between a tooth flank and a leaf is a line contact, for which the maximum pressure is [8]

$$p_H = \sqrt{\frac{F_n E^*}{\pi b R^*}}, \quad \frac{1}{E^*} = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}, \quad (16)$$

with F_n the normal force, b the face width, R^* the relative radius of curvature at the contact and E^* the contact modulus of brass on steel. Yielding begins below the surface when the maximum shear stress, $0.30 p_H$ in line contact, reaches the shear yield stress. With the Tresca criterion this gives a first-yield pressure of

$$p_{H,\text{yield}} = \frac{R_{p0.2}}{2 \times 0.30} = 1.67 R_{p0.2} \approx 670 \text{ MPa} \quad (17)$$

for brass with $R_{p0.2} = 400 \text{ MPa}$. The simulated 913 MPa at full wind, on teeth 0.25 mm wide, exceeds it by more than a third. The tips of the brass teeth would have been plastically deformed every time the watch was fully wound.

Equation (16) shows the remedy: for a given force and geometry $p_H \propto b^{-1/2}$. The width needed to bring the pressure to the limit is

$$b \geq 0.25 \left(\frac{913}{670} \right)^2 = 0.46 \text{ mm}. \quad (18)$$

The barrel teeth were widened to 0.50 mm, which gives

$$p_H = 913 \sqrt{\frac{0.25}{0.50}} = 646 \text{ MPa}, \quad (19)$$

and the second pinion was lengthened from 0.40 to 0.60 mm so that it covers the wider teeth with an allowance for end shake.

The margin against first yield is $670/646 = 1.04$. This is below the factor of 1.2 adopted as the criterion for contact in Table 19, so the barrel teeth do not meet the criterion at their present width. Meeting it would require $p_H \leq 670/1.2 = 558 \text{ MPa}$, that is a width of

$$b \geq 0.25 \left(\frac{913}{558} \right)^2 = 0.67 \text{ mm}. \quad (20)$$

The width of 0.50 mm is retained in the present model as a recorded deviation, and a width of 0.70 mm is recommended for the next revision (Section 12). Three things should be said about the present margin. First, the load case is severe: full wind, no losses and a friction coefficient of 0.20. Second, the figure of 646 MPa is a scaling of the simulated result; the simulation was not rerun at the new width. Third, it depends directly on the temper of the barrel brass (Section 3.4). The other three meshes are far below the limit and their widths were left as they were, since those are set by stiffness and by what can be cut, not by stress.

7.4 Sensitivity to Manufacturing Deviations

Figure 6 plots the torque variation of each Monte Carlo sample against its centre-distance error. The dependence is close to linear. Table 14 quantifies it.

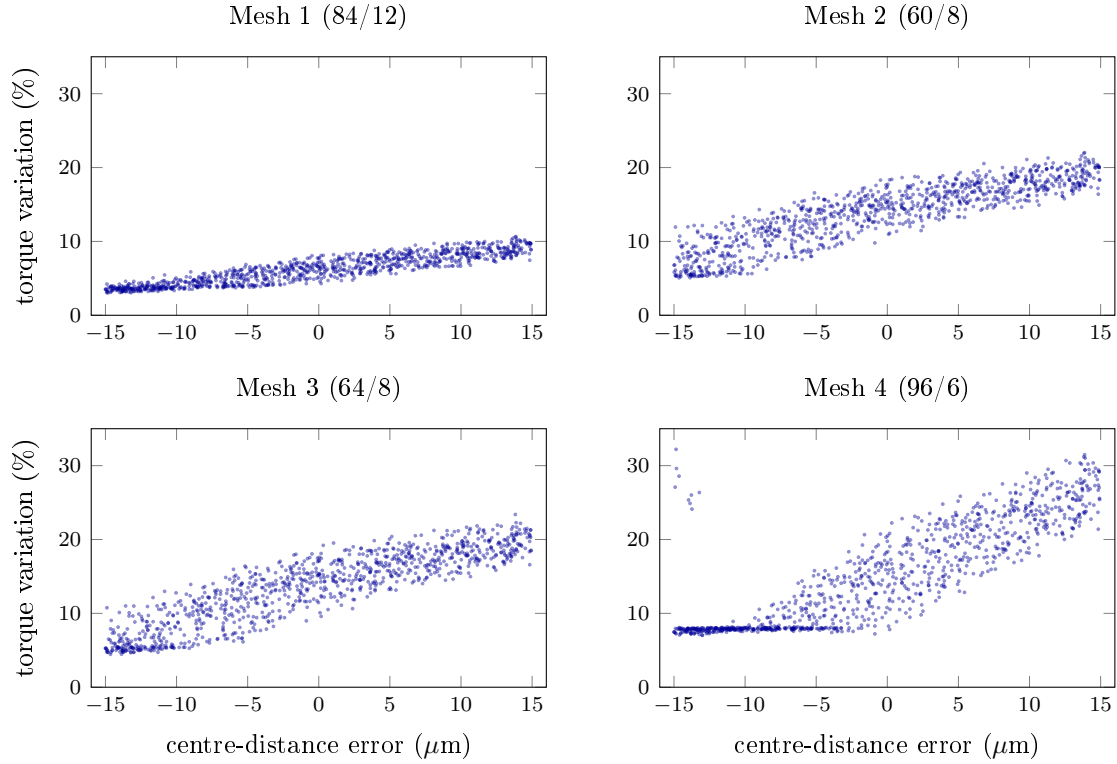


Figure 6: Torque variation against centre-distance error for the 1000 Monte Carlo samples of each mesh, plotted from the archived simulator output. The vertical spread at a given abscissa is the effect of the two diameter errors.

Table 14: Correlation of torque variation with each error source, and the effect of narrowing the centre-distance tolerance (worst sample within the stated band).

Mesh	Correlation coefficient with			Worst torque variation (%) within		
	centre distance	wheel $\varnothing$	pinion $\varnothing$	± 0.015	± 0.010	± 0.005
1 (84/12)	+0.92	-0.31	+0.09	10.6	9.7	8.9
2 (60/8)	+0.89	-0.34	+0.15	22.0	21.1	19.1
3 (64/8)	+0.90	-0.32	+0.14	23.4	21.2	20.1
4 (96/6)	+0.87	-0.21	+0.11	32.2	29.0	25.2

The centre distance dominates. Its correlation with the torque variation is 0.87 to 0.92 in every mesh: a mesh set too deep runs smoothly, and a mesh set too shallow ripples and loses efficiency, because the contact moves further out along the pinion flank and more of the action takes place before the line of centres. An oversized wheel has the same effect as a reduced centre distance, hence the negative sign in its column. The pinion diameter hardly matters.

No sample collided and no sample lost drive, even at ± 0.015 mm. The tolerance on centre distance is therefore not set by whether the gears work, but by how well they run. I specified ± 0.010 mm for the first three meshes and ± 0.005 mm for the fourth, where narrowing the band reduces the worst ripple from 32 % to 25 % and raises the worst instantaneous efficiency from 85.5 % to 87.3 %. These are the variables `tol_a_13` and `tol_a_4`.

One limitation of the tool should be noted. It reports an “active window” of contact that is always exactly one pitch of the driving wheel (4.286° , 6.000° , 5.625° and 3.750° for the four meshes). That is a property of how the window is defined and not a measurement of contact ratio. The evidence that drive is continuous is the Monte Carlo result that no sample lost drive.

8 Spring Design

The mechanism has two springs with opposite jobs. The mainspring is the energy store: it should deliver as much torque as the material allows, as evenly as possible, for as many turns as possible. The hairspring is the time standard: its stiffness, together with the inertia of the balance, is the only thing that sets the rate. This section derives the dimensions and constants of both.

8.1 Mainspring

8.1.1 Material

The mainspring is specified in Nivaflex 45/5, a cobalt-nickel-chromium alloy (Co 45, Ni 21, Cr 18, Fe 5, W 4, Mo 4, Ti 1, Be 0.2) that is cold rolled and then aged. It is the standard alloy for watch mainsprings because it combines a very high elastic limit with corrosion resistance and freedom from magnetism. The manufacturer gives a modulus of 220 GPa, a density of 8500 kg/m^3 and a tensile strength up to about 3000 MPa in the fully worked and aged condition [4, 5]; the calculations here use $R_{p0.2} = 2475 \text{ MPa}$. The first version of the model had used 200 GPa, a generic steel value, which understated the stiffness by 10 %.

8.1.2 Available Space and Spring Length

The spring lives in the annulus between the arbor core, of radius r , and the inner wall of the drum, of radius R . The wall radius follows from the barrel teeth: it is the root radius of the teeth less the wall thickness left under them,

$$R = \frac{m_{b2}(Z_b - 2h_f)}{2} - w = \frac{0.140(84 - 2 \times 1.20)}{2} - 0.43 = 5.282 \text{ mm}, \quad (21)$$

and the arbor core is taken as one third of it, $r = R/3 = 1.761 \text{ mm}$, the classical proportion.

A strip of thickness t and length L occupies an area Lt in plan whether it is wound tightly on the arbor or lying against the wall. The number of turns the barrel can make between these two states is largest when the spring fills half of the free annulus [14]:

$$L = \frac{\pi(R^2 - r^2)}{2t}. \quad (22)$$

The two limiting states are shown in Figure 7. When fully wound the coils form a solid ring on the arbor of outer radius R_i , and when let down a ring against the wall of inner radius R_o :

$$R_i = \sqrt{r^2 + \frac{Lt}{\pi}}, \quad R_o = \sqrt{R^2 - \frac{Lt}{\pi}}, \quad N = \frac{R_i - r}{t} - \frac{R - R_o}{t}. \quad (23)$$

The first term of N is the number of coils when wound and the second the number when let down; their difference is the number of turns the barrel makes. With $t = 0.095 \text{ mm}$,

$$L = 410.1 \text{ mm}, \quad R_i = R_o = 3.937 \text{ mm}, \quad N = 22.91 - 14.16 = 8.75 \text{ turns}.$$

The requirement is $40/7 = 5.71$ turns. The margin is deliberate, for two reasons: the model ignores the length taken by the hook and the outer end, which typically costs 5 to 10 % of the turns, and only the upper part of the torque curve is useful (Section 9.2).

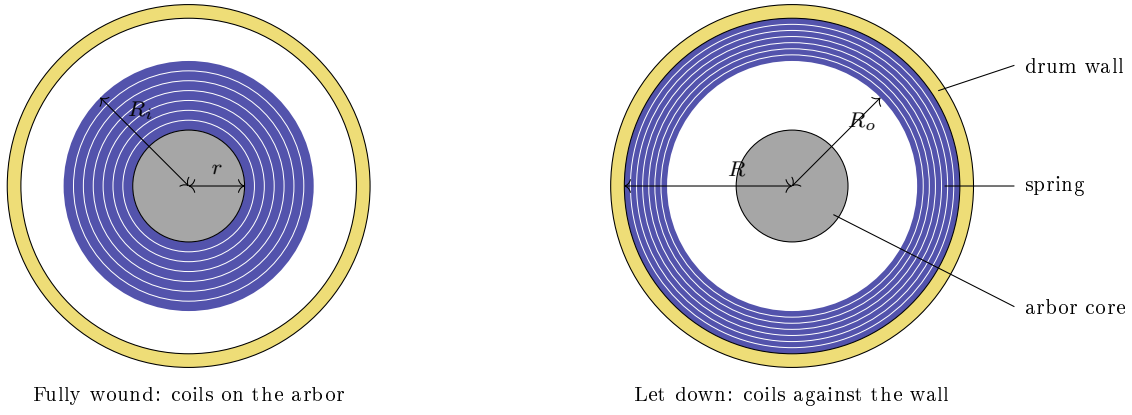


Figure 7: The two limiting states of the mainspring in the barrel, to scale: $r = 1.761$ mm, $R = 5.282$ mm and $R_i = R_o = 3.937$ mm. The spring occupies the same plan area Lt in both. The rings are drawn schematically; the real spring has 22.9 coils when wound and 14.2 when let down, and the difference of 8.75 is the number of turns the barrel makes.

8.1.3 Stiffness and Torque Characteristic

A spiral strip whose ends are rotated relative to each other is, to first order, a beam in pure bending. A moment M produces a relative rotation ML/EI , so the stiffness per radian is

$$K = \frac{EI}{L} = \frac{Ebt^3}{12L}, \quad (24)$$

where b is the height of the strip. The highest moment the strip can carry is limited by the bending stress at its surface, $\sigma = 6M/bt^2$:

$$M_{\max} = \frac{\sigma_d bt^2}{6}. \quad (25)$$

The torque after the barrel has run down by n turns from full wind is then taken as linear,

$$M(n) = M_{\max} - 2\pi K n. \quad (26)$$

This model replaced an earlier one, and the difference determined whether the design met its reserve requirement. The earlier model set $M_{\max} = 2\pi KN$, which is the torque of a spring that is completely relaxed when let down and is then wound through N turns. A real mainspring is not relaxed in the barrel. It is formed with a reverse curvature, so that even when let down it is bent and presses on the wall, and its torque at full wind is limited by the stress the strip can bear, not by the number of turns. Equation (25) states that directly. The slope K is the same in both models; what changes is the level of the whole curve.

The design stress was set at $\sigma_d = 2000$ MPa, which is $0.81 R_{p0.2}$ and gives a safety factor of 1.24 on the proof strength and 1.55 on the tensile strength. A mainspring is one of the few machine elements routinely designed this close to yield, because any unused strength is unused energy. With $b = 1.40$ mm and $t = 0.095$ mm:

$$K = \frac{220\,000 \times 1.40 \times 0.095^3}{12 \times 410.05} = 0.0537 \text{ N mm/rad}, \quad (27)$$

$$M_{\max} = \frac{2000 \times 1.40 \times 0.095^2}{6} = 4.212 \text{ N mm}. \quad (28)$$

Table 15 gives the torque at the points of interest, using the barrel period of 7 hours to convert time to turns.

Table 15: Mainspring torque from Equation (26), compared with the first version of the design.

Quantity	Relation	First version	Current design
Barrel inner radius R (mm)		4.400	5.282
Strip height b (mm)		1.00	1.40
Length L (mm)	Equation (22)	284.5	410.1
Barrel turns N	Equation (23)	7.29	8.75
Stiffness K (N mm/rad)	Equation (24)	0.0502	0.0537
Torque at full wind (N mm)		2.300	4.212
Torque after 24 h (N mm)	$n = 24/7$	1.098	3.056
Torque after 40 h (N mm)	$n = 40/7$	0.297	2.285
Torque when let down (N mm)	$n = N$	0	1.261
Bending stress at full wind (MPa)	$6M_{\max}/bt^2$	1529	2000

8.1.4 Selection of Dimensions

The three dimensions of the strip act on different things, which made the choice systematic.

Thickness is the most sensitive. At a fixed stress the torque grows with t^2 while the number of turns falls with $1/t$, and the stress in the coils wrapped on the arbor grows with t/r . The value $t = 0.095$ mm keeps $N = 8.75$ in the enlarged barrel. Raising it to 0.100 mm would give 11 % more torque but fewer turns and a higher arbor stress, so it was kept.

Height acts linearly on the torque and on nothing else. The first version, with $b = 1.00$ mm, delivered too little torque to keep the balance swinging (Section 9.2). Working back from the amplitude targets of 200° at 24 hours and 180° at 40 hours gives a requirement of $b \geq 1.32$ mm, since the amplitude at a given time varies as $\sqrt{b}$ and the 40-hour target governs: $1.40 \times (180/185.4)^2 = 1.32$ mm. A height of 1.40 mm was adopted. The cost is a taller barrel: 1.88 mm instead of 1.35 mm.

Length is not chosen freely; it follows from the barrel radius through Equation (22). This is why the barrel module was raised from 0.120 to 0.140 mm: a larger barrel holds a longer spring, a longer spring is softer, and a softer spring loses less torque per hour. The loss of torque over 40 hours is 46 % of $M_{\max}$ in the current design, against 87 % in the first.

8.1.5 Bending Stress at the Arbor

The model above treats the stress as uniform along the strip, which is true only of the free coils. Coils that are wound onto the arbor are forced to its curvature by contact, whatever the torque. If the strip has a free curvature κ_0 at that point, the bending stress imposed on it is

$$\sigma_a = \frac{Et}{2} \left(\frac{1}{r} - \kappa_0 \right). \quad (29)$$

For a strip that is straight when free this gives $220\,000 \times 0.095 / (2 \times 1.761) = 5935$ MPa, twice the tensile strength. A strip that is straight when free would therefore fail at the arbor. In practice the inner coils are given a permanent curvature in manufacture. Requiring $\sigma_a \leq \sigma_d$ gives the largest free radius the inner coils may have:

$$R_{\text{set}} = \frac{1}{\kappa_0} \leq \left(\frac{1}{r} - \frac{2\sigma_d}{Et} \right)^{-1} = 2.66 \text{ mm}. \quad (30)$$

This check was missing from the first version of the design. It does not change any dimension, but it becomes a line in the purchasing specification of the spring (Section 12).

8.1.6 Model Limitations

The linear law is an idealisation. A measured torque curve is steeper near both ends, where coils are peeling off the wall or stacking onto the arbor, and friction between the coils lowers the delivered torque by something like 5 to 15 %. A nonlinear finite element analysis of the spring in its barrel was planned to replace the model and has not been run. Until it is, or until a spring is measured, the torque values in Table 15 are estimates, and the sensitivity of the amplitude to a $\pm 5\%$ error in torque is given in Table 18.

8.2 Hairspring

8.2.1 Required Stiffness

The balance and hairspring form a torsional oscillator with natural frequency

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{I}} \quad \Rightarrow \quad k = 4\pi^2 I f^2. \quad (31)$$

The frequency is fixed by the rate: $f = \text{bph}/7200 = 4 \text{ Hz}$, since there are two beats per oscillation and 3600 seconds per hour. The inertia is a property of the balance as drawn. With a rim of 8.0 mm outside and 7.2 mm inside diameter and 0.30 mm height in Glucydur, three arms and the staff with its rollers, the CAD mass properties give 3.588 mg cm^2 , and the equations file carries the rounded value 3.60 mg cm^2 .

Units need care here. Watchmakers quote inertia in mg cm^2 , and $1 \text{ mg cm}^2 = 10^{-10} \text{ kg m}^2 = 10^{-7} \text{ N mm s}^2$. Hence

$$k = 3.60 \times 10^{-7} \times (2\pi \times 4)^2 = 2.274 \times 10^{-4} \text{ N mm/rad} = 3.969 \times 10^{-6} \text{ N mm/deg}. \quad (32)$$

8.2.2 Length, Number of Turns and Pitch

The hairspring obeys the same beam relation as the mainspring, Equation (24), now solved for the length that gives the required stiffness:

$$L = \frac{E b t^3}{12 k}, \quad n = \frac{L}{\pi (r_0 + r_1)}, \quad \text{pitch} = \frac{r_1 - r_0}{n}, \quad (33)$$

where r_0 and r_1 are the inner and outer radii of the spiral, so that $\pi(r_0 + r_1)$ is the mean length of a turn. The spring is specified in Nivarox, the iron-nickel alloy used for almost all balance springs, with $E = 190 \text{ GPa}$. A strip of $b = 0.10 \text{ mm}$ by $t = 0.025 \text{ mm}$, starting on a collet of radius 0.45 mm and ending at 2.60 mm, gives

$$L = \frac{190\,000 \times 0.10 \times 0.025^3}{12 \times 2.274 \times 10^{-4}} = 108.80 \text{ mm}, \quad n = 11.35 \text{ turns}, \quad \text{pitch} = 0.189 \text{ mm},$$

and so a clear gap of 0.164 mm between adjacent coils. The outer coil lies 1.0 mm inside the balance rim.

The cross-section was chosen so that the spring has a reasonable number of turns in the space inside the rim. Fewer turns would mean a shorter spring that is stressed more highly and breathes less evenly; more would bring the coils too close together.

The cubic dependence on thickness is the reason hairsprings are never made to a drawing alone. Since $f \propto \sqrt{k} \propto t^{3/2}$, an error of 1 % in thickness, a quarter of a micrometre, changes the rate by 1.5 %, which is over 21 minutes per day. In practice each spring is matched to its balance by cutting it to the length at which the pair beats at the right frequency. The length computed here is therefore the nominal value from which that adjustment starts.

8.2.3 Bending Stress

In pure bending the moment, and so the surface stress, is uniform along the spring. At amplitude θ the moment is $k\theta$ and

$$\sigma = \frac{6k\theta}{bt^2}. \quad (34)$$

At 300° this is 114 MPa and at 330° , taken as an upper bound, 126 MPa, a strain of 0.066 %. Against a proof strength of about 1300 MPa the safety factor is 10. The stress is fully reversed and is applied 1.26×10^9 times in ten years, so fatigue is the relevant criterion. I found no published S-N data for Nivarox; against an estimated endurance strength of 450 MPa the safety factor is 3.6. The energy stored at 300° is $\frac{1}{2}k\theta^2 = 3.12 \mu\text{J}$.

8.2.4 Thermal Compensation

Both the stiffness of the spring and the inertia of the balance change with temperature. Since $k \propto Ebt^3/L$ scales with temperature as $\gamma_s + 3\alpha_s$, where $\gamma_s = (1/E)dE/dT$ is the thermoelastic coefficient of the spring and α_s its expansion coefficient, and $I \propto r^2$ scales as $2\alpha_b$,

$$\frac{\Delta f}{f} = \frac{1}{2}(\gamma_s + 3\alpha_s - 2\alpha_b) \Delta T. \quad (35)$$

With $\alpha_b \approx 17 \times 10^{-6} \text{ K}^{-1}$ for the copper-beryllium balance and $\alpha_s \approx 7.5 \times 10^{-6} \text{ K}^{-1}$ for the spring, the rate is independent of temperature when $\gamma_s = 2 \times 17 - 3 \times 7.5 = +11.5 \times 10^{-6} \text{ K}^{-1}$. An ordinary steel has γ near $-250 \times 10^{-6} \text{ K}^{-1}$ and would lose about ten seconds per day per kelvin. The whole point of the Nivarox family is that its thermoelastic coefficient is small and can be adjusted by heat treatment, so the spring grade has to be ordered to suit the balance alloy. Each $1 \times 10^{-6} \text{ K}^{-1}$ of residual mismatch costs 0.043 seconds per day per kelvin.

8.2.5 Model Limitations

The stiffness and stress above are reliable, because they rest only on beam theory and the modulus. What the model cannot say anything about is isochronism, the independence of rate from amplitude. A flat Archimedean spiral pinned at both ends does not open and close concentrically: its centre of mass moves as it breathes, which loads the pivots sideways and makes the rate depend on amplitude and on the position of the watch. Real springs are given shaped end curves to reduce this. Assessing it needs a nonlinear finite element model of the spring and, in the end, a timing machine.

9 Escapement and Oscillator

9.1 Escapement Geometry

The escapement was laid out by the classical construction for a 15-tooth wheel, with every angle a variable of the equations file. Figure 8 shows the resulting layout to scale, and Table 16 lists the values and the reasoning behind each. A motion-study recording of the result is at https://youtu.be/M3oqJB_sfAk [2].

Centre distance. The two stones span 60° of the escape wheel, that is $2\frac{1}{2}$ tooth pitches of 24° , so that one stone is always in a position to receive a tooth when the other releases one. The pallet axis is placed where the tangents to the wheel at the two locking points meet. For a wheel of radius R_e and a span of 60° this gives

$$a_{\text{pallet}} = \frac{R_e}{\cos 30^\circ} = \frac{2.200}{0.8660} = 2.540 \text{ mm}, \quad r_{\text{lock}} = a_{\text{pallet}} \sin 30^\circ = 1.270 \text{ mm}, \quad (36)$$

where r_{lock} is the distance from the pallet axis to each locking point. Placing the axis on the tangents means that each stone meets its tooth squarely at the moment of locking.

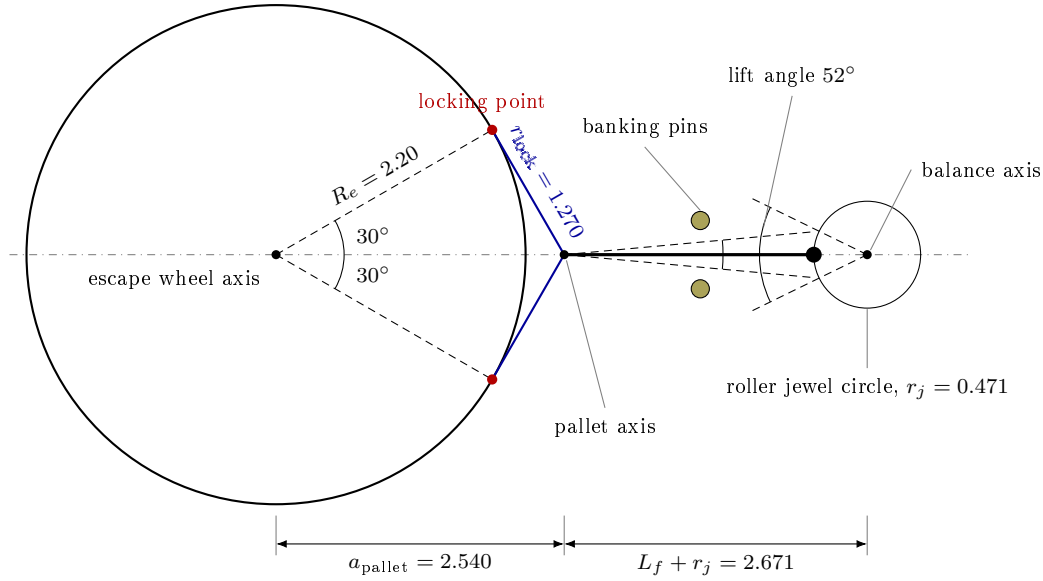


Figure 8: Layout of the escapement, drawn to scale from the equations file (dimensions in mm). The pallet axis lies at the intersection of the tangents to the escape wheel at the two locking points, which are 60° apart. The lever is shown in its mid position (solid) and against each banking pin (dashed). The roller jewel is engaged with the fork while the balance is within 26° of rest. The balance rim, of radius 4.0 mm about the balance axis, is omitted for clarity.

Lever angles. The lever turns through 10.5° in total, 5.25° to each side of the line of centres, and the banking pins are placed to stop it there. The total is made up as

$$\underbrace{1.5^\circ}_{\text{lock}} + \underbrace{2.5^\circ}_{\text{impulse on tooth}} + \underbrace{6.0^\circ}_{\text{impulse on stone}} + \underbrace{0.5^\circ}_{\text{slide}} = 10.5^\circ. \quad (37)$$

A lock of 1.5° at a radius of 1.270 mm is an overlap of only

$$1.270 \times 1.5 \times \frac{\pi}{180} = 0.033 \text{ mm} \quad (38)$$

between tooth and stone. This is the most delicate dimension in the watch. It explains the tolerance of $\pm 0.005 \text{ mm}$ on the escapement centre distances, which alone changes the lock by $\pm 0.23^\circ$.

Draw. The locking face is inclined by 12° so that the pressure of the tooth pulls the stone into the wheel and holds the lever on its banking pin. For this to work the inclination must exceed the angle of friction between tooth and stone, $\arctan 0.15 = 8.5^\circ$ for oiled steel on ruby; otherwise friction alone could hold the lever wherever a shock left it.

Drop. After a tooth leaves one stone the wheel turns freely through 1.7° , about 0.065 mm at the tooth tip, before the next tooth lands. This clearance is what allows for errors in the tooth spacing and the stone positions. It is also pure loss: of the 12° the wheel turns per beat, 1.7° , or 14 %, does no work on the balance.

Lift angle and roller jewel. The balance is engaged with the fork over a lift angle of 52° , 26° either side of rest. The radius at which the roller jewel must sit follows from the triangle formed by the pallet axis, the balance axis and the jewel at the moment it leaves the fork, when the lever is at $\lambda/2 = 5.25^\circ$ and the balance at $\beta/2 = 26^\circ$. Let L_f be the length of the fork, defined as the distance from the pallet axis to the centre of the jewel when the jewel is on the line of centres, so that the distance between the axes is $L_f + r_j$. The side of the triangle opposite the lever angle is r_j , and the side opposite the remaining angle, $180^\circ - (\lambda/2 + \beta/2)$, is the distance

between the axes. The sine rule then gives

$$r_j = \frac{L_f \sin(\lambda/2)}{\sin(\lambda/2 + \beta/2) - \sin(\lambda/2)} = \frac{2.20 \times \sin 5.25^\circ}{\sin 31.25^\circ - \sin 5.25^\circ} = 0.471 \text{ mm}, \quad (39)$$

and so a distance of 2.671 mm between the pallet and balance axes.

Table 16: Escapement parameters.

Parameter	Value	Basis
Escape wheel	15 teeth, $R_e = 2.20 \text{ mm}$	classical Swiss lever count; one tooth per oscillation
Pallet span	60° ($2\frac{1}{2}$ teeth)	alternate engagement of the two stones
Centre distance, wheel to pallet	2.540 mm	tangential locking, $R_e / \cos 30^\circ$
Lock	1.5°	usual range 1.5 to 2° ; must exceed the free travel of the guard dart (0.75°)
Draw	12°	greater than the friction angle of 8.5°
Impulse	$2.5^\circ + 6.0^\circ$	divided between the tooth and the stone
Slide (run to banking)	0.5°	as small as tolerances allow, since it is lost motion
Total lever angle	10.5°	sum of the above; banking at $\pm 5.25^\circ$
Drop	1.7° of the wheel	usual range 1.5 to 2°
Lift angle	52°	usual range 50 to 53°
Roller jewel radius	0.471 mm	from the lift angle and lever angle
Guard dart clearance	0.02 mm	dart reaches the safety roller after 0.75° of lever travel

Safety action. While the balance is away on its free swing nothing but the draw holds the lever. A shock could throw it across, and the returning roller jewel would then strike the back of the fork. A guard dart on the lever, facing a safety roller of radius 0.34 mm on the balance staff, prevents this. If the lever is knocked, the dart meets the edge of the roller after 0.75° of travel, which is half the lock, so the tooth is still locked and the draw pulls the lever back. A crescent cut in the roller lets the dart pass at the one moment it should, when the jewel is in the fork. The clearance of 0.02 mm between dart and roller is the tightest functional clearance in the model.

Forces. The forces in the escapement are minute. At full wind the tooth presses on the locking face with $T_E/R_e = 4.39 \times 10^{-4}/2.20 \approx 2 \times 10^{-4} \text{ N}$, the weight of about twenty milligrams. Static strength is never in question here. What matters is the condition of the surfaces, the lubrication, and the impact when a tooth lands after the drop (Section 10.10).

9.2 Balance Amplitude

9.2.1 Energy Balance

How far the balance swings is decided by an energy balance. An oscillator of quality factor Q loses a fraction $2\pi/Q$ of its stored energy in each period. In steady state the escapement must supply exactly that. In one period the escape wheel advances by one tooth, an angle $\varphi = 2\pi/Z_e$, under a torque T_E , and a fraction η_e of that work reaches the balance:

$$\frac{2\pi}{Q} \cdot \frac{k\theta^2}{2} = \eta_e T_E \varphi \quad \implies \quad \theta = \sqrt{\frac{Q\eta_e T_E \varphi}{\pi k}}, \quad (40)$$

with the torque at the escape wheel obtained from the barrel torque M through the train,

$$T_E = \frac{M\eta_t}{R_{b4}R_{4e}} = \frac{M\eta_t}{6720}. \quad (41)$$

Equation (40) contains the whole design in one line. The amplitude grows only with the square root of the torque, so doubling the mainspring torque raises the amplitude by 41 %. It falls with the square root of the hairspring stiffness, which is fixed by the inertia of the balance and the frequency. A heavy balance at a high frequency is a better timekeeper and needs a much stronger barrel.

9.2.2 Predicted Amplitude

With $Q = 250$, $\eta_e = 0.30$, $\eta_t = 0.70$ and $k = 2.274 \times 10^{-4}$ N mm/rad, Equation (40) and the torque curve of Table 15 give the amplitudes in Table 17. Figure 9 shows the torque and amplitude over the whole run.

Table 17: Predicted amplitude of the balance.

	Full wind	After 24 h	After 40 h	Hours above 150°
Requirement	$\leq 315^\circ$	$\geq 200^\circ$	$\geq 150^\circ$ (target 180°)	≥ 40
First version of the design	196°	135°	70°	19
Current design	252°	214°	185°	56

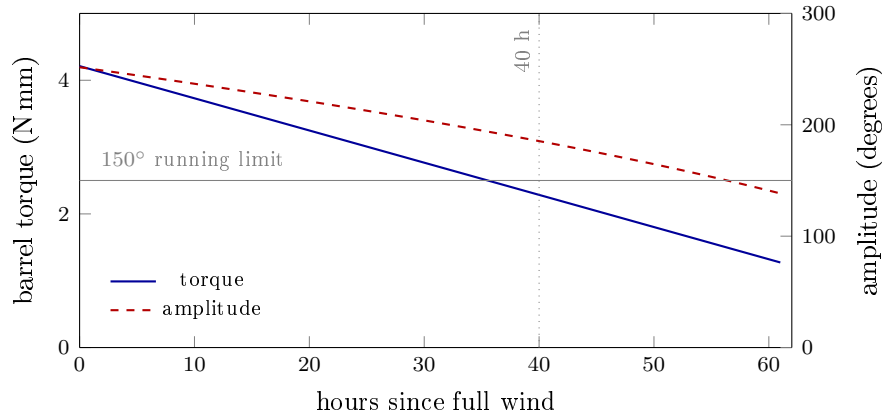


Figure 9: Barrel torque from the linear model of Equation (26) and the balance amplitude from Equation (40), for the current design at nominal assumptions. The curves end at 61.3 hours, where the spring has used all of its 8.75 turns.

The first version of the design fails outright: the watch would have run for less than a day. This single result drove the largest change in the project, the enlargement of the barrel and the mainspring described in Section 8.1.

The number of hours for which the amplitude stays above 150° follows by inverting Equation (40) for the barrel torque needed at that amplitude,

$$M_{150} = \frac{\theta_{150}^2 \pi k R_{b4} R_{4e}}{Q \eta_e \varphi \eta_t} = 1.496 \text{ N mm}, \quad (42)$$

and then asking how many turns the barrel makes before its torque falls to that value:

$$t_{150} = \frac{M_{\max} - M_{150}}{2\pi K} \times 7 \text{ h} = 56.4 \text{ h}. \quad (43)$$

The spring cannot run longer than its turns allow, $8.75 \times 7 = 61.3$ hours, which is also where its torque reaches the let-down value.

Table 18: Sensitivity of the predicted amplitude and running time to the assumptions. The running time is capped at 61.3 hours by the turns of the spring.

Case	Full wind	24 h	40 h	Hours above 150°
Nominal (Q 250, η_e 0.30, η_t 0.70)	252°	214°	185°	56.4
$Q = 200$	225°	192°	166°	48.6
$Q = 300$	276°	235°	203°	61.3
$\eta_e = 0.25$	230°	196°	169°	50.2
$\eta_e = 0.35$	272°	232°	200°	60.8
$\eta_t = 0.80$	269°	229°	198°	60.3
Mainspring torque -5%	245°	207°	177°	52.0
Mainspring torque $+5\%$	258°	222°	194°	60.7
All pessimistic (Q 200, η_e 0.25, η_t 0.65)	198°	169°	146°	37.3
All optimistic (Q 300, η_e 0.35, η_t 0.80)	318°	271°	234°	61.3

9.2.3 Sensitivity Analysis

The three efficiencies in Equation (40) are taken from literature and none has been measured on this mechanism, so the result has to be read with its sensitivity (Table 18).

The design meets the 40-hour requirement for any single assumption at its unfavourable value, including the lower quality factor of 200 that the literature gives for an ordinary watch. Only when all three are unfavourable together does it fall just short, at 146° after 40 hours. At the other extreme the full-wind amplitude reaches 318°, just over the limit. If a built movement showed that, the remedy would be a slightly weaker spring, which in the model means lowering the design stress σ_d .

The upper limit itself can be reasoned as follows. After its impulse the roller jewel leaves the fork at 26° past rest. Swinging on, it would meet the outside of the fork again after turning a further $360^\circ - 2 \times 26^\circ$, that is at an amplitude of about 334°. The limit of 315° keeps a margin of roughly 20° below that.

9.2.4 Model Limitations

Equation (40) treats Q as a constant. In a real watch it depends on amplitude, because air drag is not linear, and on position, because pivot friction is higher when the watch is on edge. The equation also takes the escapement efficiency as given. The motion study models the impulse and the drop explicitly and so gives an independent estimate of η_e . In the exported run, which was made without contact friction, the escapement delivered about 0.6 of the work of the escape wheel to the balance. That is the efficiency of the geometry with drop and impact losses only, and it is consistent with a value of about 0.30 once friction is included, so the assumed value is retained (Section 11.5).

10 Design of Machine Elements and Strength

This section checks each element of the mechanism against the loads it carries. The approach is that of a machine design course: identify the load, choose the model (beam, shaft, thick cylinder, plate, contact), compute the stress, and compare it with the strength of the material at the appropriate number of cycles. Where the element is sized by the check, the formula is inverted to give the required dimension. The step-by-step working is in Appendix E.

10.1 Load Cases and Acceptance Criteria

Two load cases are used throughout.

Running. The torque at each arbor is the full-wind value with no losses (Table 8). This overstates the load everywhere except at the barrel and is conservative.

Shock. Every moving part is subjected to an acceleration of $5000g$, and the inertial force $m n_g g$ on an arbor is shared by its two pivots. The origin of this figure is as follows.

A watch that falls from a height $h = 1$ m reaches the floor at

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 1} = 4.43 \text{ m/s}, \quad (44)$$

which is the impact velocity of the pendulum hammer in the ISO 1413 test [9]. The fall itself is harmless. What loads the parts is the deceleration at the end of it, and that depends entirely on the distance d over which the watch is brought to rest:

$$a = \frac{v^2}{2d}, \quad \text{so that} \quad d = 0.2 \text{ mm gives } a = \frac{4.43^2}{2 \times 0.0002} = 4.9 \times 10^4 \text{ m/s}^2 = 5000g. \quad (45)$$

A steel case meeting a hard floor stops within a fraction of a millimetre, taken up by local deformation of the case and the floor, and this is why accelerations of thousands of g arise from so modest a fall. The same event described as a half-sine pulse of duration τ has a peak of $\pi v/2\tau$, which is $5000g$ for $\tau = 0.14$ ms. A stopping distance of 0.3 mm instead of 0.2 mm would give about $3300g$, and published descriptions of the test quote peaks between about $3000g$ and $5000g$. The value used here is therefore at the upper end of the plausible range. It is an assumption about the stopping distance and not a property of the design, and every shock stress in this section scales in direct proportion to it: at $3000g$ all the shock safety factors would be higher by a factor of 1.67 .

Two simplifications in the way the load is applied should also be stated. The parts are assumed to experience the full acceleration of the case, with no attenuation by the case, the movement holder or the bearings. And the load is applied statically, although the pulse lasts only a fraction of a millisecond. For stiff parts such as pivots the static treatment is adequate. For the balance on its arms it is not conservative: the static deflection of 0.116 mm found in Table 24 corresponds to a natural period of $2\pi\sqrt{\delta/\bar{a}} = 0.31$ ms, comparable with the duration of the pulse, so the dynamic response may exceed the static one by a factor of up to about 1.7 .

The pass criteria are given in Table 19. For fatigue, an S-N line is drawn on logarithmic axes between $0.9R_m$ at 10^3 cycles and the endurance strength S_e at 10^8 cycles, and read at the number of cycles the part sees in ten years [15].

Table 19: Pass criteria.

Load case	Required safety factor	Remark
Running, static	2.0 on yield	
Fatigue	1.5	notch factor $K_f = 1.5$ at fillets
Shock, hardened steel	2.0 on yield	little ductility in reserve
Shock, brass and Glucydur	1.5 on yield	ductile
Contact	1.2 on first yield	first yield at $p_H = 1.67R_{p0.2}$

10.2 Tooth Bending Strength

The classical Lewis method treats a tooth as a cantilever loaded at its tip and uses a tabulated form factor for involute teeth. No such table exists for ogival teeth, but the idea carries over directly, because the profile is known exactly. With the tangential force F_t applied at radius r_L , a section at radius r of chordal thickness $s(r)$ and width b carries the bending stress

$$F_t = \frac{T}{r_p}, \quad \sigma(r) = \frac{6F_t(r_L - r)}{bs(r)^2}, \quad \sigma_F = \max_{r_f \leq r \leq r_p} \sigma(r). \quad (46)$$

The critical section is found by evaluating $\sigma(r)$ along the real profile, from the root circle up to the pitch circle, and keeping the largest value. The wheel tooth is loaded at its tip, $r_L = r_a$, the

worst position. The pinion leaf is loaded at its pitch circle, where the wheel tip meets it. Fatigue is then assessed as

$$SF_{\text{fat}} = \frac{S_N(N)}{1.2 K_f \sigma_F}, \quad (47)$$

where the factor 1.2 allows for uneven load sharing and dynamic effects.

Table 20: Tooth and leaf bending on the NIHS profiles, full-wind torque without losses. Brass wheels, steel pinions.

Mesh	F_t (N)	Wheel tooth			Pinion leaf		
		σ_F (MPa)	SF yield	SF fatigue	σ_F (MPa)	SF yield	SF fatigue
1	0.716	46	8.6	4.6	86	18.6	8.5
2	0.167	35	11.4	4.9	84	19.1	7.6
3	0.0251	6.3	63	22	15	106	36
4	0.0032	1.5	270	76	4.5	358	101

The number of load cycles in ten years rises from 1.3×10^4 for a barrel tooth to 8.4×10^7 for a leaf of the escape pinion, and the S-N strength is read at the count for each part. Bending does not govern anywhere. The most highly stressed tooth, on the barrel, has a fatigue safety factor of 4.6, and by the third mesh the stresses are a few megapascals. Inverting Equation (46) for the width that would just meet the criterion gives about 0.06, 0.01 and 0.003 mm for the second, third and fourth wheels. Their actual thicknesses of 0.15 to 0.18 mm are set by the need for a blank stiff enough to be cut and handled. This is a general feature of watch gearing: after the first mesh, the teeth are sized by manufacture and not by load. The barrel teeth are sized by contact pressure, as shown in Section 7.3.

10.3 Pivot Strength

A pivot is a short cantilever. It carries a radial load F whose line of action is at a distance a from the shoulder where the pivot meets the arbor, roughly the middle of the jewel hole. The bending stress at the shoulder, and the diameter needed for an allowable stress σ_{allow} with a safety factor S , are

$$\sigma_b = \frac{32 F a}{\pi d^3}, \quad d_{\text{min}} = \left(\frac{32 F a S}{\pi \sigma_{\text{allow}}} \right)^{1/3}. \quad (48)$$

In running, F is the reaction to the two mesh forces on the arbor, found by treating the arbor as a beam simply supported at its pivots. Under shock,

$$F_{\text{shock}} = \frac{m n_g g}{2}. \quad (49)$$

The arm is taken as $a = h/2 + 0.02 \text{ mm}$ for a jewel hole of height h , with $h = 0.10 \text{ mm}$ for the train and 0.08 mm for the escapement and balance. Jewels are not modelled, so these are typical proportions and not dimensions of the design. The arbor masses are estimated from the CAD geometry.

Two results in Table 21 changed or qualified the design.

Second arbor. When the mainspring was strengthened, the running load on the second arbor rose by 59 %. With the original $\varnothing 0.10 \text{ mm}$ pivot the stress would have been $198 \times (0.12/0.10)^3 = 342 \text{ MPa}$. Equation (48) with $S = 2$ on the endurance strength gives $d_{\text{min}} = 0.095 \text{ mm}$, so 0.10 mm would just have passed in bending; the pivot was raised to 0.12 mm mainly to lower the bearing pressure, discussed next.

Balance. The balance is the heaviest moving part relative to its pivots. At $5000 g$ its pivots reach 829 MPa, a safety factor of 1.93, just under the required 2.0. Solving Equation (48) for a

Table 21: Pivot bending under running load and under a 5000 g shock. Steel: $R_{p0.2} = 1600$ MPa, $S_e = 800$ MPa.

Arbor	d (mm)	mass (mg)	Running			Shock		
			F (N)	σ_b (MPa)	SF fat.	F (N)	σ_b (MPa)	SF yield
Second	0.12	30.0	0.479	198	4.0	0.735	303	5.3
Third	0.10	21.0	0.119	85	9.4	0.514	367	4.4
Fourth	0.08	15.3	0.0239	33	24	0.376	524	3.1
Escape	0.07	8.8	0.0022	4	> 100	0.215	383	4.2
Pallet	0.07	3.2	≈ 0	0		0.079	140	11.4
Balance	0.08	28.3	≈ 0	0		0.695	829	1.93

factor of 2 gives $d_{\min} = 0.081$ mm. Increasing the pivot diameter is not an acceptable remedy, since balance pivot friction is the main loss of the oscillator. Every modern watch solves this with shock-absorbing settings, in which the jewel is held by a spring and gives way before the pivot breaks. The model has rigid bearings, and this is the one structural requirement it does not meet.

The other pivots are at sizes that are routine to make and comfortably strong. Deflection of every arbor under running load is below $0.2 \mu\text{m}$, fifty times less than the centre-distance tolerance, and the Euler buckling load of the most slender arbor, the fourth at $\varnothing 0.36 \times 2.65$ mm, is about 240 N. Neither stiffness nor stability is a concern.

10.4 Pivot Bearing Pressure

A pivot in a jewel hole is a conformal line contact: a cylinder of radius R_1 inside a hole of slightly larger radius R_2 . The Hertz half-width b_H and peak pressure are [8]

$$b_H = \sqrt{\frac{4FR'}{\pi\ell E^*}}, \quad p_{\max} = \frac{2F}{\pi b_H \ell}, \quad R' = \frac{R_1 R_2}{R_2 - R_1}, \quad (50)$$

with ℓ the height of the hole. For wear and for survival of the oil film the projected pressure $F/(d\ell)$ is also used. For the second pivot with a diametral clearance of 0.010 mm, $p_{\max} = 543$ MPa and the projected pressure is 40 MPa in a jewel 0.10 mm high. Specifying a jewel hole height of 0.15 mm for this arbor lowers the projected pressure to 27 MPa. The contact stress is far below what hardened steel and ruby can carry; the concern at this pivot is wear over ten years, treated in Section 10.12.

10.5 Arbor Strength

Each arbor carries bending from the mesh forces and torsion between its wheel and its pinion. For a solid round shaft

$$\sigma_b = \frac{32M}{\pi d^3}, \quad \tau = \frac{16T}{\pi d^3}, \quad \sigma_{vM} = \sqrt{\sigma_b^2 + 3\tau^2}, \quad \frac{1}{SF_{\text{fat}}} = \frac{K_f \sigma_b}{S_e(N)} + \frac{\sqrt{3}\tau}{R_m}. \quad (51)$$

The last expression is the Goodman criterion for a rotating shaft, in which the bending stress is fully reversed and the torsion is steady. The results are in Table 22. The second arbor is the only one with stresses worth mentioning, and its safety factor on yield is 17. The winding square of the barrel arbor, 0.90 mm across flats, carries the full barrel torque in torsion at $\tau = T/(0.208 s^3) = 27.8$ MPa.

10.6 Wheel-to-Arbor Joints

In the CAD model wheel and arbor are one body. In reality the brass wheel has to be fixed to the steel arbor, and the obvious way is a press fit. Lamé's solution for a thick-walled hub of

Table 22: Arbor bodies at full-wind torque.

Arbor	σ_b (MPa)	τ (MPa)	σ_{vM} (MPa)	SF yield	SF fatigue	Deflection (μm)
Second, $\varnothing 0.40$	46.0	47.9	94.8	17	11	0.20
Third, $\varnothing 0.40$	9.2	6.4	14.4	111	57	0.03
Fourth, $\varnothing 0.36$	2.4	1.1	3.1	> 500	> 200	0.04
Escape, $\varnothing 0.16$	2.4	0.8	2.8	> 500	> 190	0.02

outer diameter D pressed onto a solid shaft of diameter d with a diametral interference δ gives the interface pressure, the torque the joint can carry over a length L , and the hoop stress at the bore of the hub [15]:

$$p = \frac{\delta}{d \left[\frac{1}{E_h} \left(\frac{D^2 + d^2}{D^2 - d^2} + \nu_h \right) + \frac{1 - \nu_s}{E_s} \right]}, \quad T_{\text{cap}} = \mu p \pi d L \frac{d}{2}, \quad \sigma_\theta = p \frac{D^2 + d^2}{D^2 - d^2}. \quad (52)$$

The interference is bounded on both sides. It must be large enough that the joint does not slip, with a factor of 2 on the torque, and small enough that the hub does not yield, with a factor of 1.5. Table 23 gives the resulting window for each joint.

Table 23: Interference window for a press fit of each wheel on its arbor ($\mu = 0.15$). Interferences are diametral, in μm .

Joint	Torque (N mm)	δ against slip	δ at hub yield	Window
Second wheel	0.602	1.32	1.68	0.36
Third wheel	0.0802	0.18	1.68	1.50
Fourth wheel	0.0100	0.03	1.50	1.47
Escape wheel	6.3×10^{-4}	≈ 0	1.58	1.58
Balance on staff	1.3×10^{-3}	≈ 0	2.40	2.40

For the second wheel the window is $0.36 \mu\text{m}$ wide. No turning or reaming process holds a bore of 0.40 mm to a third of a micrometre, so a plain press fit cannot be relied on to carry the torque of the second wheel. This agrees with established watchmaking practice, in which train wheels are riveted. The pinion is turned with a short shoulder and a thin collar, the wheel is seated on the shoulder and the collar is spread over it with a punch. The joint then carries torque through the deformed metal and not through a controlled interference. The lightly loaded joints further down the train would hold as press fits, but one process for all wheels is simpler.

10.7 Spokes and Balance Arms

The wheels are crossed out, leaving a rim joined to the hub by five or six spokes. Each spoke is a beam built in at the hub and guided at the rim. Under torque each of the n spokes carries a tangential force at the rim and bends in the plane of the wheel; under an axial shock it carries a share of the rim's inertial force and bends out of the plane:

$$F_s = \frac{T}{n r_{\text{rim}}}, \quad M = \frac{F l}{2}, \quad \sigma_{\text{torque}} = \frac{6M}{t w^2}, \quad \sigma_{\text{shock}} = \frac{6M}{w t^2}, \quad (53)$$

where l is the length of the spoke, w its width in the plane of the wheel and t the thickness of the wheel.

All pass. The balance is again the critical part: its heavy rim on three arms gives 664 MPa under shock, a factor of 1.51 on the strength of Glucydur, and the rim deflects by 0.12 mm . That deflection is elastic, but it means that at least 0.12 mm of axial clearance has to be kept under the balance rim so that it cannot strike the pallet or the bridge during a shock.

Table 24: Spokes and balance arms under full-wind torque and under a 5000 *g* axial shock. Stresses in MPa.

Part	Spokes	σ_{torque}	SF fatigue	σ_{shock}	SF shock	Rim deflection (mm)
Second wheel	5	39.0	3.2	164	2.44	0.018
Third wheel	5	5.0	25	116	3.45	0.009
Fourth wheel	6	1.2	> 100	125	3.20	0.013
Escape wheel	5	0.1	> 1000	88	18.3	0.001
Balance	3	0.1	> 1000	664	1.51	0.116

10.8 Barrel Lid and Base

The lid and the base of the barrel are thin annular plates, and during an axial shock the whole weight of the mainspring bears on one of them. The spring weighs

$$m_s = \rho L b t = 8500 \times 10^{-9} \times 410.1 \times 1.40 \times 0.095 \text{ kg} = 464 \text{ mg}, \quad (54)$$

so at 5000 *g* the plate carries $W = 464 \times 10^{-6} \times 5000 \times 9.81 = 22.8 \text{ N}$, spread over the annulus as a uniform pressure q .

The deflection $w(r)$ of a thin plate under axisymmetric loading satisfies $\nabla^4 w = q/D$, with the general solution [16]

$$w = \frac{q r^4}{64D} + C_1 + C_2 r^2 + C_3 \ln r + C_4 r^2 \ln r, \quad D = \frac{E t^3}{12(1 - \nu^2)}. \quad (55)$$

The model is shown in Figure 10. The four constants follow from the edge conditions: no moment and no shear at the free inner edge around the arbor, zero deflection at the outer edge, and there either zero slope (clamped) or zero moment (simply supported). The radial and tangential moments M_r and M_θ then give the stress $\sigma = 6 \max(|M_r|, |M_\theta|)/t^2$. I solved the four conditions numerically for each case; the results are in Table 25.

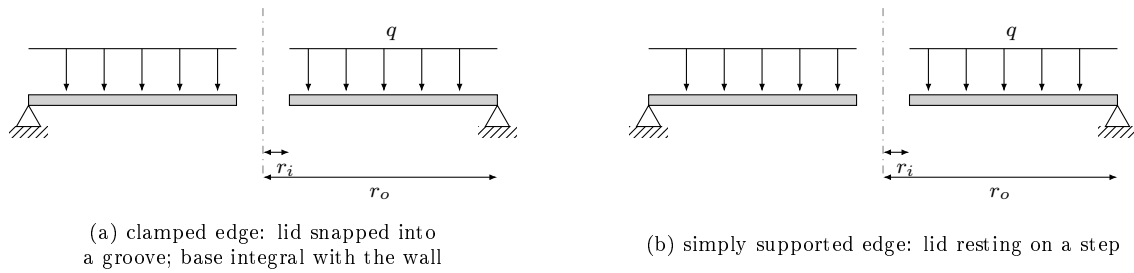


Figure 10: Model of a barrel plate under axial shock, in section. The plate is an annulus with a free inner edge at r_i and carries the inertial load of the mainspring as a uniform pressure q . The two outer edge conditions considered differ by a factor of nearly three in peak stress.

The first lid, 0.10 mm thick and simply resting on its seat, would have been stressed to more than four times its yield strength. It would have dished permanently at the first hard knock and jammed the spring. Because the stress varies as $1/t^2$, doubling the thickness reduces it fourfold, to 436 MPa, which is still not enough. The remaining factor comes from the edge condition: a clamped edge reduces the peak moment by a factor of nearly three. The lid is therefore specified 0.20 mm thick and snapped into a groove in the drum wall, so that the wall restrains its edge against rotation, and the base was thickened to 0.18 mm. The barrel grew from 1.75 to 1.88 mm in height as a result.

The conclusion depends on the groove actually clamping the lid. If it only locates it, the factor of 0.92 applies.

Table 25: Barrel plates under the weight of the mainspring at 5000 *g*. Outer radius 5.282 mm, inner radius 0.60 mm, brass with $R_{p0.2} = 400$ MPa.

Plate	Outer edge	Thickness (mm)	σ (MPa)	SF
Lid, first design	resting on a step (simply supported)	0.10	1745	0.23
Lid	resting on a step (simply supported)	0.20	436	0.92
Lid, current design	snapped into a groove (clamped)	0.20	158	2.53
Base, earlier	integral with the wall (clamped)	0.15	282	1.42
Base, current design	integral with the wall (clamped)	0.18	196	2.05

10.9 Mainspring Attachments

The inner end of the spring hooks onto the arbor through an eye punched in the strip, and the outer end bears on the drum wall. At full wind the hook carries $F = M_{\max}/r = 4.212/1.761 = 2.39$ N. The hook itself, 0.12×0.06 mm in section over the 1.40 mm height of the spring, sees 14 MPa in shear and 43 MPa in bending, which is negligible. The strip is another matter. The eye removes 0.15 mm of its height, and if the full moment is assumed to act on the net section the stress is

$$\sigma = \frac{6 \times 4.212}{(1.40 - 0.15) \times 0.095^2} = 2240 \text{ MPa}, \quad (56)$$

a factor of only 1.10 on the proof strength. The assumption is conservative, because the innermost coil is supported by the arbor and carries less than the full moment, but this is the point of the spring that most needs the finite element check that has not yet been done.

10.10 Locking Impact

At the end of each drop a tooth of the escape wheel strikes the locking face of a stone. The force is tiny in static terms but it arrives as an impact on a sharp corner, eight times a second: 2.5×10^9 impacts in ten years, or about 1.7×10^8 on each tooth. The energy to be absorbed is the work done by the escape torque over the drop,

$$U = T_E \Delta\varphi_{\text{drop}} = 4.39 \times 10^{-4} \times 1.7 \times \frac{\pi}{180} = 1.30 \times 10^{-5} \text{ N mm}, \quad (57)$$

using the full-wind escape torque with train losses. It is absorbed elastically by two springs in series: the tooth bending as a cantilever, $k_{\text{tooth}} = 3EI/l^3 \approx 82$ N/mm, and the local contact, $k_{\text{contact}} \approx \pi LE^*/8$, where $L = 0.15$ mm is the thickness of the escape wheel. The peak force and the resulting line-contact pressure on a locking corner of radius R are

$$\frac{1}{k} = \frac{1}{k_{\text{tooth}}} + \frac{1}{k_{\text{contact}}}, \quad F = \sqrt{2kU}, \quad p_H = \sqrt{\frac{F E^*}{\pi RL}}. \quad (58)$$

The contact is some hundred times stiffer than the tooth, so $k \approx 81$ N/mm and $F = 0.046$ N, about 230 times the static locking force.

Table 26: Contact pressure at locking impact as a function of the radius of the tooth's locking corner. First-yield limit of the hardened steel: $1.67 \times 1600 = 2670$ MPa.

Corner radius (mm)	Peak force (N)	p_H (MPa)	SF
0.005	0.046	1705	1.57
0.010	0.046	1206	2.22
0.020	0.046	853	3.13

The corner radius is caught between two requirements. It must stay well below the lock depth of 0.033 mm, or the tooth would not lock securely, and it must not be so sharp that it flattens.

A radius of 0.005 to 0.010 mm, polished, satisfies both, and this goes into the drawing of the escape wheel. The compliance of the tooth matters a great deal to this result. If the tooth were treated as rigid, the contact alone would have to absorb the energy, the force would come out about ten times higher, and the check would wrongly fail.

10.11 Hairspring Attachments

The hairspring is pinned into the collet at its inner end and into the stud at its outer end. Both are stress raisers on a part that sees 1.26×10^9 fully reversed cycles. With the nominal stress of 126 MPa at 330° , a stress concentration factor of 1.5 gives 189 MPa and a fatigue safety factor of 2.4 on the estimated endurance strength; a factor of 2.0 gives 251 MPa and 1.8. Both pass. The forces involved are very small: the stud reaction is 5×10^{-4} N.

10.12 Wear Assessment

For sliding contacts that are not limited by stress, life is limited by wear. Archard's law [17] gives the depth removed after a sliding distance s under a pressure p on a surface of hardness H :

$$h = \frac{K p s}{H}. \quad (59)$$

The dimensionless wear coefficient K depends strongly on lubrication. I bracketed it between 10^{-9} , for a well-lubricated polished contact, and 10^{-7} , for a degraded one, with $H \approx 6900$ MPa for steel at 700 HV.

Table 27: Wear depth after ten years from Archard's law.

Contact	Pressure (MPa)	Sliding distance (m)	Wear depth (μm)	
			$K = 10^{-9}$	$K = 10^{-7}$
Second pivot	26.6	33	0.0001	0.013
Third pivot	11.9	207	0.0004	0.036
Fourth pivot	2.99	1320	0.0006	0.058
Escape pivot	0.39	1.85×10^4	0.001	0.10
Balance pivot (watch on edge)	0.022	8.87×10^5	0.003	0.28
Escape tooth impulse face	44	1.68×10^4	0.11	10.8

The pivots wear by fractions of a micrometre even with poor lubrication. The escape teeth are different: they combine the highest sliding pressure with a long sliding distance, and with degraded lubrication the predicted wear of $11 \mu\text{m}$ is a third of the lock. The escape wheel and pallet stones are therefore the parts that set the service interval. They need an oil that stays in place, which in practice means treating the surfaces with a coating (epilame) that stops the oil from spreading, and servicing every five to seven years and not ten.

10.13 Summary of Results

Table 28 collects the governing check for each element and compares its margin with the criterion of Table 19. Two elements do not meet their criterion: the barrel teeth in contact, and the balance pivots under shock. A third, the mainspring, is deliberately designed outside the general criterion, as all mainsprings are.

Table 28: Governing check, margin and criterion for each element.

Element	Governing check	Margin	Required	Outcome
Barrel teeth	contact pressure at full wind	1.04	1.2	<i>not met</i> at 0.50 mm width; 0.67 mm required; hard temper required
Other teeth and leaves	bending fatigue	≥ 4.6	1.5	met; sized by manufacture
Train pivots	bending under shock	≥ 3.1	2.0	met
Second pivot	bearing pressure	27 MPa	none set	$\varnothing 0.12$, jewel hole 0.15 high
Balance pivots	bending under shock	1.93	2.0	<i>not met</i> ; shock settings required
Arbors	bending and torsion, fatigue	≥ 11	1.5	met
Wheel joints	press-fit window	$0.36 \mu\text{m}$		not manufacturable; rivet
Spokes, balance arms	bending under shock	≥ 1.51	1.5	met with no margin for the balance; keep 0.12 mm clearance under the rim
Barrel lid and base	plate bending under shock	2.53, 2.05	1.5	met, provided the lid is clamped
Mainspring	bending at full wind; eye	1.24; 1.10	by design	below the general criterion as intended; eye to be verified
Hairspring	fatigue at the pinned ends	≥ 1.8	1.5	met, on an estimated endurance strength
Escape teeth	locking impact	1.6 to 2.2	1.2	met; corner radius 0.005 to 0.010; lubrication

11 Motion Simulation

11.1 Objectives and Study Structure

The calculations so far treat the mechanism one element at a time. The motion studies ask the question that none of them can: does the assembled mechanism tick? They were run in SolidWorks Motion, a rigid multibody solver with contact [18], as three studies of increasing scope. Each uses the smallest assembly that can answer its question (Table 29).

Table 29: The three motion studies.

Study	Assembly	Drive	Question
A	full train, 13 parts	motor on the fourth wheel, 1 rev/min	Are the ratios and the directions of rotation right?
B	escapement only: escape wheel, pallet fork, banking pins, balance	constant torque on the escape wheel	Does the escapement lock, unlock, give impulse and keep the balance swinging?
C	full train, 13 parts	constant torque on the barrel drum	Does the train deliver that torque through four meshes, and does the seconds hand advance at the right rate?

A rigid-body solver cannot bend a part, so neither spring can act as itself. Each is replaced by its effect. The hairspring becomes a linear torsional spring between the balance and the frame, with the stiffness of Equation (31). The mainspring becomes a torque on the barrel drum. The spring parts remain in the assembly for appearance only and take no part in the dynamics.

One practical limit governs everything that follows. SolidWorks Motion becomes unstable

when it has to resolve the repeated impacts of an escapement: the integrator is driven to its minimum step at an impact and stops with an error, and in this model that happened after little more than a second of simulated time, even with the settings described below. Only short studies could therefore be run. This is the reason that the studies of the escapement and of the complete train cover a little over one second, about eleven beats, and not the seconds or minutes that a settled amplitude or a rate measurement would need.

The gear meshes are modelled as kinematic gear mates, not as contacts between teeth. This is deliberate. Tooth contact was analysed separately and in more detail in Section 7, and contact between hundreds of teeth would multiply the cost of the simulation without adding to what it can show. Gear mates are frictionless, so the train losses are accounted for by reducing the driving torque by η_t .

11.2 Study A: Kinematic Verification

With the fourth wheel driven at 1 rev/min for 10 seconds, the angular displacement of each part must be that given by the tooth counts: 60° for the fourth wheel, 7.5° for the third, 1° for the second, 0.143° for the barrel and 960° for the escape wheel. The directions must alternate along the train. Seen from above, the fourth wheel and the seconds hand turn clockwise, the third wheel anticlockwise, the second clockwise, and the barrel and the escape wheel anticlockwise.

This study is trivial as physics, but it catches the errors that would make the later studies meaningless: a gear mate entered with the wrong ratio or the wrong sense, and, through an interference check run at the same time, wrong diameters or levels. Studies B and C depend on the directions established here, since the torque on the escape wheel has to press a locked tooth into its stone and not pull it away.

11.3 Study B: Isolated Escapement

11.3.1 Rationale for a Separate Assembly

When a steel tooth lands on a ruby stone the contact lasts for microseconds, and the integrator has to take steps shorter than that to resolve it. In the full assembly every such step also solves four gear mates and thirteen bodies. In a separate assembly of four parts, built from the same part files, a run that would take hours takes minutes, and the setup can be debugged quickly. Once the escapement runs on its own, Study C only adds the train.

11.3.2 Model Definition

- *Hairspring.* A torsional spring about the balance staff, $k = 3.969 \times 10^{-6}$ N mm/deg, with its free angle at the position where the roller jewel is centred in the fork.
- *Losses of the balance.* A viscous damper in parallel with the spring. Nothing else in the model represents pivot friction and air drag on the balance, and without it the balance would swing too freely and the amplitude would read high. For an oscillator of inertia I and angular frequency ω , a quality factor Q corresponds to

$$c = \frac{I\omega}{Q} = \frac{3.60 \times 10^{-7} \times 2\pi \times 4}{250} = 3.62 \times 10^{-8} \text{ N mm s/rad} = 6.32 \times 10^{-10} \text{ N mm/(deg/s)}. \quad (60)$$

- *Drive.* A constant torque on the escape wheel equal to the half-reserve value $T_E = 3.384 \times 10^{-4}$ N mm from Table 8. A constant torque is physically right: the mainspring pushes on the train all the time, including while the escapement is locked.
- *Contacts.* Three contact pairs and no others: escape wheel with pallet fork (teeth on stones), pallet fork with balance (fork and dart on roller jewel and safety roller), and pallet fork with banking pins.

- *Start.* The lever resting on one banking pin, a tooth resting on the locking face of the stone on that side, and the balance turned 140° away from rest and released. The angle is measured from the mid position of engagement, in which the roller jewel is centred in the fork slot on the line joining the pallet and balance axes; this is also the position at which the hairspring exerts no torque. A balance held at its release angle is in the state of a running watch at the extreme of a swing. The release angle is an initial condition and not a design value: the amplitude at which the balance settles is set by the energy balance of Equation (40) and does not depend on where it starts. Any release angle below the rebanking limit is valid. A release below the settled amplitude shows the escapement building the amplitude up, and a release near the expected amplitude reaches the steady state sooner.

11.3.3 Solver and Contact Parameters

Table 30 lists every setting that differs from the defaults, with the reason for it. Most were determined from runs that failed with the default values.

The two numbers in Table 30 that are quoted as estimates can be checked. A tooth of effective mass 3 mg arriving at 70 mm/s carries 7.4×10^{-6} N mm of kinetic energy. Equating this with the energy stored in the contact, $k_c x^{2.5}/2.5$, gives a penetration of $x = 0.32 \mu\text{m}$. The contact stiffness linearised at that depth is $1.5 k_c \sqrt{x} \approx 270$ N/mm, for which the critical damping $2\sqrt{k m}$ is 1.8×10^{-3} N s/mm. The value used is 56 % of that.

The step size sets the cost. Half a second of simulated time at a maximum step of $10 \mu\text{s}$ is at least 50 000 steps, and considerably more in practice because the step shortens at every impact.

11.3.4 Staged Verification

A contact simulation that fails gives little indication of why. The study was therefore brought up in stages, each with a known expected result, so that a failure could be attributed to one thing.

1. *Spring only*, all contacts and the torque suppressed. The balance must oscillate with a period of 0.25 s about the position where the jewel is centred in the fork. This verifies the stiffness, the inertia and the free angle of the spring.
2. *Fork and pins added*. Released from 140° , an undamped balance reaches the start of the lift, 26° before rest, when $\cos(2\pi ft) = 26/140$, that is at $t = 0.055$ s, and passes through rest a quarter of a period after release, at 0.0625 s. At that moment the jewel must enter the fork and carry the lever across to the other pin, with the dart passing through the crescent of the safety roller.
3. *Torque and tooth contact added*, run for 30 ms only. The lever must stay pressed against its pin, held by the draw; the escape wheel must not turn backwards; and the contact force on the locked tooth must settle at $T_E/R_e = 1.5 \times 10^{-4}$ N.
4. *Full run* of 0.5 s without friction, then with friction.

When the solver stopped, the time at which it stopped identified the cause. A failure within the first few milliseconds meant a problem with the locked tooth under load: static friction left on, or the tooth resting on the corner or the impulse face of the stone and not on its locking face. A failure near 55 ms, when the jewel reaches the fork, meant a problem with the jewel meeting the fork: the lever off its pin, or the dart rubbing on the rim of the safety roller instead of passing through the crescent.

11.4 Study C: Complete Train

Study C applies the same escapement model in the full assembly and moves the drive to the other end of the train.

Table 30: Solver and contact settings of Studies B and C, and the reason for each.

Setting	Value	Reason
<i>Integrator</i>		
Integrator type	WSTIFF	The default GSTIFF and its variant SI2_GSTIFF use coefficients that assume the step size changes slowly. At every impact the step has to fall from $10\ \mu\text{s}$ to below $1\ \mu\text{s}$, and both stopped there with a minimum-step-size error. WSTIFF recomputes its coefficients whenever the step changes and ran through.
Maximum step	$10\ \mu\text{s}$	A tooth tip moves at roughly $70\ \text{mm/s}$ when it lands. In a step of $200\ \mu\text{s}$ it would travel $14\ \mu\text{m}$, far into the stone, before the solver detected contact. In $10\ \mu\text{s}$ it travels $0.7\ \mu\text{m}$, comparable to the contact penetration.
Initial step	$1\ \mu\text{s}$	The run starts with a tooth already in contact under load.
Minimum step	$0.1\ \mu\text{s}$	The smallest value the software accepts.
Maximum iterations	50	Allows the corrector to converge through an impact.
Jacobian re-evaluation	every iteration	The system matrix changes abruptly when a contact opens or closes.
Accuracy	10^{-4}	Corresponds to about $1\ \mu\text{m}$, already coarser than the contact penetration; loosening it lets teeth pass through stones.
<i>Contact</i>		
Geometry	precise contact, high 3D resolution	The locking overlap is $0.033\ \text{mm}$. Contact on a faceted approximation of the stones would not resolve it.
Contact properties	entered by hand	Picking a material pair from the library fills in values meant for machine-sized parts: for greased steel a penetration of $0.1\ \text{mm}$, three times the entire lock.
Stiffness k_c	$10^4\ \text{N/mm}$	With the force law $F = k_c x^e$, a landing tooth then penetrates about 0.3 to $0.4\ \mu\text{m}$: small against the lock, large enough for the solver to resolve.
Force exponent e	1.5	The Hertzian value for contact between curved elastic bodies.
Maximum damping $c_{\max}$	$10^{-3}\ \text{Ns/mm}$	About half the critical damping of a tooth, with an effective mass near $3\ \text{mg}$ at its tip, on the linearised contact stiffness. Impacts settle without bouncing and without an artificially sticky contact.
Penetration for full damping d	$0.5\ \mu\text{m}$	Just above the penetration actually reached, so that damping acts during every impact.
<i>Friction</i>		
Static friction	off	A held contact, such as a locked tooth or a lever on its pin, sits at zero sliding speed. With static friction on, the stick-slip transition chatters there and stops the integrator.
Kinetic friction μ_k	0.12 teeth on stones; 0.15 jewel in fork; 0.20 lever on pins	Oiled steel on ruby, ruby on steel, and dry steel on steel. Set per contact pair, since the three pairs are different materials.
Transition velocity v_k	$1\ \text{mm/s}$	Well below the sliding speeds during impulse.
<i>Output</i>		
Frames per second	2000 to 5000	Affects only how often results are stored, not the integration. A beat lasts $125\ \text{ms}$ but the unlocking and impulse within it only a few milliseconds.
Gravity	off	The orientation of the watch is not being studied.

Drive. A constant torque on the barrel drum of

$$\eta_t M_{\text{run}} = 0.70 \times 3.248 = 2.274 \text{ N mm}, \quad (61)$$

where M_{run} is the barrel torque at half reserve, 20 hours after full wind. Divided by the train ratio this is $2.274/6720 = 3.384 \times 10^{-4} \text{ N mm}$ at the escape wheel, the same value that Study B applied directly. The efficiency is applied here because the gear mates are frictionless.

A constant torque is an accurate representation over the duration of a run. The barrel turns at $360^\circ/(7 \times 3600 \text{ s}) = 0.0143^\circ$ per second, so in five seconds of simulation it turns 0.07° and the torque of the spring changes by one part in fifty thousand.

Everything else is as in Study B: the same torsional spring and damper on the balance, the same three contact pairs with the same parameters, the same start position and the same integrator settings. No torque is applied to the escape wheel; it now arrives through the gears. The barrel arbor, the banking pins and the two spring parts are fixed. The drum with its lid, the three wheels with the seconds hand, the escape wheel, the pallet fork and the balance are free to turn.

The cost is several minutes of computation per simulated second, dominated by the contacts. The gear mates add little.

11.5 Results

The angular displacement of each part was exported from Studies A and C and is archived under `docs/csv_movement`. No curves were exported from Study B, so the results below are those of the kinematic check and of the complete train. Two recordings accompany the results. The first shows the escapement in motion at close range, https://youtu.be/M3oqJB_sfAk [2]. The second shows Study C, the complete train driven from the barrel, https://www.youtube.com/watch?v=dhshj_wPEz8 [19]; Figure 11 reproduces two frames from it.

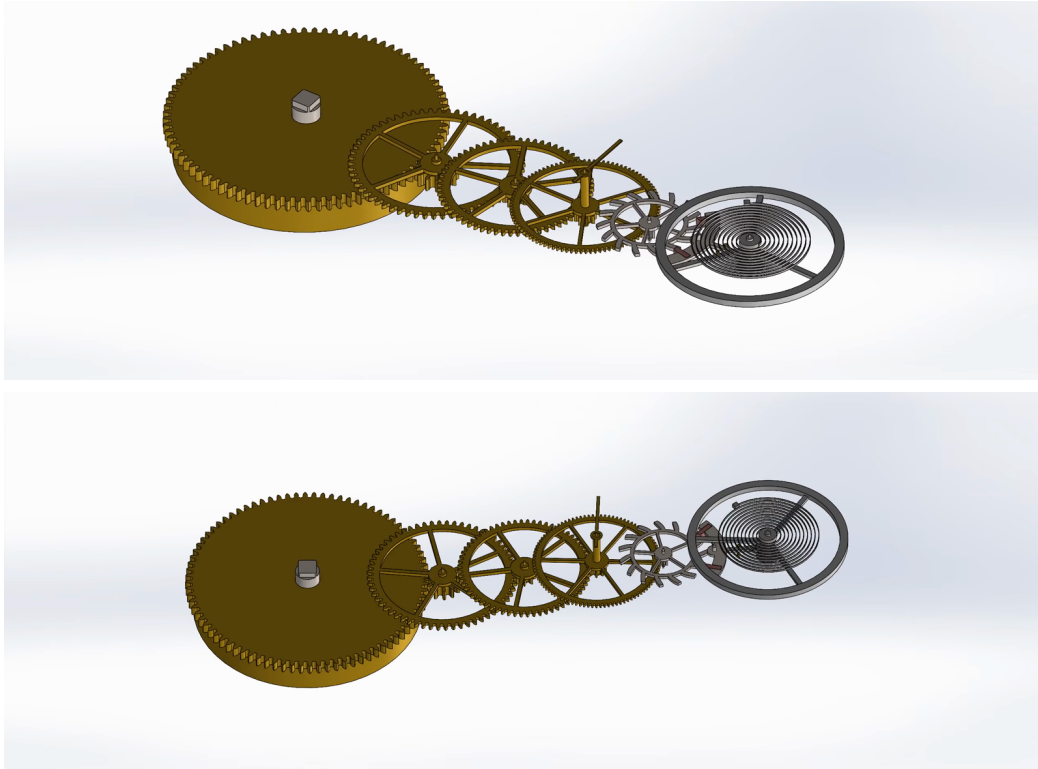


Figure 11: Two frames from the recording of Study C [19]: the complete train running under the torque applied to the barrel, seen from two camera positions. The seconds hand on the fourth arbor has advanced between the frames.

11.5.1 Study A

Study A was run for 11 seconds with the fourth wheel driven at 1 rev/min. Table 31 compares the mean angular rate of each exported part with the value required by the tooth counts. The agreement is to the precision of the output, and the signs alternate along the train as required. The escape wheel was not among the exported parts in this study; its ratio to the fourth wheel is confirmed by Study C below.

Table 31: Study A: angular rate of each part with the fourth wheel driven at 1 rev/min, over 11 s. Positive is anticlockwise seen from above.

Part	Rotation in 11 s	Measured rate (deg/s)	Required rate (deg/s)	Deviation
Fourth wheel	-66.0000°	-6.000000	-6	$< 10^{-9}$
Third wheel	$+8.2500^\circ$	$+0.750000$	$+6/8 = 0.75$	$< 10^{-9}$
Second wheel	-1.1000°	-0.100000	$-0.75/7.5 = -0.1$	$< 10^{-9}$
Barrel	$+0.157143^\circ$	$+0.0142857$	$+0.1/7 = 0.0142857$	$< 10^{-9}$

11.5.2 Study C: Motion of the Train

The exported run of Study C covers 1.397 s of simulated time, sampled at intervals of at most 0.2 ms, and contains eleven beats. The balance was released 140° from its rest position, the position in which the roller jewel is centred in the fork. It first passed through rest at 0.0625 s, exactly a quarter of the design period, and the escape wheel made its first step at 0.065 s. Figure 12 shows the response and Table 32 sets the measured quantities against those predicted by the design.

Table 32: Study C: measured quantities against the design predictions.

Quantity	Predicted	Measured
Balance period	0.2500 s	0.2498 s, constant to ± 0.0001 s over ten half periods
Beat rate	8 per second	8.01 per second (mean interval 124.8 ms)
Escape wheel advance per beat	12°	alternately 12.29° and 11.71° ; exactly 24.00° per pair
Direction of the escape wheel	anticlockwise, with a recoil of about 0.18° at unlocking	anticlockwise; recoil of 0.27° and 0.12° on alternate beats
Escape wheel while locked	stationary	stationary to within 10^{-5} degrees
Natural frequency in the free arcs	4.000 Hz	4.004 Hz
Ratio escape wheel to fourth wheel	16	16.000000
Ratio escape wheel to third wheel	128	127.99999
Ratio escape wheel to second wheel	960	959.9999
Fourth wheel advance	0.75° per beat, 6° per second, clockwise	alternately 0.768° and 0.732° , clockwise; 6.009° per second over ten beats
Barrel speed	0.01429° per second	0.01431° per second up to 1.19 s
Balance amplitude	221° when settled	not settled: 146° rising to 175° (Table 33)
Pallet fork position; force on the locked tooth	$\pm 5.25^\circ$; 1.5×10^{-4} N	not exported

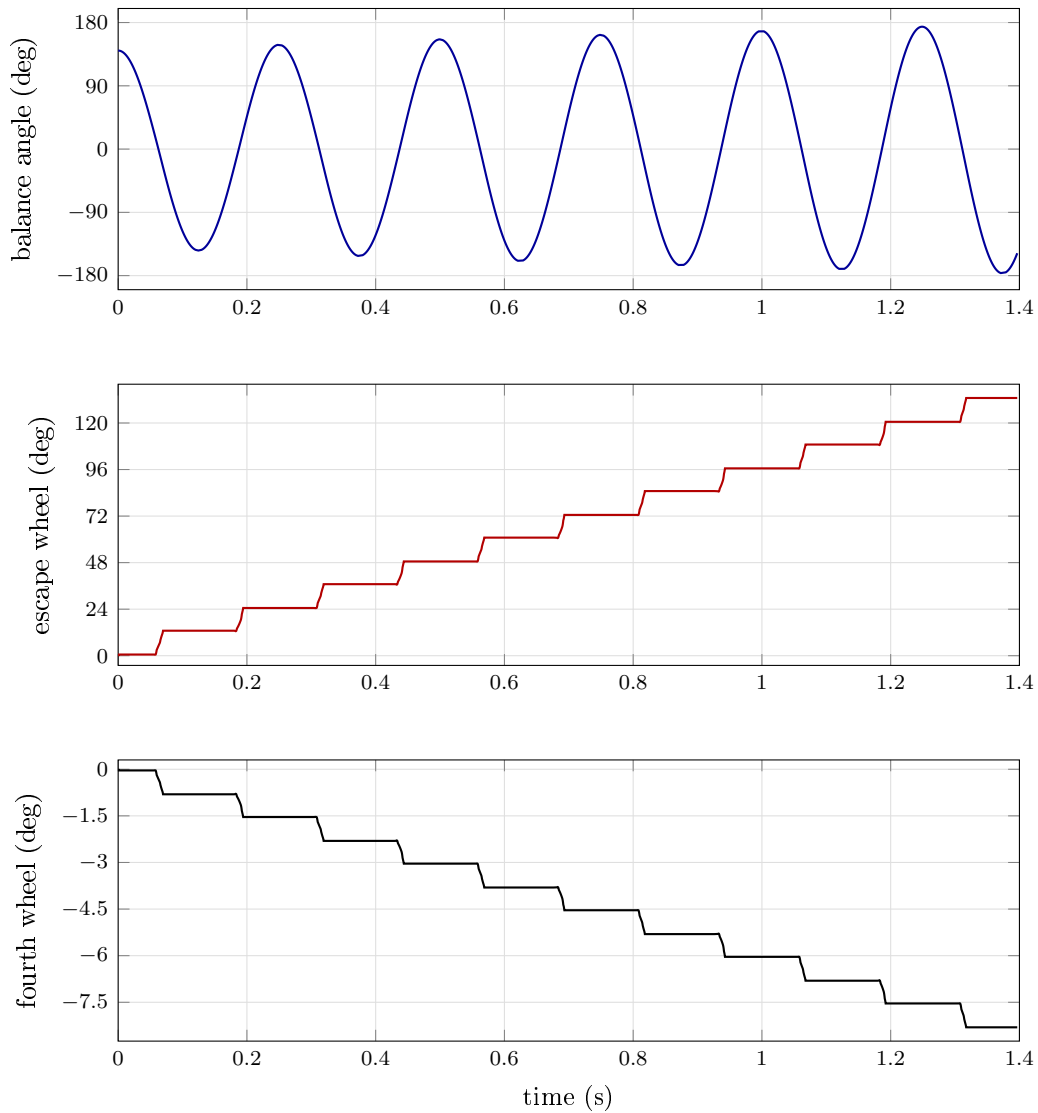


Figure 12: Study C: measured response of the complete train driven from the barrel. Top: balance angle about its rest position; the amplitude is still growing at the end of the run. Middle: escape wheel, advancing by one step per beat. Bottom: fourth wheel, which carries the seconds hand; each grid interval of 1.5° is two beats, a quarter of a second.

Timekeeping. The balance period, measured between successive passages through the rest position in the same direction, is 0.24976 s, shorter than the design value by 0.095 %. Taken as a rate, that is a gain of about 82 seconds per day.

Most of this gain does not come from the escapement. Fitting $\ddot{\theta} = -\omega_0^2 \theta$ to the free arcs, where nothing touches the balance, gives a natural frequency of 4.004 Hz and not 4.000 Hz: the ratio of stiffness to inertia in the simulated oscillator is about 0.2% above the design value. The stiffness was entered to three figures, and the inertia that the solver uses is that of the CAD balance, about 3.59 mg cm^2 , while the hairspring was sized for 3.60. This is the kind of error that regulation removes in a real watch, and in the model it would be removed by setting `B_I` to the measured inertia.

The period also shortens slightly as the amplitude grows, from 249.79 to 249.73 ms while the amplitude rises from 150° to 175° , a gain of about 19 seconds per day over that range. This is a first indication of the isochronism error of the escapement, but it was measured during a transient and should not be taken as a rate prediction. The fourth wheel advanced 7.5° in the ten beats between the first and last complete steps, which took 1.248 s: a mean rate of 6.009°

per second against the 6° per second of a seconds hand.

Kinematic consistency. The second, third and fourth wheels follow the escape wheel at ratios of 960, 128 and 16 to within 10^{-7} throughout the run, so the gear mates transmit the motion of the escapement back through the train without error. The barrel follows at the ratio of 6720 until $t = 1.19\text{s}$. During the step that occurs at that moment it advances by an additional 0.0028° relative to the second wheel and then continues at the correct ratio. The amount is negligible, less than a thousandth of a barrel tooth pitch, but it is a departure from an ideal gear mate and is recorded here as a numerical artefact of the solver.

Unequal steps. The escape wheel does not advance by equal steps of 12° . The steps alternate between 12.29° and 11.71° , and the intervals between beats alternate between 126.0ms and 123.7ms. The sum over a pair of beats is exactly 24° , one tooth pitch, as it must be. The alternation shows that the two stones do not lock the wheel at positions exactly half a pitch apart: the locking position on one stone is 0.29° of wheel rotation later than the ideal. If the impulse arcs on the two stones are equal, the difference is taken up by the free travel of the wheel, and the drop is about $1.7^\circ + 0.29^\circ$ on one stone and $1.7^\circ - 0.29^\circ$ on the other. In watchmaking terms the escapement has unequal inside and outside drop, and the unequal intervals correspond to a beat error of about 1.2ms. Both are within what a real escapement tolerates, but they are a property of the geometry as modelled, and the positions of the stones in the pallet body should be reviewed against the escape wheel to equalise them.

Locking, recoil and lock depth. Between beats the escape wheel does not move at all: its angle is constant to within 10^{-5} degrees for about 85ms of every 125ms, so the lock holds without creep, and the overshoot when a tooth lands is below 0.02° .

At each unlocking the wheel first turns backwards, by 0.27° on one stone and 0.12° on the other. This recoil is the intended effect of the draw: as the stone is withdrawn along a locking face inclined at 12° , it pushes the tooth back by the lock depth times $\tan 12^\circ$, which for the design lock of 0.033mm is 0.0071mm at the tooth tip, or 0.18° of wheel rotation. The mean of the two measured values, 0.19° , agrees with this.

Their difference does not: scaled by the same relation, the lock is about 2.2° of lever angle on one stone and about 0.9° on the other, against a design value of 1.5° on both. The shallower lock leaves little margin over the 0.75° of free travel allowed to the guard dart. This is the same asymmetry that produces the unequal steps, seen from the locking side, and it confirms that the positions of the stones, or of the banking pins, need correction.

Unlocking begins when the balance is 23° to 24° before its rest position and the tooth has landed on the other stone by 25° to 26° after it, consistent with the design lift angle of 52° .

Duration of a step. Each advance of the escape wheel, from the start of unlocking to the landing of the next tooth, takes 11 to 13ms out of a beat of 125ms, and becomes shorter as the amplitude grows. For the remaining time the wheel is locked and stationary, which is the behaviour of a detached escapement.

11.5.3 Study C: Amplitude and Energy Delivered

Conditions of the exported run. The run differs from the full specification of Table 30 in one respect: friction was switched off on all three contact pairs. This is the configuration of the first full run in the staged procedure. The torsional spring was entered as $3.97 \times 10^{-6} \text{ N mm/deg}$ with its damper enabled. The run ends at 1.4s because the solver does not remain stable for longer: in practice the integrator fails with a minimum-step error within about a second of simulated time, so a longer record could not be obtained.

Growth of the amplitude. The amplitude did not reach a steady value within the run. Table 33 gives the amplitude of successive half swings, taken as half the angle between consecutive turning points.

Table 33: Study C: balance amplitude over successive half swings.

Half swing	1	2	3	4	5	6	7	8	9	10
Amplitude (deg)	146.4	150.3	154.1	157.8	161.0	164.1	167.0	169.8	172.7	175.4
Increase (deg)		3.9	3.8	3.7	3.2	3.1	2.9	2.8	2.9	2.8

The amplitude rises by about 3° per beat and the increments fall slowly. The approach to a steady amplitude has a time constant of the order of Q/ω , about ten seconds, so a run of 1.4 s covers only its beginning and the steady amplitude cannot be read from it.

Damping actually present. Whether the damper acts can be tested on the data. Between impulses the balance is free, and its total energy $\frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}k\theta^2$ should then be constant without a damper and decay at the rate ω/Q with one. Evaluating this energy over the eleven free arcs of the run, with the velocity obtained by differentiating the exported angle, gives a decay rate of $0.060 \pm 0.010 \text{ s}^{-1}$. The energy does decay, so the damper is acting, but more weakly than specified: the rate corresponds to $Q \approx 420$ (between 320 and 580 over the individual arcs) and not to the design value of 250, for which the rate would be 0.100 s^{-1} . The most likely explanation is that the damping coefficient, $6.3 \times 10^{-10} \text{ N mm/(deg/s)}$, was rounded when it was entered, since it lies below the precision that the input field displays.

Energy delivered per beat. The same energy, compared between one free arc and the next, gives the energy that the escapement delivers to the balance at each impulse, without any assumption about the damper. It can be set against the work done by the escape wheel in one beat at the design torque,

$$W_{\text{beat}} = T_E \frac{\varphi}{2} = 3.384 \times 10^{-4} \times 0.2094 = 7.09 \times 10^{-5} \text{ N mm.} \quad (62)$$

The net gain of the balance per beat falls from 0.57 to 0.47 of W_{beat} over the run, and the loss to the damper within each half period, at the measured rate, is 0.07 to 0.11 of W_{beat} . The fraction of the work of the escape wheel that reaches the balance is therefore

$$\eta_{e,\text{sim}} \approx 0.59 \text{ to } 0.64, \quad (63)$$

falling slowly as the amplitude grows.

Interpretation. This value is for an escapement without friction. The only losses represented are the drop, in which the wheel turns without acting on the lever, and the impacts at locking and at the banking pins. The drop alone accounts for 14% of the wheel's rotation (Section 9), and the remainder of the shortfall from unity is impact loss. A real escapement adds to these the sliding friction of the teeth on the stones and of the jewel in the fork, the friction of the lever and escape wheel pivots, and the drag of the oil, and its efficiency is about 0.30 [7]. The simulated 0.6 and the literature 0.30 are therefore consistent: the first is the geometric and impact efficiency of this escapement, and the difference between them is what friction costs. The result confirms that the geometry does not waste more than the classical proportions imply. It does not replace the assumed 0.30, which remains the basis of the amplitude and reserve predictions in Section 9.2.

Obtaining the settled amplitude within the stable run length. Since the run cannot be extended, the settled amplitude has to be found another way. The balance gains amplitude when released below its settled value and loses it when released above. A series of short runs, each of a few beats, released from different angles, therefore brackets the settled amplitude: it

is the release angle at which the amplitude neither grows nor decays, and the net gain per beat measured in each run can be interpolated to zero. With friction applied at the specified values and the damper coefficient verified, this would give the simulated amplitude at each state of wind from runs of well under a second.

11.6 Limitations of the Simulation

Power reserve and settled amplitude. About one second of running is what the contact simulation delivered before the integrator failed, and even a stable simulation at $10\mu\text{s}$ steps could deliver only a few seconds. Forty hours of running is longer by more than four orders of magnitude and is out of reach. The behaviour over the reserve comes from the energy balance of Section 9.2, evaluated at different torques, and could be checked by repeating Study B at the full-wind, 24-hour and 40-hour torques.

Elastic behaviour of the springs. The hairspring is an ideal linear spring here. Its real breathing, the side loads it puts on the balance pivots, and their effect on rate are not represented.

Bearings. Every arbor turns about a perfect axis with no clearance and no friction. Pivot friction is represented only through η_t and the damper on the balance. Side shake of the pallet arbor, which in a real watch alters the lock and the dart clearance, is absent.

Gear tooth dynamics. With gear mates the train is perfectly stiff and free of backlash. The torque ripple found in Section 7, up to $\pm 7\%$ at the escape pinion, does not reach the escapement in the simulation.

12 Manufacturability

The model is a design study, and several of its features are drawn the way CAD makes convenient and not the way a workshop would make them. This section goes through what it would take to manufacture the mechanism: the process for each part, the features that are at the limit of what can be made, the tolerances, and the changes that manufacture forces on the design.

12.1 Component Breakdown for Manufacture

The first difference is in how the parts divide. In the model each wheel is a single body with its arbor, pinion and pivots. A real wheel assembly consists of two components:

- a *pinion*, turned from steel bar in one piece with its arbor and both pivots, then cut, hardened, tempered and polished;
- a *wheel*, blanked from hard brass strip, cut, crossed out and riveted onto a shoulder of the pinion.

The pallet fork is likewise a steel lever into which the two ruby stones are set and fixed with shellac, on a separate arbor. The balance is a rim with its arms, riveted to a steel staff that also carries the roller with its jewel, and the hairspring is pinned to a collet that is pushed onto the staff. The thirteen CAD parts therefore correspond to between twenty-five and thirty manufactured components, before any plates, bridges or jewels are counted. A production drawing set would start by splitting the bodies along these lines.

12.2 Process Routes and Critical Features

Table 34 gives, for each group of parts, the usual process, the feature that is hardest to make and the risk attached to it.

Three entries of the table are discussed further.

Spring procurement. The dimensions of a mainspring do not determine its torque, because the torque depends on the reverse curvature formed into the strip and on the heat treatment. What the design requires of the spring is functional, and this report supplies the numbers: a

Table 34: Manufacturing route, critical feature and main risk for each group of parts.

Part	Process	Critical feature	Main risk and recommendation
Brass wheels, m 0.065 to 0.120	blank from hard strip; cut teeth in a stack with a form cutter or hob; cross out; rivet to pinion	pointed ogival tips; module down to 0.065 mm (practical minimum about 0.05)	damaged tips and runout. One NIHS 20-30 cutter is needed per mesh, since the tooth proportions differ with the pinion.
Steel pinions and arbors	Swiss-type automatic turning; cut leaves; harden and temper; polish leaves; burnish pivots	escape pinion: 6 leaves of 0.072 mm thickness on a $\varnothing 0.182$ mm root; pivots $\varnothing 0.07$ to 0.12	flank finish and concentricity of pinion to pivots. Pivots of $\varnothing 0.06$ mm are routine in the industry, so the sizes are feasible.
Barrel drum and lid	turn from hard brass bar; cut teeth on the rim; snap lid into a groove	0.43 mm of wall under the tooth roots; groove that clamps the lid	contact pressure on tooth tips; lid dishing if the groove does not clamp. Specify a hard temper.
Barrel arbor	turn; mill square; form hook; harden	hook 0.12×0.06 mm	hook radius must suit the spring eye.
Mainspring	bought: rolled Nivaflex strip, formed to a reverse curve, aged	inner coils preset to a radius of at most 2.66 mm	torque scatter of about $\pm 5\%$ between springs. Buy to a torque specification and accept by test.
Escape wheel	blank and finish, or wire erosion with $\varnothing 0.02$ to 0.03 mm wire, or LIGA; harden; polish	club teeth; locking corner radius 0.005 to 0.010 mm; impulse face angle	corner too round makes locking unsafe, too sharp flattens. Polished locking and impulse faces, epilame, oil.
Pallet fork	blank or wire-erode the lever; set and shellac the stones; fit dart and arbor	guard dart 0.05 mm wide; stone positions	dart bends in handling; 0.05 mm is at the limit for a blanked steel feature. Consider a pressed-in dart pin of $\varnothing 0.06$ mm.
Pallet stones and roller jewel	bought: ground and polished synthetic ruby	stones 0.25×0.80 mm with impulse faces at 17.3° and 24.7° ; jewel $\varnothing 0.14$ mm	face angle within about $\pm 0.25^\circ$. Standard catalogue items.
Banking pins	turned steel pins pressed into the plate	position relative to the pallet axis	they set the lock. In a real movement they are often solid walls milled in the plate, or eccentric for adjustment.
Balance	turn rim and arms in Glucydur; rivet to staff; poise	poise (centre of mass on the axis); pivots $\varnothing 0.08$ mm	imbalance causes rate to vary with position. Poise by removing material from the rim.
Hairspring	bought: drawn and rolled Nivarox strip, coiled, heat-set; pinned to collet and stud	thickness 0.025 mm, on which stiffness depends cubically	rate error and thermal mismatch. Order the grade to suit the balance alloy and match each spring to its balance.

torque of $4.21 \text{ N mm} \pm 5\%$ at full wind and at least 1.26 N mm when let down, 8.75 turns in a barrel of 5.282 mm inner radius on an arbor of 1.761 mm , and inner coils preset to a radius of no more than 2.66 mm . The hairspring is likewise specified by its stiffness, $2.274 \times 10^{-4} \text{ N mm/rad}$, and the thermoelastic coefficient it needs, about $+11.5 \times 10^{-6} \text{ K}^{-1}$ for this balance.

Lithographic fabrication of the escapement. The escape wheel and the lever are flat parts with features of 0.005 to 0.05 mm whose shape matters to the micrometre. They are exactly what LIGA (electroformed nickel-phosphorus in a lithographic mould) and deep reactive ion etching of silicon are used for in current watchmaking. Either process would produce the club teeth and the guard dart directly from the CAD outline, with sidewalls smooth enough to need no polishing. Silicon would also remove the wear and magnetism concerns attached to the steel escape wheel, and its low density would reduce the inertia that the escapement has to accelerate eight times a second.

Tooling. Because the NIHS proportions depend on the pinion leaf count, the four wheels need four different tooth forms, and the pinions likewise. This is normal in watchmaking but it is a real tooling cost for a small series. For prototypes, wire erosion of both wheels and pinions directly from the generated profiles avoids cutters altogether.

12.3 Tolerances and Surface Finish

The general tolerance standard, ISO 286 [20], provides the notation, but its standard grades are too coarse for features below 0.1 mm . The values in Table 35 are those that follow from the analysis in this report, together with usual watch practice for the bearing clearances. The bearing values would have to be agreed with the jewel supplier.

Table 35: Functional tolerances and clearances.

Feature	Tolerance or clearance	Origin
Centre distance, meshes 1 to 3	$\pm 0.010 \text{ mm}$	Monte Carlo study: running quality (Section 7)
Centre distance, mesh 4	$\pm 0.005 \text{ mm}$	Monte Carlo study: torque ripple falls from 32 to 25 %
Centre distance, escape wheel, pallet and balance	$\pm 0.005 \text{ mm}$	a lock of 1.5° is 0.033 mm ; $\pm 0.005 \text{ mm}$ changes it by $\pm 0.23^\circ$
Lock	$1.5^\circ \pm 0.3^\circ$	set at assembly by the banking
Guard dart to safety roller	$0.020 \pm 0.010 \text{ mm}$	must stay positive in every position
Pivot diameter	$0 / -0.002 \text{ mm}$, burnished	usual practice
Side shake (diametral)	0.008 to 0.012 mm escapement and balance; 0.010 to 0.015 train	keeps contact pressure moderate and the lever geometry within $\pm 0.2^\circ$
End shake	0.02 to 0.04 mm	room for an oil film and thermal expansion
Hairspring coil gap	$\geq 0.10 \text{ mm}$ at 330°	nominal gap is 0.164 mm
Clearance under the balance rim	$\geq 0.12 \text{ mm}$	rim deflection under shock (Table 24)
Mainspring to base and lid	0.05 mm each side	as modelled

Surface finish is functional in a watch, not cosmetic. Pivots are burnished to a roughness R_a of about 0.02 to $0.05 \mu\text{m}$, which also work-hardens the surface. Jewel holes are polished. Pinion leaves and the impulse and locking faces of the escapement need about $0.1 \mu\text{m}$ or better. At a diameter of 0.07 mm a stylus instrument cannot measure roughness in any practical way, and

optical areal methods are used instead; ISO 21920 [21] governs how the requirement is indicated on the drawing.

12.4 Design Modifications Required for Manufacture

Several findings of the analysis are, in effect, manufacturing decisions. They are collected here.

1. *Rivet the wheels.* The interference window for a press fit of the second wheel is $0.36\ \mu\text{m}$ (Section 10.6). Each pinion needs a riveting shoulder and collar, which the model does not yet have.
2. *Clamp the barrel lid.* The lid passes the shock check only if its edge is restrained against rotation (Section 10.8). The groove has to be designed as a snap fit that achieves this, and the result confirmed by test.
3. *Widen the barrel teeth and specify the brass temper.* The barrel teeth have a margin of 1.04 on first yield at $R_{p0.2} = 400\ \text{MPa}$, below the criterion of 1.2. A width of 0.70 mm, with the second pinion lengthened to 0.80 mm, meets it. Bar in a softer temper would not provide even the present margin.
4. *Control the locking corner.* A radius of 0.005 to 0.010 mm, polished after hardening (Section 10.10).
5. *Fit shock settings to the balance.* The balance pivots have a factor of 1.93 at 5000 g with rigid jewels. A spring-mounted jewel and end stone at each end of the staff is standard and necessary. Its design includes the 0.12 mm of clearance under the rim.
6. *Use a taller jewel on the second arbor.* A hole height of 0.15 mm brings the projected pressure down to 27 MPa.
7. *Reconsider the last mesh.* The 6-leaf escape pinion is the hardest gear in the train to make and the worst in action: lowest efficiency, largest ripple, tightest tolerance. The ratio of 16 can equally be made as 112/7 or 128/8. At the same module the fourth wheel would grow from 6.24 to 7.28 or 8.32 mm pitch diameter, which the in-line layout accommodates easily, or the module could be reduced to about 0.055 mm to keep the diameter. An 8-leaf pinion would share its proportions with the two before it.

13 Limitations and Future Work

This section states what the work has not established and sets out the steps that would close each gap.

13.1 Limitations of the Present Work

The following has not been done. Each item is also a direction for future work.

- Nothing has been made or measured. Every result is a calculation or a simulation.
- The frame is missing. Without plates, bridges and jewels, the side shake and end shake of each arbor, the real positions of the axes and the stiffness of their supports are unknown. The tolerance values above are requirements on a frame that does not exist yet.
- The springs have been analysed only by beam theory. The mainspring torque curve, the stress at its eye and the isochronism of the hairspring are open.
- The efficiencies η_e and η_t and the quality factor Q are literature values. The exported run of Study C gives $\eta_e \approx 0.6$ without contact friction, and is too short to give a settled amplitude; Q and the pivot losses can only come from a physical movement.

- The endurance strengths of brass, hardened steel and Nivarox at 10^8 to 10^9 cycles are estimates.
- The motion studies are short. SolidWorks Motion becomes unstable on the repeated impacts of the escapement and could not be run for more than about a second of simulated time, so the studies of the escapement and of the train cover about eleven beats. They show that the mechanism runs correctly; they cannot show a settled amplitude or a rate.
- The NIHS standard text was not consulted; the generator was trusted.
- The simulated contact pressure on the barrel teeth was scaled to the new width, not recomputed.

13.2 Future Work

In rough order of how directly each item builds on what exists:

1. *Find the settled amplitude by bracketing.* The solver does not remain stable beyond about a second, so a long run is not available. Release the balance from several angles in short runs, with contact friction at the specified values and the damper coefficient verified, and find the angle at which the amplitude neither grows nor decays. Repeat at the full-wind, 24-hour and 40-hour torques, and export the pallet fork angle and the contact force as well. This gives a simulated amplitude curve to set against Figure 9 and an escapement efficiency that includes sliding friction.
2. *Improve the stability of the contact simulation,* so that runs of several seconds become possible. The candidates are the measures already identified in the build procedure: separate contact bodies for the stones, the roller jewel and the safety roller, hinge mates in place of redundant axis and distance mates, and a smaller maximum step.
3. *Widen the barrel teeth.* Change BR_teeth_t from 0.50 to 0.70 mm and P2_L from 0.60 to 0.80 mm in the equations file, so that the barrel mesh meets the contact criterion with a factor of about 1.23 (Section 7.3). All dependent dimensions follow on rebuild. The change reduces the axial clearance between the barrel teeth and the underside of the second wheel from 0.16 to 0.06 mm, so an interference check of the assembly is required afterwards.
4. *Equalise the lock and the drop.* Review the positions of the two pallet stones and of the banking pins so that the escape wheel advances by equal steps of 12° and locks equally on both stones; the simulation shows steps of 12.29° and 11.71° and locks of about 2.2° and 0.9° .
5. *Set the balance inertia from the model.* Replace the rounded value of B_I with the inertia reported by the CAD mass properties, so that the hairspring is sized for the balance as drawn; the simulated oscillator runs at 4.004 Hz.
6. *Rerun the contact analysis of the barrel mesh at a width of 0.50 mm,* to replace the scaled value of 646 MPa with a computed one.
7. *Finite element analysis of the mainspring:* a two-dimensional nonlinear static model with contact between the coils, the arbor and the drum, wound by rotating the arbor. It yields the real torque curve, which replaces Equation (26) as the input to Study C, and the stress at the eye.
8. *Finite element analysis of the hairspring:* a nonlinear static model for the stiffness and the shape of the breathing coils, followed by a dynamic model with the balance to compare the period at 135° and 270° . The second is a direct test of isochronism and would show whether shaped end curves are needed.

9. *Design the frame*: main plate, bridges, jewels with their settings, and shock absorbers for the balance. With these, bearing clearances can be put into the motion study and a proper tolerance stack carried out from the plate to the lock.
10. *Split the bodies into manufacturable components* and add the riveting shoulders, the lid groove and the collet and stud details.
11. *Build the hour and minute display train*, the motion works (Section 5), after generating and checking tooth profiles for pinion-driving meshes, and extend the second arbor to carry the cannon pinion.
12. *Fold the train*. The in-line layout is 23 mm long. Arranged around the barrel as in a real calibre it would fit within a round movement; the centre distances stay the same, and only the angles between them change.
13. *Evaluate a 112/7 or 128/8 last mesh* in the contact simulator against the present 96/6.
14. *Prototype*. A scaled-up model (the scale factor S exists for this) could be made by wire erosion or by additive manufacture, to check assembly and to observe the action of the escapement. A full-scale movement would then go on a timing machine in six positions, at full wind and after 24 hours, which is the only way to measure amplitude, rate and their variation.

14 Standards Compliance and Design Status

14.1 Standards Compliance

Table 36 records how the design stands against the standards that apply to a watch movement. The distinction that matters is between engineering compatibility, which a design calculation can show, and compliance, which for most of these standards means passing a test on a finished, cased and regulated watch. Nothing in this report certifies anything.

Table 36: Status of the design against the relevant standards.

Standard	Subject	Status in this work
NIHS 20-30 [12]	dimensions of ogival tooth profiles	Applied to all four meshes through generated profiles, reproduced by construction (Section 6). The standard text was not read.
ISO 6336 [11]	load capacity of involute gears	Not applicable: ogival teeth and pinions of 6 to 12 leaves are outside its scope. A section search on the real profile was used instead.
ISO 1413 [9]	shock-resistant watches	Not met as modelled. There are no shock absorbers, and the balance pivots have a factor of 1.93 at the assumed 5000 <i>g</i> .
ISO 764 [22]	magnetic-resistant watches	Cannot be assessed. The steel escape wheel and lever are magnetisable; the hairspring and mainspring alloys are not.
ISO 3159 [23]	chronometers: rate tests over 15 days	Cannot be assessed. It is a test on a finished watch in five positions and at three temperatures.
ISO 286 [20]	tolerance notation	Used as notation only; its grades are too coarse below 0.1 mm.
ISO 21920 [21]	indication of surface texture	Applicable to the drawings, which do not exist yet.

14.2 Subsystem Status

Table 37: Status of each subsystem at the close of this work.

Subsystem	Established	Open
Gear train	ratios exact; NIHS geometry verified; no collision or loss of drive in 4000 tolerance samples; bending margins ≥ 4.6	barrel tooth contact pressure to be recomputed at the final width; riveted joints to be detailed
Barrel	plates and hooks pass with the revised thicknesses	lid groove must clamp; brass temper to be specified
Mainspring	dimensions, turns and stress consistent; torque specification written	torque curve and eye stress need finite element analysis or test
Escapement	classical proportions verified; safety action checked; period, beat rate and train ratios confirmed in simulation	exported run made without contact friction and too short for a settled amplitude; unequal drop on the two stones; wear sets the service interval
Balance and hairspring	stiffness, length and stress consistent; margins adequate	shock protection; isochronism; thermal matching of the spring
Arbors and pivots	strong and stiff at routine sizes	bearings, clearances and frame not modelled

15 Conclusion

This project set out to design the basic mechanism of a mechanical watch from its requirements and to check that the result works. The outcome is a parametric model of thirteen parts, driven by 393 linked equations, with a train ratio of 6720 that gives 28 800 beats per hour, a second wheel turning once per hour and a fourth wheel once per minute. The tooth form on all four meshes follows NIHS 20-30 and reproduces independently generated profiles to within rounding on every working surface.

The most useful part of the work was finding out where the first design was wrong. A model can be complete, fully constrained and free of interference and still fail to function. The original barrel could not have driven the balance for a day. Its torque model assumed a spring that is relaxed when let down, and correcting that model, together with enlarging the barrel and the spring, moved the predicted running time above 150° from 19 hours to 56. Checking each element in turn then produced a series of smaller corrections of the same kind: barrel teeth twice as wide, to bring the contact pressure below yield; a thicker lid that has to be clamped at its edge; a thicker pivot on the second arbor; and the recognition that the wheels cannot be pressed on and that the balance cannot do without shock protection. None of these was visible in the geometry. Each appeared only when a number was calculated and compared with a limit.

The motion studies show the assembled mechanism locking, unlocking and sustaining the balance, first with the escapement alone and then with the whole train driven from the barrel. In the exported run of the complete train the balance period is 0.2498 s against a design value of 0.2500 s, the escape wheel makes 8.01 beats per second, and every wheel follows it at its exact ratio.

The same run shows two things the calculations did not: the escape wheel advances in alternate steps of 12.29° and 11.71°, which indicates unequal drop on the two stones, and, in a run made without contact friction, the escapement delivers about 0.6 of its work to the balance, which is the efficiency of the geometry before friction and is consistent with the 0.30 assumed for

a real escapement. The solver does not remain stable beyond about a second of simulated time, so the amplitude was still rising when the run ended. Obtaining a stable run at all depended on treating the simulation parameters as an engineering problem in their own right: an integrator that tolerates abrupt changes of step, a step size derived from the speed of a landing tooth, and contact properties scaled to parts whose critical overlap is $33\text{ }\mu\text{m}$.

The limits of the work are stated in Section 13 and can be summarised as follows: nothing has been built, the frame and bearings are not designed, the springs rest on beam theory, and three efficiencies come from literature. Within those limits, every dimension in the model now has a reason that can be traced to a requirement, a formula and a number, and the appendices that follow give each of those calculations in full.

Acknowledgments

The tooth profiles, the contact analysis and the Monte Carlo study in this report were produced with the Watchmaking Gear Simulator published by hdrpano [13]. I did not write that software; I am grateful that it is openly available, because the NIHS standards themselves are not. The project owes its concept to Bartosz Ciechanowski's illustrated article on the mechanical watch [1]. It was my reference for how the mechanism works, the source of the starting requirements, and the inspiration for the overall design and for the way the motion studies are framed. I used it by reading it; no model or code from it is reproduced here.

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A Worked Calculations: Gear Train

The appendices work each calculation of the report with the numbers substituted. Where the result is also a variable of the equations file, the calculation ends with the value that SolidWorks evaluates for that variable, so that the hand calculation and the model can be compared directly.

A.1 Rate and Transmission Ratios

Balance frequency. Two beats per oscillation and 3600 seconds per hour:

$$f = \frac{\text{bph}}{2 \times 3600} = \frac{28\,800}{7200} = 4.000 \text{ Hz.}$$

Value in the equations file: **f_balance** = 4.

Escape wheel speed. One tooth passes per oscillation, so the wheel makes f/Z_e revolutions per second:

$$n_e = \frac{4}{15} \text{ rev/s} = 16 \text{ rev/min.}$$

The fourth wheel must turn at 1 rev/min, so $R_{4e} = 16$, obtained as $Z_4/p_e = 96/6 = 16$.

Train ratio.

$$R_{b4} = \frac{Z_b}{p_2} \cdot \frac{Z_2}{p_3} \cdot \frac{Z_3}{p_4} = \frac{84}{12} \times \frac{60}{8} \times \frac{64}{8} = 7 \times 7.5 \times 8 = 420.$$

$$R_{b4}R_{4e} = 420 \times 16 = 6720.$$

Value in the equations file: **R_b4** = 420, **R_4e** = 16.

Rate check. Working forwards from the fourth wheel at 60 rev/h:

$$\text{bph} = 60 \times R_{4e} \times Z_e \times 2 = 60 \times 16 \times 15 \times 2 = 28\,800.$$

Value in the equations file: **bph_check** = 28 800.

Periods. The fourth wheel turns once per minute. Then

$$\begin{aligned} \text{third wheel: } & 1 \text{ min} \times \frac{64}{8} = 8 \text{ min,} \\ \text{second wheel: } & 8 \text{ min} \times \frac{60}{8} = 60 \text{ min,} \\ \text{barrel: } & 60 \text{ min} \times \frac{84}{12} = 420 \text{ min} = 7.00 \text{ h.} \end{aligned}$$

Value in the equations file: **T_2_min** = 60, **T_br1_h** = 7.

Barrel turns for the reserve.

$$n_{40} = \frac{40 \text{ h}}{7 \text{ h/turn}} = 5.71 \text{ turns.}$$

Original second wheel. For comparison, $Z_2 = 54$ gave $R_{b4} = 7 \times 6.75 \times 8 = 378$, a second wheel period of $8 \times 54/8 = 54$ minutes and a barrel period of 6.30 hours.

A.2 Centre Distances and Axis Positions

$$a = \frac{m(Z+p)}{2}$$

$$\begin{aligned} a_{b2} &= \frac{1}{2} \times 0.140 \times (84 + 12) = 6.720 \text{ mm,} & a_{23} &= \frac{1}{2} \times 0.120 \times (60 + 8) = 4.080 \text{ mm,} \\ a_{34} &= \frac{1}{2} \times 0.100 \times (64 + 8) = 3.600 \text{ mm,} & a_{4e} &= \frac{1}{2} \times 0.065 \times (96 + 6) = 3.315 \text{ mm.} \end{aligned}$$

The axis positions are the running sums: 6.720, 10.800, 14.400 and 17.715 mm. Adding the escapement distances of Section 9, 2.540 and 2.671 mm, gives the pallet axis at 20.255 mm and the balance axis at 22.926 mm. Value in the equations file: **X_2** to **X_Bal**.

A.3 Angular Speeds

Starting from the fourth wheel, $\omega_4 = 2\pi/60 = 0.10472 \text{ rad/s}$:

$$\begin{aligned}\omega_e &= \omega_4 \times 16 = 1.6755 \text{ rad/s}, & \omega_3 &= \omega_4/8 = 1.3090 \times 10^{-2} \text{ rad/s}, \\ \omega_2 &= \omega_3/7.5 = 1.7453 \times 10^{-3} \text{ rad/s}, & \omega_b &= \omega_2/7 = 2.4933 \times 10^{-4} \text{ rad/s}.\end{aligned}$$

A.4 Torque at Full Wind, Lossless

This is the load case for strength. From $M_{\max} = 4.2117 \text{ N mm}$ at the barrel (Appendix C), each stage divides the torque by its ratio:

$$\begin{aligned}T_2 &= 4.2117/7 = 0.6017 \text{ N mm}, & T_3 &= 0.6017/7.5 = 0.08022 \text{ N mm}, \\ T_4 &= 0.08022/8 = 0.010028 \text{ N mm}, & T_e &= 0.010028/16 = 6.267 \times 10^{-4} \text{ N mm}.\end{aligned}$$

The tangential force at each mesh is the torque of the driving wheel divided by its pitch radius:

$$\begin{aligned}F_1 &= \frac{4.2117}{5.880} = 0.716 \text{ N}, & F_2 &= \frac{0.6017}{3.600} = 0.167 \text{ N}, \\ F_3 &= \frac{0.08022}{3.200} = 0.0251 \text{ N}, & F_4 &= \frac{0.010028}{3.120} = 0.00321 \text{ N}.\end{aligned}$$

A.5 Torque and Power at Half Reserve, with Losses

This is the load case for the escapement. The barrel torque 20 hours after full wind is (Appendix C)

$$M_{\text{run}} = 4.2117 - 0.33719 \times \frac{20}{7} = 3.2483 \text{ N mm}.$$

The assumed train efficiency of 0.70 is distributed as the simulated mesh efficiency of each stage times an equal share of the remaining pivot loss:

$$\begin{aligned}\eta_{\text{gears}} &= 0.975 \times 0.962 \times 0.963 \times 0.954 = 0.8617, \\ \eta_{\text{pivots}} &= \frac{0.70}{0.8617} = 0.8124, \\ \eta_{\text{pivot, stage}} &= 0.8124^{1/4} = 0.9494.\end{aligned}$$

Then

$$\begin{aligned}T_2 &= 3.2483 \times \frac{1}{7} \times 0.975 \times 0.9494 = 0.4295 \text{ N mm}, \\ T_3 &= 0.4295 \times \frac{1}{7.5} \times 0.962 \times 0.9494 = 0.05230 \text{ N mm}, \\ T_4 &= 0.05230 \times \frac{1}{8} \times 0.963 \times 0.9494 = 0.005977 \text{ N mm}, \\ T_E &= 0.005977 \times \frac{1}{16} \times 0.954 \times 0.9494 = 3.384 \times 10^{-4} \text{ N mm}.\end{aligned}$$

The same end value follows directly from $T_E = M_{\text{run}} \eta_t / (R_{b4} R_{4e}) = 3.2483 \times 0.70 / 6720$. Value in the equations file: $\mathbf{E_T} = 3.3836 \times 10^{-4}$.

The power at each arbor is $P = T\omega$:

$$\begin{aligned}P_b &= 3.2483 \times 2.4933 \times 10^{-4} = 8.10 \times 10^{-4} \text{ N mm/s} = 0.81 \mu\text{W}, \\ P_2 &= 0.4295 \times 1.7453 \times 10^{-3} = 0.75 \mu\text{W}, & P_3 &= 0.05230 \times 1.3090 \times 10^{-2} = 0.68 \mu\text{W}, \\ P_4 &= 0.005977 \times 0.10472 = 0.63 \mu\text{W}, & P_e &= 3.384 \times 10^{-4} \times 1.6755 = 0.57 \mu\text{W}.\end{aligned}$$

The ratio $P_e/P_b = 0.70$, as it must.

A.6 Hour and Minute Display Train

Required: a reduction of 12 in two stages on a common pair of axes with one module.

$$\frac{30}{10} \times \frac{32}{8} = 3 \times 4 = 12, \quad 10 + 30 = 40 = 8 + 32.$$

$$a_{mw} = \frac{1}{2} \times 0.12 \times (10 + 30) = 2.400 \text{ mm} = \frac{1}{2} \times 0.12 \times (8 + 32).$$

Pitch diameters: cannon pinion 1.20, minute wheel 3.60, minute pinion 0.96, hour wheel 3.84 mm.
Value in the equations file: $R_{mw} = 12$, $a_{mw} = a_{mw_chk} = 2.4$.

B Worked Calculations: NIHS 20-30 Geometry

The first mesh is worked in full, the barrel tooth and the leaf of the second pinion, both at $m = 0.140 \text{ mm}$. Coordinates are in millimetres with the origin at the centre of the gear. The results for all four meshes are tabulated at the end.

B.1 Barrel Tooth ($Z = 84$)

Factors. $h_a = 1.40$, $h_f = 1.20$, $s^* = 1.60257$, $R_o = 2.00 \text{ m}$.

Circles.

$$\begin{aligned} r_p &= \frac{0.140 \times 84}{2} = 5.880, \\ r_a &= 5.880 + 1.40 \times 0.140 = 6.076, \\ r_f &= 5.880 - 1.20 \times 0.140 = 5.712. \end{aligned}$$

Value in the equations file: $BR_da = 12.152$, $BR_df = 11.424$.

Angles. The angular pitch is $2\pi/84$. The tooth occupies an arc s^*m on the pitch circle, an angle $s^*m/r_p = 2s^*/Z$. The gap takes the rest, and half of the gap is

$$\delta = \frac{1}{2} \left(\frac{2\pi}{Z} - \frac{2s^*}{Z} \right) = \frac{\pi - s^*}{Z} = \frac{3.14159 - 1.60257}{84} = 0.018322 \text{ rad} = 1.0498^\circ.$$

The centreline of the tooth is at half a pitch from the centreline of the gap, $\pi/84 = 2.1429^\circ$.
Value in the equations file: $BR_half_gap = 1.04976$, $BR_half_pitch = 2.14286$.

Pitch point and tip. With the y axis along the centreline of the gap,

$$\begin{aligned} P &= r_p(\sin \delta, \cos \delta) = (0.10773, 5.87901), \\ T &= r_a(\sin \frac{\pi}{Z}, \cos \frac{\pi}{Z}) = (0.22719, 6.07175). \end{aligned}$$

Ogive centre. The chord and its midpoint are

$$|PT| = \sqrt{0.11946^2 + 0.19274^2} = 0.22676, \quad M = (0.16746, 5.97538).$$

The centre lies on the perpendicular bisector at a distance

$$h = \sqrt{R_o^2 - \frac{1}{4}|PT|^2} = \sqrt{0.280^2 - 0.11338^2} = 0.25602$$

from M . The unit vector along the chord is $(0.5268, 0.8500)$, and the perpendicular pointing away from the gap, into the tooth, is $(0.8500, -0.5268)$. Hence

$$C = M + h(0.8500, -0.5268) = (0.38506, 5.84050).$$

In the frame of the generated file, which is centred on the tooth and not on the gap, this point is $(-0.16641, 5.85082)$ for the same flank. The file gives $(-0.1664121, 5.8508178)$. The two agree to within 10^{-7} mm .

Root fillet. The fillet is centred on the gap centreline at $(0, y_c)$ with radius $\rho = y_c - r_f$, so that it touches the root circle, and it is tangent to the ogive from outside:

$$|C - (0, y_c)| = R_o + (y_c - r_f).$$

Squaring, $C_x^2 + C_y^2 - 2C_y y_c + y_c^2 = (R_o - r_f)^2 + 2(R_o - r_f)y_c + y_c^2$, and the y_c^2 terms cancel:

$$y_c = \frac{|C|^2 - (R_o - r_f)^2}{2(C_y + R_o - r_f)} = \frac{34.25976 - (0.280 - 5.712)^2}{2(5.84050 + 0.280 - 5.712)} = \frac{4.75314}{0.81701} = 5.81774.$$

$$\rho = 5.81774 - 5.712 = 0.10574 \text{ mm} = 0.755 m.$$

Verification: $|C - (0, y_c)| = \sqrt{0.38506^2 + 0.02277^2} = 0.38574$ and $R_o + \rho = 0.280 + 0.10574 = 0.38574$.

The generated profile uses $\rho = 0.1079 \text{ mm} = 0.771 m$, with the fillet centre slightly off the gap centreline. The difference in the root outline is below $1 \mu\text{m}$ for this wheel.

B.2 Second Pinion Leaf ($n = 12$)

Factors. $h_a = 0.72$, $h_f = 1.80$, $p_l = 1.11823$, $R_t = 0.85 m$.

Circles.

$$\begin{aligned} r_p &= \frac{0.140 \times 12}{2} = 0.840, \\ r_a &= 0.840 + 0.72 \times 0.140 = 0.9408, \\ r_f &= 0.840 - 1.80 \times 0.140 = 0.588. \end{aligned}$$

Value in the equations file: $\text{P2_da} = 1.8816$, $\text{P2_df} = 1.176$.

Flank angle.

$$\alpha = \frac{p_l}{n} = \frac{1.11823}{12} = 0.093186 \text{ rad} = 5.3392^\circ, \quad \sin \alpha = 0.093051, \quad \cos \alpha = 0.995661.$$

Value in the equations file: $\text{P2_half_leaf} = 5.3392$.

Flank end point. The tip arc has radius $R_t = 0.85 \times 0.140 = 0.119$. Let $F_t = t(\sin \alpha, \cos \alpha)$ be its point of tangency with the flank, and $\mathbf{n} = (-\cos \alpha, \sin \alpha)$ the unit normal to the flank pointing into the leaf. The centre of the arc is $C = F_t + R_t \mathbf{n}$, and the tip $T = (0, r_a)$ must be at distance R_t from it:

$$|T - F_t - R_t \mathbf{n}|^2 = R_t^2.$$

Expanding, with $F_t \cdot \mathbf{n} = 0$ and $T \cdot \mathbf{n} = r_a \sin \alpha$, $T \cdot F_t = r_a t \cos \alpha$,

$$r_a^2 + t^2 - 2r_a t \cos \alpha - 2R_t r_a \sin \alpha = 0,$$

whose smaller root is

$$\begin{aligned} t &= r_a \cos \alpha - \sqrt{2r_a R_t \sin \alpha - r_a^2 \sin^2 \alpha} \\ &= 0.93672 - \sqrt{0.020835 - 0.007664} = 0.93672 - 0.11477 = 0.82195. \end{aligned}$$

$$F_t = 0.82195 \times (0.093051, 0.995661) = (0.07648, 0.81839).$$

The generated file gives $(0.0764837, 0.8183851)$.

Root fillet. The centreline of the gap between leaves is at $\pi/n = 15^\circ$ from the centreline of the leaf, so it makes an angle $15^\circ - 5.3392^\circ = 9.6608^\circ$ with the flank. A circle centred on it at radius y_c is at perpendicular distance $y_c \sin(9.6608^\circ)$ from the flank, and this must equal its radius $\rho = y_c - r_f$:

$$y_c = \frac{r_f}{1 - \sin(\pi/n - \alpha)} = \frac{0.588}{1 - 0.167816} = 0.70657, \quad \rho = 0.70657 - 0.588 = 0.11857 \text{ mm.}$$

The generated file gives 0.1185740.

Leaf thickness on the pitch circle. The pitch radius, 0.840, lies above the top of the flank, at radius 0.82195, so the pitch circle crosses the tip arc and not the flank. The tip arc is centred at $(-0.04200, 0.82946)$; its intersection with the pitch circle is at $x = 0.07679$. The chordal thickness is 0.1536 mm, or 1.10 m .

B.3 Clearances and Backlash of the First Mesh

$$\begin{aligned} c_w &= a - r_{a,w} - r_{f,p} = 6.720 - 6.076 - 0.588 = 0.056 \text{ mm,} \\ c_p &= a - r_{a,p} - r_{f,w} = 6.720 - 0.9408 - 5.712 = 0.067 \text{ mm,} \\ j &= \pi m - s_w^* m - s_p^* m = 0.4398 - 0.2244 - 0.1539 = 0.062 \text{ mm.} \end{aligned}$$

B.4 Results for All Meshes

Table 38: Principal radii of all wheels and pinions (mm).

Mesh	Wheel				Pinion		
	r_p	r_a	r_f	δ	r_p	r_a	r_f
1 (84/12, m 0.140)	5.880	6.076	5.712	1.0498°	0.840	0.9408	0.588
2 (60/8, m 0.120)	3.600	3.756	3.444	1.4911°	0.480	0.5784	0.276
3 (64/8, m 0.100)	3.200	3.330	3.070	1.3985°	0.400	0.4820	0.230
4 (96/6, m 0.065)	3.120	3.198	3.029	0.9556°	0.195	0.2483	0.091

Table 39: Constructed points compared with the generated profiles (mm). Wheel coordinates are in the tooth-centred frame of the generated files.

Mesh	Ogive centre, constructed	Ogive centre, file	Leaf flank top, constructed	Pinion fillet ρ
1	(−0.166412, 5.850818)	(−0.166412, 5.850818)	(0.076483, 0.818385)	0.118574
2	(−0.142512, 3.562893)	(−0.142512, 3.562893)	(0.064878, 0.459044)	0.091817
3	(−0.118783, 3.169095)	(−0.118783, 3.169095)	(0.054065, 0.382537)	0.076514
4	(−0.077283, 3.093466)	(−0.077283, 3.093466)	(0.034465, 0.175104)	0.043484

C Worked Calculations: Mainspring

Inputs. $E = 220\,000$ MPa, $b = 1.40$ mm, $t = 0.095$ mm, $\sigma_d = 2000$ MPa; barrel module 0.140 mm, 84 teeth, dedendum factor 1.20, wall under the roots 0.43 mm.

C.1 Available Space

$$R = \frac{0.140 \times (84 - 2 \times 1.20)}{2} - 0.43 = 5.712 - 0.43 = 5.282 \text{ mm,} \quad r = \frac{R}{3} = 1.7607 \text{ mm.}$$

Value in the equations file: `R_barrel_in = 5.282`, `r_arbor = 1.76067`.

C.2 Spring Length

$$L = \frac{\pi(R^2 - r^2)}{2t} = \frac{\pi(27.8995 - 3.1000)}{2 \times 0.095} = \frac{77.910}{0.190} = 410.05 \text{ mm.}$$

Value in the equations file: `ms_L` = 410.054.

C.3 Number of Turns

The plan area of the strip divided by π is $Lt/\pi = 410.05 \times 0.095/\pi = 12.3998 \text{ mm}^2$.

$$R_i = \sqrt{r^2 + Lt/\pi} = \sqrt{3.1000 + 12.3998} = 3.9370 \text{ mm},$$

$$R_o = \sqrt{R^2 - Lt/\pi} = \sqrt{27.8995 - 12.3998} = 3.9370 \text{ mm}.$$

The two are equal because the spring fills exactly half of the annulus. The number of coils in each state is the radial thickness of the ring divided by the strip thickness:

$$n_{\text{wound}} = \frac{3.9370 - 1.7607}{0.095} = 22.908, \quad n_{\text{let down}} = \frac{5.282 - 3.9370}{0.095} = 14.158,$$

$$N = 22.908 - 14.158 = 8.750 \text{ turns.}$$

Value in the equations file: `MS_N` = 8.75025. Required: 5.71 turns.

C.4 Stiffness

$$K = \frac{E b t^3}{12L} = \frac{220\,000 \times 1.40 \times 8.5738 \times 10^{-4}}{12 \times 410.05} = \frac{264.07}{4920.6} = 0.053666 \text{ N mm/rad.}$$

Per turn of the barrel, $2\pi K = 0.33719 \text{ N mm}$. Value in the equations file: `MS_K` = 0.0536661.

C.5 Torque Characteristic

$$M_{\text{max}} = \frac{\sigma_d b t^2}{6} = \frac{2000 \times 1.40 \times 0.009025}{6} = 4.2117 \text{ N mm.}$$

$$M(24 \text{ h}) = 4.2117 - 0.33719 \times \frac{24}{7} = 4.2117 - 1.1561 = 3.0556 \text{ N mm,}$$

$$M(40 \text{ h}) = 4.2117 - 0.33719 \times \frac{40}{7} = 4.2117 - 1.9268 = 2.2848 \text{ N mm,}$$

$$M_0 = 4.2117 - 0.33719 \times 8.7503 = 4.2117 - 2.9505 = 1.2611 \text{ N mm.}$$

Value in the equations file: `MS_Mmax` = 4.21167, `MS_M24` = 3.05557, `MS_M40` = 2.28484, `MS_M0` = 1.26114.

Safety factors at full wind.

$$\frac{R_{p0.2}}{\sigma_d} = \frac{2475}{2000} = 1.24, \quad \frac{R_m}{\sigma_d} = \frac{3095}{2000} = 1.55.$$

Stored energy. With a linear torque law the work delivered over the full N turns is the mean torque times the angle:

$$W = \frac{M_{\text{max}} + M_0}{2} \times 2\pi N = \frac{4.2117 + 1.2611}{2} \times 2\pi \times 8.750 = 150.4 \text{ N mm} = 0.150 \text{ J.}$$

Comparison with the earlier model. With $R = 4.400$, $b = 1.00$, $E = 200\,000$: $L = 284.5 \text{ mm}$, $N = 7.29$, $K = 0.0502$, and $M_{\text{max}} = 2\pi K N = 2\pi \times 0.0502 \times 7.29 = 2.30 \text{ N mm}$, falling to zero when let down.

C.6 Bending Stress at the Arbor

A strip bent from a free curvature κ_0 to a curvature $1/r$ has a surface strain $(t/2)(1/r - \kappa_0)$. For a strip that is straight when free:

$$\sigma_a = \frac{E t}{2r} = \frac{220\,000 \times 0.095}{2 \times 1.7607} = 5935 \text{ MPa.}$$

Requiring $\sigma_a \leq \sigma_d$:

$$\kappa_0 \geq \frac{1}{r} - \frac{2\sigma_d}{E t} = \frac{1}{1.7607} - \frac{4000}{20\,900} = 0.56797 - 0.19139 = 0.37658 \text{ mm}^{-1},$$

$$R_{\text{set}} \leq \frac{1}{0.37658} = 2.655 \text{ mm.}$$

Value in the equations file: `MS_r_set` = 2.65548.

C.7 Barrel Height and Spring Mass

$$H = t_{\text{base}} + b + \text{clearance} + t_{\text{lid}} = 0.18 + 1.40 + 0.10 + 0.20 = 1.88 \text{ mm.}$$

$$m_s = \rho L b t = 8.5 \text{ mg/mm}^3 \times 410.05 \times 1.40 \times 0.095 \text{ mm}^3 = 463.6 \text{ mg.}$$

Value in the equations file: `BR_H` = 1.88.

D Worked Calculations: Oscillator, Hairspring and Amplitude

D.1 Hairspring Stiffness

Unit conversion. $1 \text{ mg cm}^2 = 10^{-6} \text{ kg} \times 10^{-4} \text{ m}^2 = 10^{-10} \text{ kg m}^2$. Since $1 \text{ kg m}^2 = 1 \text{ N m s}^2 = 10^3 \text{ N mm s}^2$, it follows that $1 \text{ mg cm}^2 = 10^{-7} \text{ N mm s}^2$.

$$I = 3.60 \text{ mg cm}^2 = 3.60 \times 10^{-7} \text{ N mm s}^2, \quad \omega = 2\pi f = 2\pi \times 4 = 25.133 \text{ rad/s.}$$

$$k = I\omega^2 = 3.60 \times 10^{-7} \times 631.65 = 2.2740 \times 10^{-4} \text{ N mm/rad.}$$

Per degree: $k \times \pi/180 = 3.9688 \times 10^{-6} \text{ N mm/deg}$. Value in the equations file: `B_k` = 2.27396×10^{-4} , `B_k_deg` = 3.9688×10^{-6} .

D.2 Hairspring Length, Number of Turns and Pitch

Inputs: $E = 190\,000 \text{ MPa}$, $b = 0.10 \text{ mm}$, $t = 0.025 \text{ mm}$, $r_0 = 0.45 \text{ mm}$, $r_1 = 2.60 \text{ mm}$.

$$L = \frac{E b t^3}{12k} = \frac{190\,000 \times 0.10 \times 1.5625 \times 10^{-5}}{12 \times 2.2740 \times 10^{-4}} = \frac{0.296875}{2.7287 \times 10^{-3}} = 108.80 \text{ mm.}$$

$$n = \frac{L}{\pi(r_0 + r_1)} = \frac{108.80}{\pi \times 3.05} = 11.354 \text{ turns,} \quad \text{pitch} = \frac{2.60 - 0.45}{11.354} = 0.1894 \text{ mm.}$$

Gap between coils: $0.1894 - 0.025 = 0.164 \text{ mm}$. Value in the equations file: `HS_L` = 108.795, `HS_turns` = 11.3543, `HS_pitch` = 0.189356.

D.3 Hairspring Stress and Stored Energy

At $\theta = 300^\circ = 5.2360 \text{ rad}$:

$$M = k\theta = 2.2740 \times 10^{-4} \times 5.2360 = 1.1907 \times 10^{-3} \text{ N mm,}$$

$$\sigma = \frac{6M}{b t^2} = \frac{6 \times 1.1907 \times 10^{-3}}{0.10 \times 6.25 \times 10^{-4}} = 114.3 \text{ MPa.}$$

At $330^\circ = 5.7596 \text{ rad}$: $\sigma = 125.7 \text{ MPa}$, strain $\sigma/E = 6.6 \times 10^{-4}$.

$$SF_{\text{yield}} = \frac{1300}{125.7} = 10.3, \quad SF_{\text{fatigue}} = \frac{450}{125.7} = 3.6.$$

Energy stored at 300° : $\frac{1}{2} k \theta^2 = \frac{1}{2} \times 2.2740 \times 10^{-4} \times 27.416 = 3.117 \times 10^{-3} \text{ N mm} = 3.12 \mu\text{J}$.

D.4 Rate Sensitivity to Hairspring Thickness

From $f \propto \sqrt{k}$ and $k \propto t^3$, $\Delta f/f = \frac{3}{2} \Delta t/t$. For $\Delta t/t = 1\%$ (that is $0.25 \mu\text{m}$): $\Delta f/f = 1.5\%$, and $0.015 \times 86400 \text{ s} = 1296$ seconds per day, about 21.6 minutes.

D.5 Thermal Coefficient

$$\frac{\Delta f}{f \Delta T} = \frac{1}{2}(\gamma_s + 3\alpha_s - 2\alpha_b) = 0 \implies \gamma_s = 2\alpha_b - 3\alpha_s = (34 - 22.5) \times 10^{-6} = +11.5 \times 10^{-6} \text{ K}^{-1}.$$

A mismatch of $1 \times 10^{-6} \text{ K}^{-1}$ gives $\frac{1}{2} \times 10^{-6} \times 86400 = 0.043$ seconds per day per kelvin.

D.6 Equivalent Viscous Damping

For a lightly damped oscillator $Q = I\omega/c$:

$$c = \frac{I\omega}{Q} = \frac{3.60 \times 10^{-7} \times 25.133}{250} = 3.619 \times 10^{-8} \text{ N mm s/rad} = 6.317 \times 10^{-10} \text{ N mm/(deg/s)}.$$

D.7 Roller Jewel Radius

In the triangle formed by the pallet axis P , the balance axis B and the jewel J at the end of engagement, the angle at P is $\lambda/2 = 5.25^\circ$, the angle at B is $\beta/2 = 26^\circ$, the side BJ is r_j and the side PB is $L_f + r_j$, where the fork length L_f is the distance from P to the jewel when the jewel is on the line of centres at mid-swing. (The side PJ at the end of engagement is slightly longer than L_f , 2.257 mm, because the jewel moves along the fork slot; it is not needed.) The angle at J is $180^\circ - 31.25^\circ$, so by the sine rule

$$\frac{r_j}{\sin 5.25^\circ} = \frac{L_f + r_j}{\sin 31.25^\circ} \implies r_j = \frac{L_f \sin 5.25^\circ}{\sin 31.25^\circ - \sin 5.25^\circ} = \frac{2.20 \times 0.091502}{0.518773 - 0.091502} = 0.4711 \text{ mm}.$$

Distance between axes: $2.20 + 0.4711 = 2.6711 \text{ mm}$. Value in the equations file: `L_jewel_r = 0.471137`, `L_pb = 2.67114`.

D.8 Amplitude at Full Wind

$$T_E = \frac{M_{\max} \eta_t}{R_{b4} R_{4e}} = \frac{4.2117 \times 0.70}{6720} = 4.3872 \times 10^{-4} \text{ N mm}, \quad \varphi = \frac{2\pi}{15} = 0.41888 \text{ rad}.$$

$$\begin{aligned} \theta &= \sqrt{\frac{Q \eta_e T_E \varphi}{\pi k}} = \sqrt{\frac{250 \times 0.30 \times 4.3872 \times 10^{-4} \times 0.41888}{\pi \times 2.2740 \times 10^{-4}}} \\ &= \sqrt{\frac{0.013783}{7.1438 \times 10^{-4}}} = 4.3924 \text{ rad} = 251.7^\circ. \end{aligned}$$

Check of the energy balance at this amplitude: stored energy $\frac{1}{2} k \theta^2 = 2.1936 \times 10^{-3} \text{ N mm}$; lost per period $(2\pi/250) \times 2.1936 \times 10^{-3} = 5.513 \times 10^{-5} \text{ N mm}$; supplied per period $0.30 \times 4.3872 \times 10^{-4} \times 0.41888 = 5.513 \times 10^{-5} \text{ N mm}$.

Since $\theta \propto \sqrt{M}$, the other points follow by scaling:

$$\begin{aligned} \theta(24 \text{ h}) &= 251.7^\circ \times \sqrt{3.0556/4.2117} = 214.4^\circ, \\ \theta(40 \text{ h}) &= 251.7^\circ \times \sqrt{2.2848/4.2117} = 185.4^\circ, \\ \theta(\text{half reserve}) &= 251.7^\circ \times \sqrt{3.2483/4.2117} = 221.0^\circ. \end{aligned}$$

Value in the equations file: `amp_full = 251.665`, `amp_24 = 214.359`, `amp_40 = 185.364`.

D.9 Power Reserve to 150° Amplitude

$$M_{150} = M_{\max} \left(\frac{150}{251.665} \right)^2 = 4.2117 \times 0.35526 = 1.4962 \text{ N mm.}$$

$$n_{150} = \frac{M_{\max} - M_{150}}{2\pi K} = \frac{4.2117 - 1.4962}{0.33719} = 8.053 \text{ turns,} \quad t_{150} = 8.053 \times 7 = 56.4 \text{ h.}$$

Value in the equations file: MS_T150 = 1.4962, PR_h = 56.37. The limit set by the turns of the spring is $8.750 \times 7 = 61.3$ hours.

Case $Q = 200$. θ scales with $\sqrt{Q}$: $251.7^\circ \times \sqrt{0.8} = 225.1^\circ$ at full wind and $185.4^\circ \times \sqrt{0.8} = 165.8^\circ$ at 40 hours. M_{150} scales with $1/Q$: $1.4962/0.8 = 1.8703 \text{ N mm}$, so $t_{150} = (4.2117 - 1.8703)/0.33719 \times 7 = 48.6$ hours.

D.10 Time to First Fork Engagement in the Motion Study

Released from rest at $\theta_0 = 140^\circ$, an undamped balance follows $\theta = \theta_0 \cos \omega t$. The jewel enters the fork at $\theta = 26^\circ$:

$$\cos \omega t = \frac{26}{140} = 0.18571, \quad \omega t = 1.3840 \text{ rad}, \quad t = \frac{1.3840}{25.133} = 0.0551 \text{ s.}$$

The balance passes through rest at $\omega t = \pi/2$, that is at $t = 0.0625 \text{ s}$, which is the time measured in Study C. For a release from 220° the same calculation gives $\cos \omega t = 0.11818$ and $t = 0.0578 \text{ s}$.

E Worked Calculations: Machine Elements

Material values: brass $E = 100 \text{ GPa}$, $\nu = 0.33$, $R_{p0.2} = 400 \text{ MPa}$, $R_m = 550 \text{ MPa}$, $S_e = 150 \text{ MPa}$; steel $E = 210 \text{ GPa}$, $\nu = 0.30$, $R_{p0.2} = 1600 \text{ MPa}$, $R_m = 2000 \text{ MPa}$, $S_e = 800 \text{ MPa}$; ruby $E = 400 \text{ GPa}$.

E.1 Fatigue Strength at Finite Life

The S-N line is straight on logarithmic axes from $0.9R_m$ at 10^3 cycles to S_e at 10^8 cycles:

$$S_N(N) = 0.9R_m \left(\frac{N}{10^3} \right)^B, \quad B = \frac{\log_{10}(S_e/0.9R_m)}{5}.$$

Brass: $0.9R_m = 495 \text{ MPa}$, $B = \log_{10}(150/495)/5 = -0.1037$. Steel: $0.9R_m = 1800 \text{ MPa}$, $B = \log_{10}(800/1800)/5 = -0.0704$.

E.2 Tooth Bending in the First Mesh

Cycles. Ten years is 87 660 hours. The barrel makes $87\,660/7 = 1.25 \times 10^4$ turns, and each barrel tooth is loaded once per turn. The second pinion makes 8.77×10^4 turns.

Barrel tooth. $F_t = 0.716 \text{ N}$ at the tip, $r_L = r_a = 6.076 \text{ mm}$, width $b = 0.50 \text{ mm}$. Evaluating Equation (46) along the profile from $r_f = 5.712$ to $r_p = 5.880$, the stress is greatest at $r = 5.795 \text{ mm}$, in the upper part of the fillet, where the chordal thickness is $s = 0.2283 \text{ mm}$:

$$\sigma_F = \frac{6 \times 0.716 \times (6.076 - 5.795)}{0.50 \times 0.2283^2} = \frac{1.2072}{0.02606} = 46.3 \text{ MPa.}$$

$$SF_{\text{yield}} = \frac{400}{46.3} = 8.6,$$

$$S_N(1.25 \times 10^4) = 495 \times 12.5^{-0.1037} = 381 \text{ MPa,}$$

$$SF_{\text{fat}} = \frac{381}{1.2 \times 1.5 \times 46.3} = 4.6.$$

Leaf of the second pinion. $F_t = 0.716$ N at the pitch circle, $r_L = 0.840$ mm, width 0.50 mm. The stress is greatest at $r = 0.664$ mm, in the root fillet, where $s = 0.1325$ mm:

$$\sigma_F = \frac{6 \times 0.716 \times (0.840 - 0.664)}{0.50 \times 0.1325^2} = \frac{0.7561}{0.008778} = 86.0 \text{ MPa.}$$

$$SF_{\text{yield}} = \frac{1600}{86.0} = 18.6,$$

$$S_N(8.77 \times 10^4) = 1800 \times 87.7^{-0.0704} = 1314 \text{ MPa,}$$

$$SF_{\text{fat}} = \frac{1314}{1.8 \times 86.0} = 8.5.$$

Width required by bending. Equation (46) is linear in $1/b$. For the barrel teeth to reach $SF_{\text{fat}} = 1.5$ the width could fall to $0.50 \times 1.5/4.6 = 0.16$ mm. Bending therefore does not set the width; contact does.

E.3 Contact Pressure on the Barrel Teeth

First yield in line contact, with $\tau_{\text{max}} = 0.30 p_H$ and the Tresca criterion $\tau_{\text{max}} = R_{p0.2}/2$:

$$p_{H,\text{yield}} = \frac{400}{2 \times 0.30} = 667 \text{ MPa} \approx 670 \text{ MPa.}$$

Simulated at $b = 0.25$ mm: 913 MPa. Since $p_H \propto b^{-1/2}$:

$$b_{\text{min}} = 0.25 \left(\frac{913}{670} \right)^2 = 0.25 \times 1.857 = 0.464 \text{ mm,}$$

$$p_H(0.50) = 913 \times \sqrt{0.5} = 646 \text{ MPa,}$$

$$SF = \frac{670}{646} = 1.04.$$

E.4 Pivot Strength

Second arbor, running. $F = 0.479$ N, $a = 0.10/2 + 0.02 = 0.07$ mm, $d = 0.12$ mm:

$$\sigma_b = \frac{32Fa}{\pi d^3} = \frac{32 \times 0.479 \times 0.07}{\pi \times 1.728 \times 10^{-3}} = \frac{1.0730}{5.4287 \times 10^{-3}} = 197.6 \text{ MPa,} \quad SF_{\text{fat}} = \frac{800}{197.6} = 4.0.$$

Second arbor, minimum diameter for $S = 2$.

$$d_{\text{min}} = \left(\frac{32 \times 0.479 \times 0.07 \times 2}{\pi \times 800} \right)^{1/3} = (8.539 \times 10^{-4})^{1/3} = 0.095 \text{ mm.}$$

Second arbor, shock. Mass 30.0 mg:

$$F = \frac{30.0 \times 10^{-6} \times 5000 \times 9.81}{2} = 0.736 \text{ N,}$$

$$\sigma_b = \frac{32 \times 0.736 \times 0.07}{5.4287 \times 10^{-3}} = 303.6 \text{ MPa,}$$

$$SF = \frac{1600}{303.6} = 5.3.$$

Balance, shock. Mass 28.3 mg, $d = 0.08$ mm, $a = 0.08/2 + 0.02 = 0.06$ mm:

$$F = \frac{28.3 \times 10^{-6} \times 5000 \times 9.81}{2} = 0.694 \text{ N,}$$

$$\sigma_b = \frac{32 \times 0.694 \times 0.06}{\pi \times 5.12 \times 10^{-4}} = 828.5 \text{ MPa,}$$

$$SF = \frac{1600}{828.5} = 1.93.$$

$$d_{\text{min}}(S = 2) = \left(\frac{32 \times 0.694 \times 0.06 \times 2}{\pi \times 1600} \right)^{1/3} = 0.081 \text{ mm.}$$

Buckling of the fourth arbor. $d = 0.36$ mm, length 2.65 mm, pinned ends:

$$I = \frac{\pi d^4}{64} = 8.245 \times 10^{-4} \text{ mm}^4, \quad P_{cr} = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 210\,000 \times 8.245 \times 10^{-4}}{2.65^2} = 243 \text{ N}.$$

E.5 Bearing Pressure of the Second Pivot

Pivot radius $R_1 = 0.060$ mm; diametral clearance 0.010 mm, so $R_2 = 0.065$ mm; hole height $\ell = 0.10$ mm.

$$R' = \frac{R_1 R_2}{R_2 - R_1} = \frac{0.060 \times 0.065}{0.005} = 0.78 \text{ mm}, \quad \frac{1}{E^*} = \frac{1 - 0.30^2}{210\,000} + \frac{1 - \nu_r^2}{400\,000}.$$

The contact modulus comes out between 149 and 151 GPa for a Poisson ratio of ruby between 0.25 and 0.29; the results change by less than 1 %. With $E^* = 151$ GPa:

$$b_H = \sqrt{\frac{4 \times 0.479 \times 0.78}{\pi \times 0.10 \times 151\,000}} = 5.61 \times 10^{-3} \text{ mm}, \quad p_{\max} = \frac{2 \times 0.479}{\pi \times 5.61 \times 10^{-3} \times 0.10} = 543 \text{ MPa}.$$

Projected pressure: $0.479/(0.12 \times 0.10) = 39.9$ MPa with a hole 0.10 mm high, and $0.479/(0.12 \times 0.15) = 26.6$ MPa with 0.15 mm.

E.6 Torsion of the Second Arbor and the Winding Square

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \times 0.6017}{\pi \times 0.40^3} = 47.9 \text{ MPa}.$$

For the square of side $s = 0.90$ mm on the barrel arbor, the maximum shear stress in torsion of a square section is $T/(0.208 s^3)$:

$$\tau = \frac{4.2117}{0.208 \times 0.729} = 27.8 \text{ MPa}.$$

E.7 Press Fit of the Second Wheel

Bore $d = 0.40$ mm, hub $D = 1.40$ mm, length $L = 0.18$ mm, $\mu = 0.15$, $T = 0.6017$ N mm.

Pressure needed against slip, with a factor of 2. From $T_{\text{cap}} = \mu p \pi d L d/2$:

$$p = \frac{2 \times 2T}{\mu \pi d^2 L} = \frac{2.4068}{0.15 \times \pi \times 0.16 \times 0.18} = 177.3 \text{ MPa}.$$

Compliance of the joint.

$$\frac{D^2 + d^2}{D^2 - d^2} = \frac{1.96 + 0.16}{1.96 - 0.16} = 1.1778, \\ \frac{1.1778 + 0.33}{100\,000} + \frac{1 - 0.30}{210\,000} = 1.5078 \times 10^{-5} + 0.3333 \times 10^{-5} = 1.8411 \times 10^{-5} \text{ MPa}^{-1}.$$

Interference.

$$\delta_{\min} = p d \times 1.8411 \times 10^{-5} = 177.3 \times 0.40 \times 1.8411 \times 10^{-5} = 1.31 \times 10^{-3} \text{ mm} = 1.31 \mu\text{m}.$$

Upper limit from yielding of the hub, with a factor of 1.5. $\sigma_\theta = 1.1778 p \leq 400/1.5 = 266.7$ MPa gives $p \leq 226.4$ MPa and

$$\delta_{\max} = 226.4 \times 0.40 \times 1.8411 \times 10^{-5} = 1.67 \mu\text{m}.$$

The usable window is $1.67 - 1.31 = 0.36 \mu\text{m}$.

E.8 Spoke Bending in the Second Wheel

Five spokes, width $w = 0.20$ mm, wheel thickness $t = 0.18$ mm, inner radius of the rim 3.144 mm, hub radius 0.70 mm, so $l = 2.444$ mm.

$$F_s = \frac{0.6017}{5 \times 3.144} = 0.03828 \text{ N},$$

$$M = \frac{F_s l}{2} = 0.04677 \text{ N mm},$$

$$\sigma = \frac{6M}{t w^2} = \frac{6 \times 0.04677}{0.18 \times 0.04} = 39.0 \text{ MPa}.$$

E.9 Barrel Lid and Base

Load: $W = 463.6 \times 10^{-6} \times 5000 \times 9.81 = 22.74$ N on an annulus from $r_i = 0.60$ mm (the arbor shaft) to $r_o = 5.282$ mm:

$$q = \frac{W}{\pi(r_o^2 - r_i^2)} = \frac{22.74}{\pi \times 27.54} = 0.2628 \text{ MPa}.$$

Flexural rigidity of the lid, $t = 0.20$ mm:

$$D = \frac{E t^3}{12(1 - \nu^2)} = \frac{100\,000 \times 0.008}{12 \times 0.8911} = 74.8 \text{ N mm}.$$

With w from Equation (55), the moments are

$$M_r = -D \left(w'' + \frac{\nu}{r} w' \right), \quad M_\theta = -D \left(\frac{1}{r} w' + \nu w'' \right),$$

and the shear force is proportional to $d(\nabla^2 w)/dr$. The four conditions are $M_r = 0$ and zero shear at r_i , $w = 0$ at r_o , and at r_o either $w' = 0$ (clamped) or $M_r = 0$ (simply supported). They form a linear system in C_1 to C_4 , which I solved numerically and then scanned for the largest moment between r_i and r_o . The results are:

Case	t (mm)	$\sigma = 6M/t^2$ (MPa)	$SF = 400/\sigma$
Lid, simply supported	0.10	1745	0.23
Lid, simply supported	0.20	436	0.92
Lid, clamped	0.20	158	2.53
Base, clamped	0.15	282	1.42
Base, clamped	0.18	196	2.05

Two checks on the solution: for a given edge condition the stress scales exactly as $1/t^2$ ($1745/436 = 4.00$ for a doubling of thickness, and $282/196 = 1.44 = (0.18/0.15)^2$), and the clamped case is lower than the simply supported one by a factor of 2.76.

Thickness required. For the clamped lid to reach $SF = 1.5$: $t \geq 0.20\sqrt{1.5/2.53} = 0.154$ mm. For the base: $t \geq 0.18\sqrt{1.5/2.05} = 0.154$ mm.

E.10 Mainspring Hook and Eye

$$F = \frac{M_{\max}}{r} = \frac{4.2117}{1.7607} = 2.392 \text{ N}.$$

Hook, 0.12 mm wide, 0.06 mm high, over the 1.40 mm height of the spring. Shear and bending, the latter with the load at the tip of the hook:

$$\tau = \frac{2.392}{0.12 \times 1.40} = 14.2 \text{ MPa}, \quad \sigma = \frac{6 \times 2.392 \times 0.06}{1.40 \times 0.12^2} = 42.7 \text{ MPa}.$$

Net section of the strip at the eye: $\sigma = 6 \times 4.2117/(1.25 \times 0.095^2) = 2240$ MPa; $SF = 2475/2240 = 1.10$.

Outer end on the drum wall: $F = 4.2117/5.282 = 0.80$ N.

E.11 Locking Impact

$$U = T_E \Delta\varphi = 4.3872 \times 10^{-4} \times 0.029671 = 1.302 \times 10^{-5} \text{ N mm.}$$

Contact stiffness with $L = 0.15 \text{ mm}$ and $E^* = 149 \text{ GPa}$: $k_{\text{contact}} = \pi \times 0.15 \times 149\,000/8 = 8777 \text{ N/mm}$. In series with $k_{\text{tooth}} = 82 \text{ N/mm}$:

$$k = \left(\frac{1}{82} + \frac{1}{8777} \right)^{-1} = 81.2 \text{ N/mm}, \quad F = \sqrt{2 \times 81.2 \times 1.302 \times 10^{-5}} = 0.0460 \text{ N.}$$

For a corner radius $R = 0.010 \text{ mm}$:

$$p_H = \sqrt{\frac{F E^*}{\pi R L}} = \sqrt{\frac{0.0460 \times 149\,000}{\pi \times 0.010 \times 0.15}} = \sqrt{1.4545 \times 10^6} = 1206 \text{ MPa},$$

$$SF = \frac{1.67 \times 1600}{1206} = 2.22.$$

The pressure scales as $R^{-1/2}$: 1705 MPa at $R = 0.005 \text{ mm}$ and 853 MPa at 0.020 mm .

If the tooth were rigid, $k = 8777 \text{ N/mm}$ and $F = \sqrt{2 \times 8777 \times 1.302 \times 10^{-5}} = 0.478 \text{ N}$, ten times larger.

E.12 Wear of the Second Pivot

Revolutions in ten years: 87 660. Sliding distance: $87\,660 \times \pi \times 0.12 = 33\,050 \text{ mm} = 33 \text{ m}$. Hardness: $700 \text{ HV} \times 9.807 = 6865 \text{ MPa}$. With $p = 26.6 \text{ MPa}$ and $K = 10^{-9}$:

$$h = \frac{K p s}{H} = \frac{10^{-9} \times 26.6 \times 33\,050}{6865} = 1.28 \times 10^{-7} \text{ mm} = 1.3 \times 10^{-4} \mu\text{m}.$$

E.13 Contact Parameters of the Motion Study

Tooth of effective mass $m = 3 \text{ mg} = 3 \times 10^{-6} \text{ kg}$ at $v = 70 \text{ mm/s}$:

$$E_k = \frac{1}{2} m v^2 = \frac{1}{2} \times 3 \times 10^{-6} \times 0.070^2 = 7.35 \times 10^{-9} \text{ J} = 7.35 \times 10^{-6} \text{ N mm}.$$

With $F = k_c x^{1.5}$, the stored energy is $k_c x^{2.5}/2.5$:

$$x = \left(\frac{2.5 \times 7.35 \times 10^{-6}}{10^4} \right)^{0.4} = 3.2 \times 10^{-4} \text{ mm} = 0.32 \mu\text{m}.$$

Linearised stiffness: $dF/dx = 1.5 k_c \sqrt{x} = 1.5 \times 10^4 \times 0.0179 = 268 \text{ N/mm} = 2.68 \times 10^5 \text{ N/m}$.

$$c_{\text{crit}} = 2\sqrt{k m} = 2\sqrt{2.68 \times 10^5 \times 3 \times 10^{-6}} = 1.79 \text{ N s/m} = 1.79 \times 10^{-3} \text{ N s/mm}.$$

The value used, 10^{-3} N s/mm , is 56 % of critical.

Travel of the tooth tip in one integrator step: $70 \text{ mm/s} \times 10 \mu\text{s} = 0.7 \mu\text{m}$; at $200 \mu\text{s}$ it would be $14 \mu\text{m}$.

F Variables of the Equations File

This appendix lists the 393 variables of `watch_global.txt.txt` with their evaluated values at $S = 1$, grouped by subsystem. Lengths are in mm, angles in degrees, torques in N mm, stiffnesses in N mm/rad and stresses in MPa. A dagger marks an independent input, that is a variable defined by a number or by the scale factor alone; all others are computed from them.

Global Parameters, Rate and Tooth Numbers

Variable	Value	Variable	Value
$S^\dagger$	1	$Z4^\dagger$	96
$bph^\dagger$	28800	$pe^\dagger$	6
$f_balance$	4	$Ze^\dagger$	15
$reserve_h^\dagger$	40	R_b4	420
$Zb^\dagger$	84	R_4e	16
$p2^\dagger$	12	bph_check	28800
$Z2^\dagger$	60	$lv1^\dagger$	0.5
$p3^\dagger$	8	T_br1_h	7
$Z3^\dagger$	64	T_2_min	60
$p4^\dagger$	8		

Modules, Pitch Diameters and Centre Distances

Variable	Value	Variable	Value
$m_b2^\dagger$	0.14	d_p2	1.68
$m_23^\dagger$	0.12	d_2	7.2
$m_34^\dagger$	0.1	d_p3	0.96
$m_4e^\dagger$	0.065	d_3	6.4
a_b2	6.72	d_p4	0.8
a_23	4.08	d_4	6.24
a_34	3.6	d_pe	0.39
a_4e	3.315	$tol_a_13^\dagger$	0.01
d_b	11.76	$tol_a_4^\dagger$	0.005

NIHS 20-30 Factors

Variable	Value	Variable	Value
$k_og^\dagger$	2	$n3_wa^\dagger$	1.3
$n1_wa^\dagger$	1.4	$n3_wf^\dagger$	1.3
$n1_wf^\dagger$	1.2	$n3_ws^\dagger$	1.57945
$n1_ws^\dagger$	1.60257	$n3_pa^\dagger$	0.82
$n1_pa^\dagger$	0.72	$n3_pf^\dagger$	1.7
$n1_pf^\dagger$	1.8	$n3_pl^\dagger$	1.12323
$n1_pl^\dagger$	1.11823	$n3_pr^\dagger$	0.95
$n1_pr^\dagger$	0.85	$n4_wa^\dagger$	1.2
$n2_wa^\dagger$	1.3	$n4_wf^\dagger$	1.4
$n2_wf^\dagger$	1.3	$n4_ws^\dagger$	1.54049
$n2_ws^\dagger$	1.58009	$n4_pa^\dagger$	0.82
$n2_pa^\dagger$	0.82	$n4_pf^\dagger$	1.6
$n2_pf^\dagger$	1.7	$n4_pl^\dagger$	1.16605
$n2_pl^\dagger$	1.12323	$n4_pr^\dagger$	1.05
$n2_pr^\dagger$	0.95		

Barrel

Variable	Value	Variable	Value
R_barrel_in	5.282	BR_base_t [†]	0.18
r_arbor	1.76067	BR_lid_t [†]	0.2
BR_z	84	BR_H	1.88
BR_m	0.14	BR_lid_step [†]	0.05
BR_f	2.8	BR_hook_h [†]	0.06
BR_d	11.76	BR_hook_w [†]	0.12
BR_da	12.152	BR_shaft_d [†]	1.2
BR_df	11.424	BR_sq [†]	0.9
BR_r_ogive	0.28	BR_arb_top [†]	0.6
BR_half_gap	1.04976	BR_arb_bot [†]	0.4
BR_half_pitch	2.14286	BR_wall [†]	0.43
BR_teeth_t [†]	0.5		

Mainspring

Variable	Value	Variable	Value
ms_t [†]	0.095	MS_Mmax	4.21167
ms_b [†]	1.4	MS_Mrun	3.24826
ms_L	410.05352	MS_sigma	2000
MS_gap [†]	0.005	MS_E [†]	220000
MS_pitch	0.1	MS_sig_d [†]	2000
MS_Ri	3.93697	MS_M0	1.26114
MS_Ro	3.93697	MS_M24	3.05557
MS_nw	22.90846	MS_M40	2.28484
MS_nu	14.15821	MS_r_set	2.65548
MS_N	8.75025	MS_T150	1.4962
MS_K	0.05367		

Second Wheel and Pinion

Variable	Value	Variable	Value
W2_z	60	W2_pivot_L [†]	0.25
W2_m	0.12	W2_arb_ext [†]	0.15
W2_f	2.6	P2_n	12
W2_d	7.2	P2_m	0.14
W2_da	7.512	P2_d	1.68
W2_df	6.888	P2_da	1.8816
W2_r_ogive	0.24	P2_df	1.176
W2_gap_ang	2.98225	P2_leaf	0.15655
W2_half_gap	1.49113	P2_r_tip	0.119
W2_half_pitch	3	P2_half_leaf	5.33915
W2_thk [†]	0.18	P2_L [†]	0.6
W2_rim [†]	0.3	W2_gap_z	0.11

Variable	Value	Variable	Value
W2_rim_id	6.288	W2_arb_front	0.33
W2_hub_d [†]	1.4	W2_arb_back	0.86
W2_spokes [†]	5	W2_piv_front	0.58
W2_spoke_w [†]	0.2	W2_piv_back	1.11
W2_arbor_d [†]	0.4	P2_half_pitch	15
W2_pivot_d [†]	0.12		

Third Wheel and Pinion

Variable	Value	Variable	Value
W3_z	64	W3_pivot_d [†]	0.1
W3_m	0.1	W3_pivot_L [†]	0.25
W3_f	2.6	P3_n	8
W3_d	6.4	P3_m	0.12
W3_da	6.66	P3_d	0.96
W3_df	6.14	P3_da	1.1568
W3_r_ogive	0.2	P3_df	0.552
W3_gap_ang	2.79701	P3_leaf	0.13479
W3_half_gap	1.3985	P3_r_tip	0.114
W3_half_pitch	2.8125	P3_half_leaf	8.04454
W3_thk [†]	0.18	P3_L [†]	0.4
W3_rim [†]	0.3	W3_gap_z	0.21
W3_rim_id	5.54	W3_arb_front	0.33
W3_hub_d [†]	1.4	W3_arb_back	0.76
W3_spokes [†]	5	W3_piv_front	0.58
W3_spoke_w [†]	0.2	W3_piv_back	1.01
W3_arbor_d [†]	0.4	P3_half_pitch	22.5

Fourth Wheel, Pinion and Seconds Hand

Variable	Value	Variable	Value
W4_z	96	P4_m	0.1
W4_m	0.065	P4_d	0.8
W4_f	2.4	P4_da	0.964
W4_d	6.24	P4_df	0.46
W4_da	6.396	P4_leaf	0.11232
W4_df	6.058	P4_r_tip	0.095
W4_r_ogive	0.13	P4_half_leaf	8.04454
W4_gap_ang	1.91118	P4_L [†]	0.4
W4_half_gap	0.95559	W4_gap_z	0.225
W4_half_pitch	1.875	W4_arb_front	1.875
W4_thk [†]	0.15	W4_arb_back	0.775
W4_rim [†]	0.25	W4_piv_front	2.075
W4_rim_id	5.558	W4_piv_back	0.975
W4_hub_d [†]	1.2	SH_L [†]	3.5
W4_spokes [†]	6	SH_tail [†]	0.8

Variable	Value	Variable	Value
W4_spoke_w [†]	0.15	SH_w [†]	0.15
W4_arbor_d [†]	0.36	SH_t [†]	0.05
W4_pivot_d [†]	0.08	SH_hub_d [†]	0.6
W4_pivot_L [†]	0.2	P4_half_pitch	22.5
P4_n	8		

Escape Wheel and Pinion

Variable	Value	Variable	Value
R_esc [†]	2.2	E_pivot_d [†]	0.07
E_z	15	E_pivot_L [†]	0.15
E_R	2.2	PE_n	6
E_pitch	24	PE_m	0.065
E_h [†]	0.55	PE_d	0.39
E_root_r	1.65	PE_da	0.4966
E_lock_face [†]	24	PE_df	0.182
E_club_ang [†]	5	PE_leaf	0.07579
E_club_rise [†]	0.055	PE_r_tip	0.06825
E_tip_d	4.51	PE_half_leaf	11.13496
E_back_R [†]	1.2	PE_L [†]	0.4
E_back_ang [†]	16	E_gap_z	0.225
E_thk [†]	0.15	E_arb_front	0.3
E_rim [†]	0.18	E_arb_back	0.775
E_rim_id	2.94	E_piv_front	0.45
E_hub_d [†]	0.9	E_piv_back	0.925
E_spokes [†]	5	E_T	3.3836×10^{-4}
E_spoke_w [†]	0.12	E_club_drop [†]	0.06
E_arbor_d [†]	0.16	PE_half_pitch	30

Pallet Fork and Banking Pins

Variable	Value	Variable	Value
span_deg [†]	60	L_jewel_cut_off [†]	0
a_pallet	2.54034	L_jewel_cut_R	0.47631
lever_angle [†]	10.5	L_jewel_cut_D	0.95262
lift_angle [†]	52	L_jewel_fil [†]	0.01
L_a	2.54034	L_slot_w	0.16
L_tan	1.27017	L_slot_depth [†]	0.25
L_lock [†]	1.5	L_horn_len [†]	0.3
L_imp_tooth [†]	2.5	L_arm_w [†]	0.3
L_imp_pallet [†]	6	L_body_w [†]	0.6
L_slide [†]	0.5	L_body_ang [†]	42
L_total	10.5	L_lever_w [†]	0.22
L_bank	5.25	L_arbor_d [†]	0.2
L_drop [†]	1.7	L_pivot_d [†]	0.07
L_draw [†]	12	L_pivot_L [†]	0.15

Variable	Value	Variable	Value
L_stone_w [†]	0.25	L_arb_up [†]	0.35
L_stone_len [†]	0.8	L_arb_dn [†]	0.3
L_plate_t [†]	0.15	L_guard_d [†]	0.05
L_ew_dz	0.19	L_guard_h [†]	0.16
L_stone_up	0.075	L_fork_end	2.3
L_stone_dn	0.295	L_head_w [†]	0.4
L_c_out	0.08313	L_horn_y [†]	0.38
L_imp_in [†]	0.078	L_horn_dx [†]	0.05
L_imp_out [†]	0.115	L_dart_x0 [†]	2
L_stone_imp_in	17.32792	BK_x [†]	1.2
L_stone_imp_out	24.70243	BK_d [†]	0.16
L_fork_len [†]	2.2	BK_y	0.30106
L_jewel_r	0.47114	L_dart_tip	2.39517
L_pb	2.67114	L_stone_h [†]	0.25
L_jewel_d [†]	0.14	L_stone_imp	21.75377

Balance and Hairspring

Variable	Value	Variable	Value
B_R_o [†]	4	B_safe_t [†]	0.08
B_R_i [†]	3.6	B_safe_z [†]	0.15
B_rim_h [†]	0.3	B_cres_r [†]	0.12
B_arms [†]	3	B_I [†]	3.6
B_arm_w [†]	0.25	B_k	2.274×10^{-4}
B_arm_t [†]	0.15	B_k_deg	3.9688×10^{-6}
B_hub_d [†]	0.8	HS_E [†]	190000
B_rim_z [†]	0.65	HS_b [†]	0.1
B_staff_d [†]	0.3	HS_t [†]	0.025
B_pivot_d [†]	0.08	HS_L	108.79531
B_pivot_L [†]	0.15	HS_r0 [†]	0.45
B_staff_top [†]	1.3	HS_r1 [†]	2.6
B_staff_bot [†]	0.4	HS_turns	11.3543
B_roll_r	0.59114	HS_pitch	0.18936
B_roll_t [†]	0.1	HS_end_ang	127.54879
B_roll_z [†]	0.18	HS_z [†]	1.05
B_jewel_len [†]	0.25	HS_collet_d [†]	0.9
B_safe_r [†]	0.34	HS_collet_h [†]	0.2

Efficiencies, Amplitude and Power Reserve

Variable	Value	Variable	Value
eta_train [†]	0.7	amp_full	251.66505
eta_gears [†]	0.8617	amp_24	214.35927
Q_bal [†]	250	amp_40	185.36352
eta_esc [†]	0.3	PR_h	56.37185

Hour and Minute Display Train

Variable	Value	Variable	Value
MW_m [†]	0.12	a_mw_chk	2.4
z_cp [†]	10	d_cp	1.2
Z_mw [†]	30	d_mw	3.6
z_mp [†]	8	d_mp	0.96
Z_hw [†]	32	d_hw	3.84
R_mw	12	X_MW	6.72
a_mw	2.4	Y_MW	-2.4

Assembly Positions

Variable	Value	Variable	Value
X_B [†]	0	Z_L5	2
X_2	6.72	Z_oB	0.25
X_3	10.8	Z_o2	0.41
X_4	14.4	Z_o3	0.91
X_E	17.715	Z_o4	1.425
X_P	20.25534	Z_oE	1.925
X_Ba1	22.92648	Z_oP	2.19
Z_L1 [†]	0	Z_oBa1	2.19
Z_L2	0.5	Z_L6	2.5
Z_L3	1	Z_L7	3
Z_L4	1.5		

Summary of Design Changes

Table 54 lists the independent inputs that changed between the first complete version of the model and the present one, with the calculation that required each change.

Table 54: Inputs that changed as a result of the analysis.

Variable	First version	Now	Reason
m_b2	0.120	0.140	barrel volume for a 40-hour reserve (Section 8.1)
ms_b	1.00	1.40	torque for the amplitude targets (Section 9.2)
MS_E	200 000	220 000	manufacturer's value for Nivaflex 45/5
MS_sig_d	none	2000	stress cap that sets the full-wind torque
BR_wall	implicit	0.43	makes the barrel radius follow the module
Z2	54	60	second wheel once per hour (Section 5)
n1_* to n4_*, k_og	one shared estimated set	one set per mesh	NIHS 20-30 (Section 6)
BR_teeth_t	0.25	0.50	contact pressure on the barrel teeth (Section 7.3)
P2_L	0.40	0.60	covers the wider barrel teeth
W2_pivot_d	0.10	0.12	bearing pressure on the second pivot (Section 10.3)
BR_lid_t	0.10	0.20	plate bending under shock (Section 10.8)
BR_base_t	0.15	0.18	plate bending under shock (Section 10.8)
tol_a_13, tol_a_4	none	0.010, 0.005	Monte Carlo study (Section 7)
eta_train, eta_esc, Q_bal	literals or none	0.70, 0.30, 250	named so that the amplitude check updates with the model